Problem 5-1

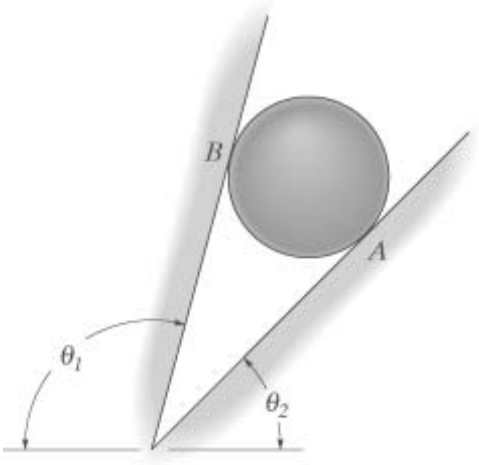
Draw the free-body diagram of the sphere of weight *W* resting between the smooth inclined planes. Explain the significance of each force on the diagram.

Given:

$W = 10$ lb

$\theta_1 = 105$ deg

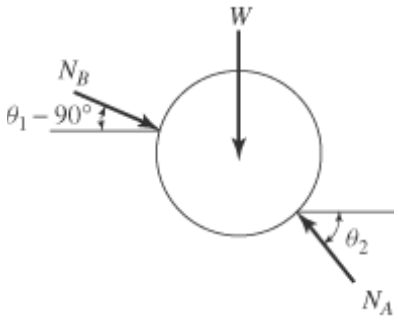
$\theta_2 = 45$ deg



Solution:

N_A, N_B force of plane on sphere.

W force of gravity on sphere.



Problem 5-2

Draw the free-body diagram of the hand punch, which is pinned at *A* and bears down on the smooth surface at *B*.

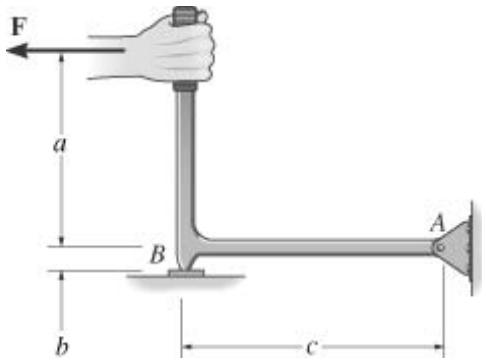
Given:

$F = 8$ lb

$a = 1.5$ ft

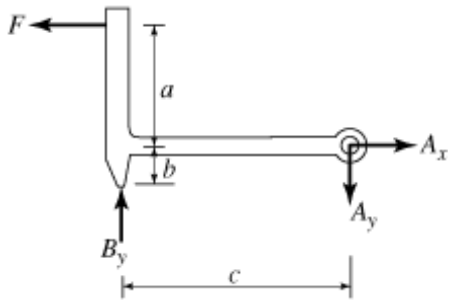
$b = 0.2$ ft

$c = 2$ ft



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Solution:



Problem 5-3

Draw the free-body diagram of the beam supported at *A* by a fixed support and at *B* by a roller. Explain the significance of each force on the diagram.

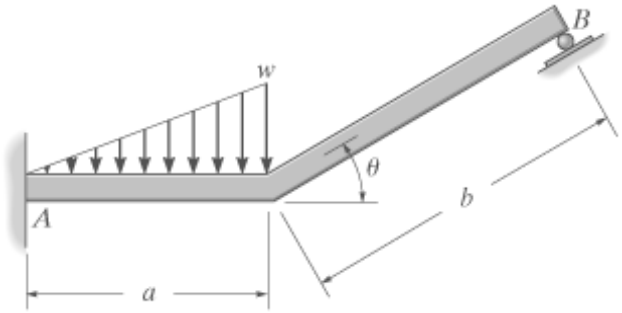
Given:

$w = 40 \ \frac{\text{lb}}{\text{ft}}$

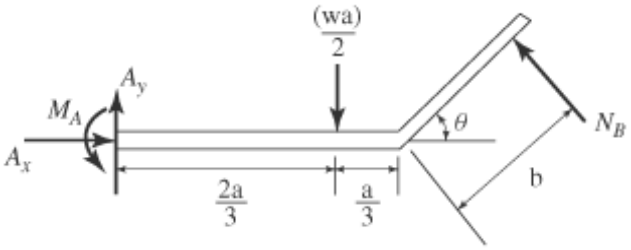
$a = 3$ ft

$b = 4$ ft

$\theta = 30$ deg



Solution:



A_x, A_y, M_A effect of wall on beam.

N_B force of roller on beam.

$\frac{wa}{2}$ resultant force of distributed load on beam.

Problem 5-4

Draw the free-body diagram of the jib crane *AB*, which is pin-connected at *A* and supported by member (link) *BC*.

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Units Used:

kN = 10^3 N

Given:

F = 8 kN

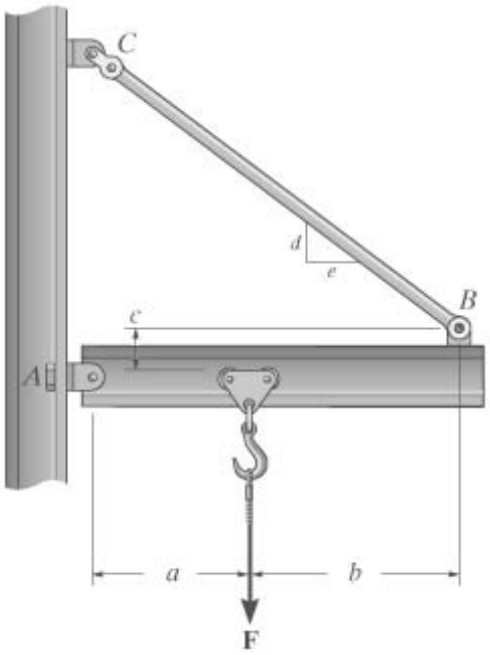
a = 3 m

b = 4 m

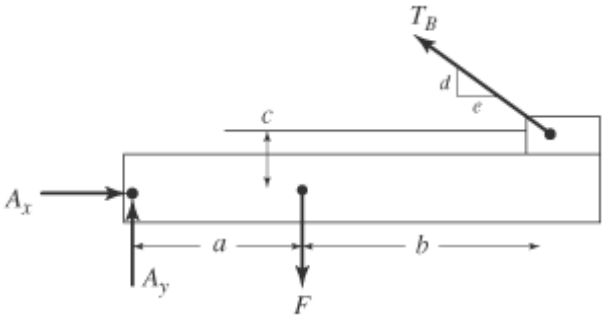
c = 0.4 m

d = 3

e = 4

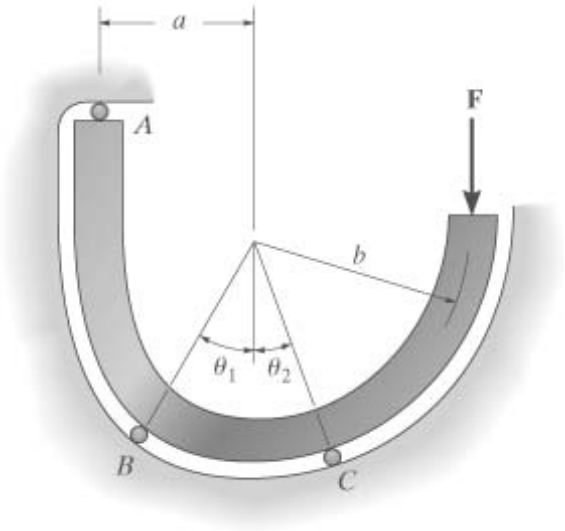


Solution:



Problem 5-5

Draw the free-body diagram of the *C*-bracket supported at *A*, *B*, and *C* by rollers. Explain the significance of each forcce on the diagram.



Given:

a = 3 ft

b = 4 ft

θ_1 = 30 deg

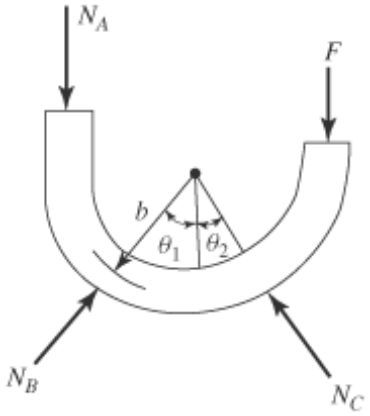
θ_2 = 20 deg

F = 200 lb

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Solution:

N_A, N_B, N_C force of rollers on beam.



Problem 5-6

Draw the free-body diagram of the smooth rod of mass M which rests inside the glass. Explain the significance of each force on the diagram.

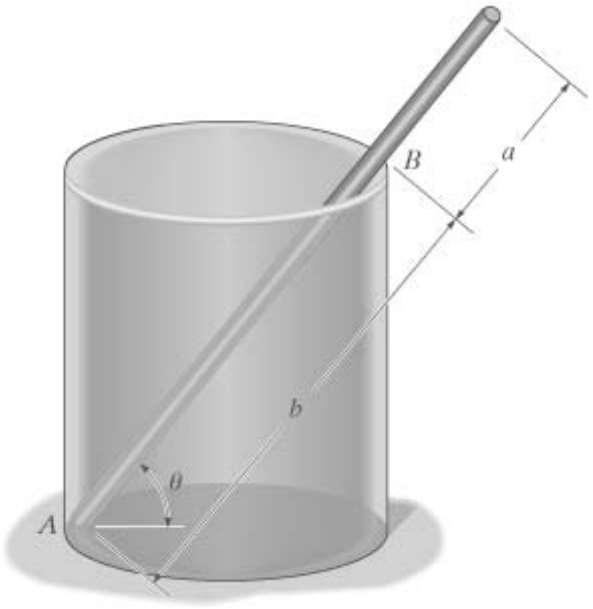
Given:

$M = 20$ gm

$a = 75$ mm

$b = 200$ mm

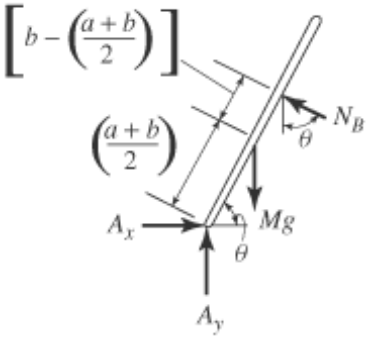
$\theta = 40$ deg



Solution:

A_x, A_y, N_B force of glass on rod.

$M(g)$ N force of gravity on rod.



Problem 5-7

Draw the free-body diagram of the "spanner wrench" subjected to the force **F**. The support at A can be considered a pin, and the surface of contact at B is smooth. Explain the significance of

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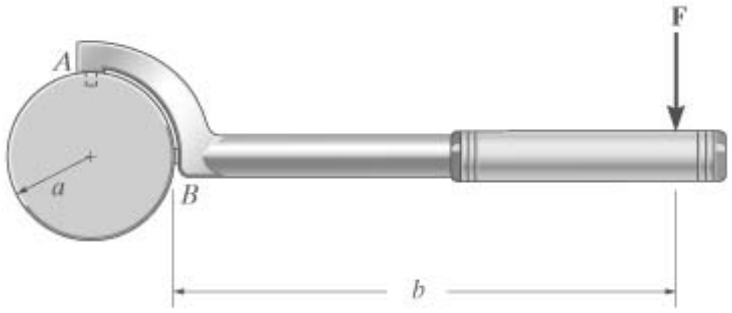
each force on the diagram.

Given:

$F = 20$ lb

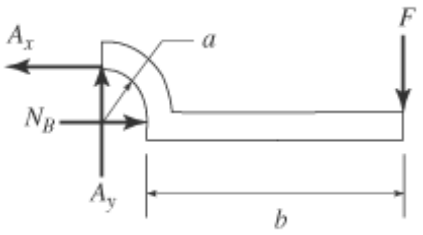
$a = 1$ in

$b = 6$ in



Solution:

A_x, A_y, N_B force of cylinder on wrench.



Problem 5-8

Draw the free-body diagram of the automobile, which is being towed at constant velocity up the incline using the cable at *C*. The automobile has a mass *M* and center of mass at *G*. The tires are free to roll. Explain the significance of each force on the diagram.

Units Used:

$\text{Mg} = 10^3$ kg

Given:

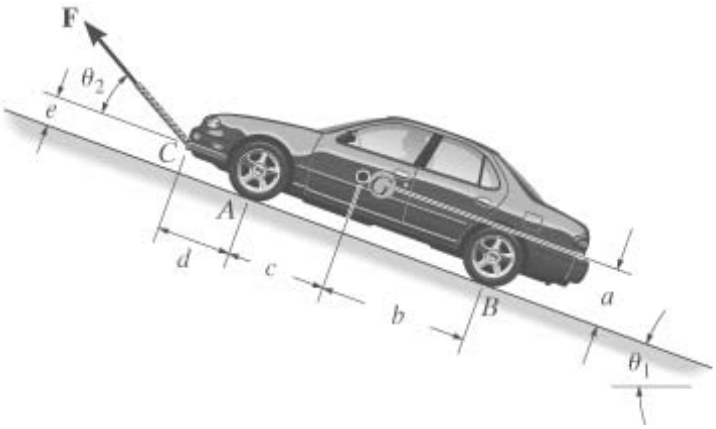
$M = 5$ Mg $\quad d = 1.50$ m

$a = 0.3$ m $\quad e = 0.6$ m

$b = 0.75$ m $\quad \theta_1 = 20$ deg

$c = 1$ m $\quad \theta_2 = 30$ deg

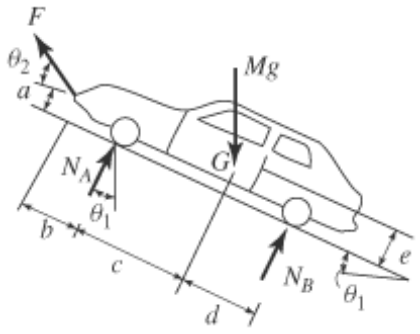
$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



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Solution:

$\mathbf{N_A}$, $\mathbf{N_B}$ force of road on car.
F force of cable on car.
Mg force of gravity on car.



Problem 5-9

Draw the free-body diagram of the uniform bar, which has mass *M* and center of mass at *G*. The supports *A*, *B*, and *C* are smooth.

Given:

$M = 100$ kg

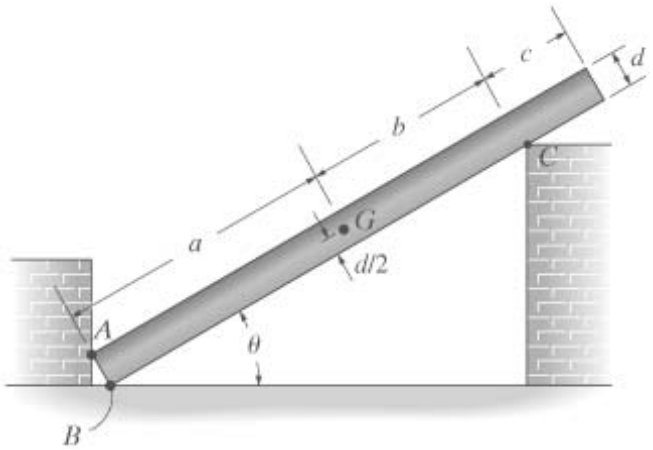
$a = 1.75$ m

$b = 1.25$ m

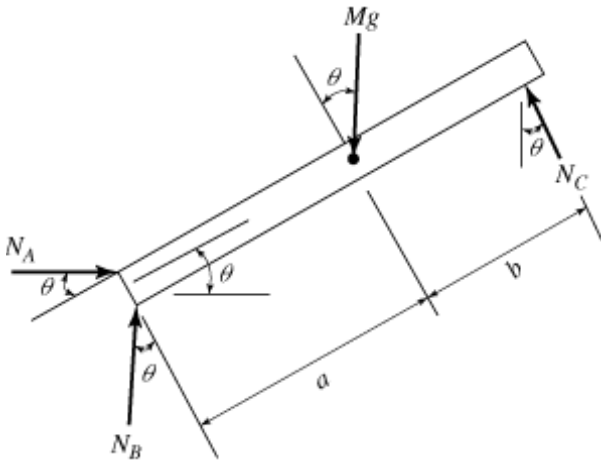
$c = 0.5$ m

$d = 0.2$ m

$g = 9.81 \ \frac{\text{m}}{\text{s}^2}$



Solution:



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Problem 5-10

Draw the free-body diagram of the beam, which is pin-connected at *A* and rocker-supported at *B*.

Given:

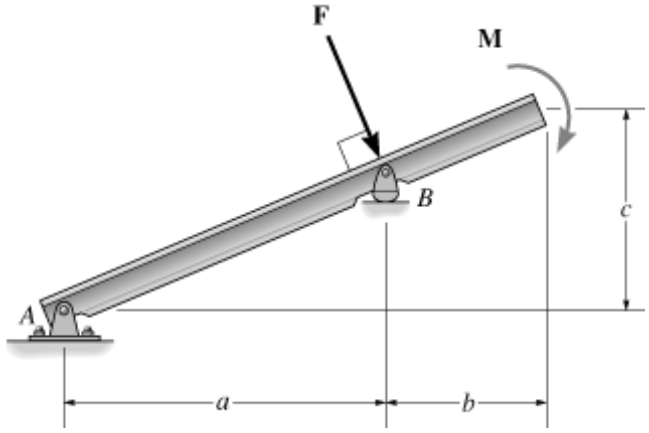
$F = 500$ N

$M = 800$ N·m

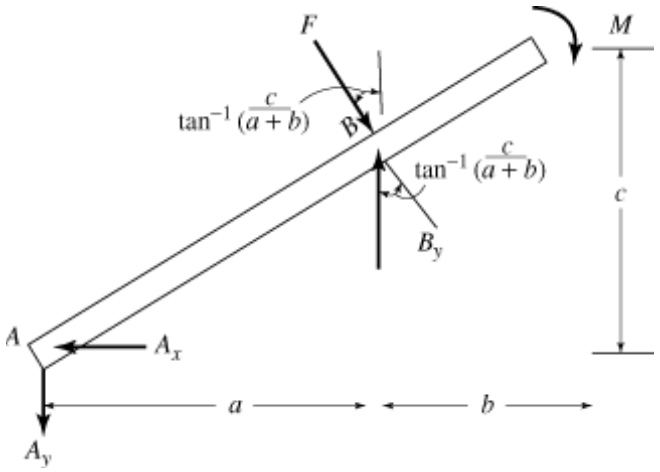
$a = 8$ m

$b = 4$ m

$c = 5$ m

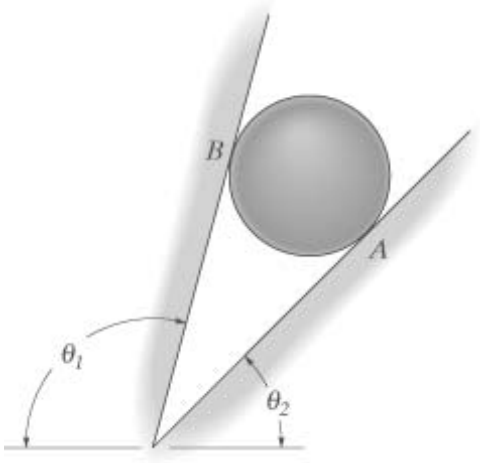


Solution:



Problem 5-11

The sphere of weight *W* rests between the smooth inclined planes. Determine the reaactions at the supports.



Given:

$W = 10$ lb

$\theta_1 = 105$ deg

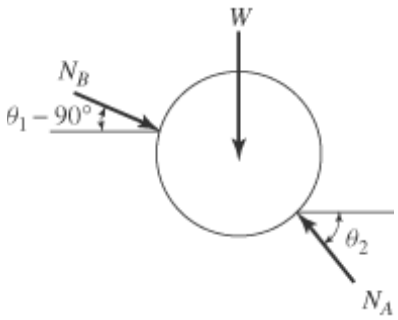
$\theta_2 = 45$ deg

Solution:

Initial guesses

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$N_A = 1$ lb $\quad N_B = 1$ lb



Given

$$N_B \cos(\theta_1 - 90\ \text{deg}) - N_A \cos(\theta_2) = 0$$

$$N_A \sin(\theta_2) - N_B \sin(\theta_1 - 90\ \text{deg}) - W = 0$$

$$\begin{pmatrix} N_A \\ N_B \end{pmatrix} = \text{Find}(N_A, N_B) \qquad \begin{pmatrix} N_A \\ N_B \end{pmatrix} = \begin{pmatrix} 19.3 \\ 14.1 \end{pmatrix} \text{lb}$$

Problem 5-12

Determine the magnitude of the resultant force acting at pin *A* of the handpunch.

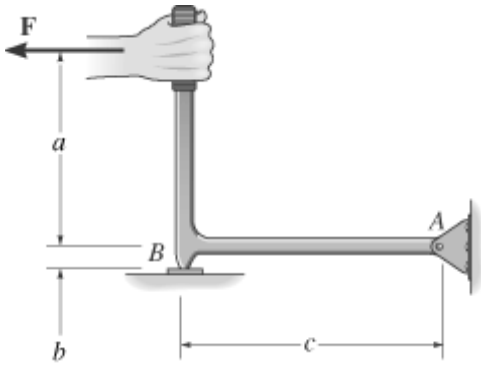
Given:

$F = 8$ lb

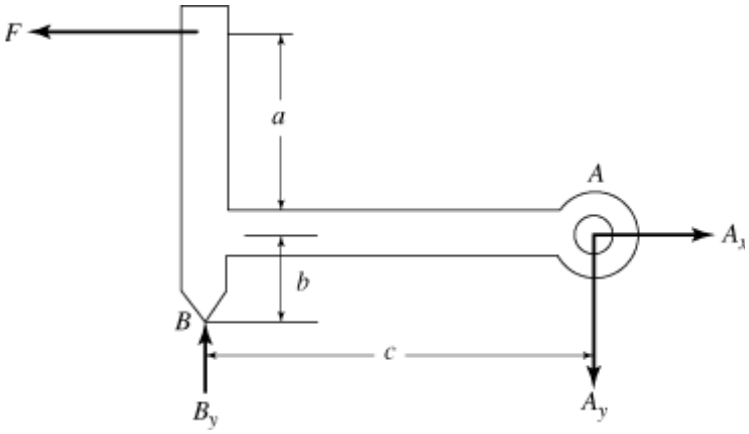
$a = 1.5$ ft

$b = 0.2$ ft

$c = 2$ ft



Solution:



$\Sigma F_x = 0;$ $\quad A_x - F = 0$ $\quad A_x = F$ $\quad A_x = 8\,\text{lb}$

$\Sigma M = 0;$ $\quad F\,a - A_y\,c = 0$ $\quad A_y = F\,\dfrac{a}{c}$ $\quad A_y = 6\,\text{lb}$

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$$F_A = \sqrt{A_x^2 + A_y^2} \qquad F_A = 10\,\text{lb}$$

Problem 5-13

The *C*-bracket is supported at *A*, *B*, and *C* by rollers. Determine the reactions at the supports.

Given:

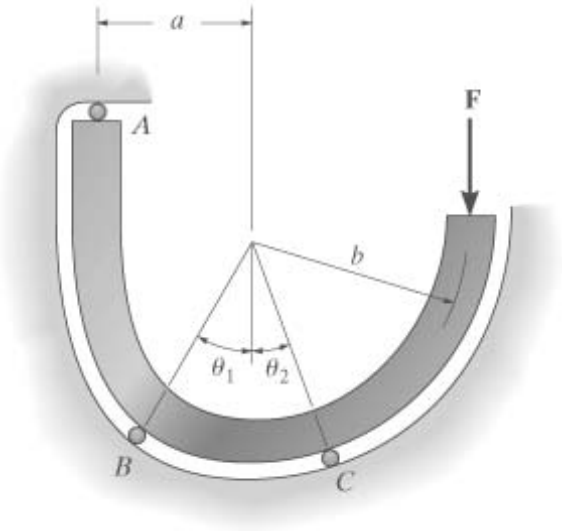
$a = 3$ ft

$b = 4$ ft

$\theta_1 = 30$ deg

$\theta_2 = 20$ deg

$F = 200$ lb



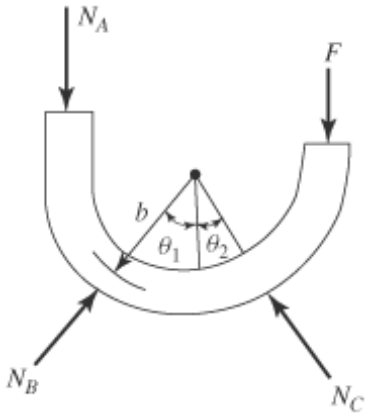
Solution:

Initial Guesses:

$N_A = 1$ lb

$N_B = 1$ lb

$N_C = 1$ lb



Given

$N_A a - F b = 0$

$N_B \sin(\theta_1) - N_C \sin(\theta_2) = 0$

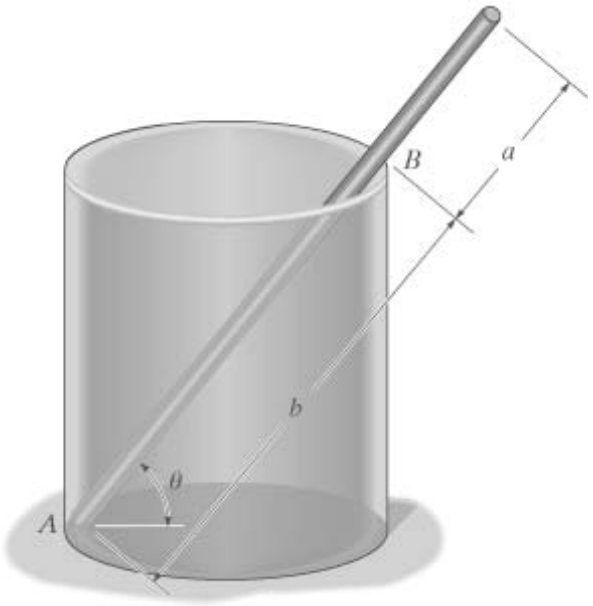
$N_B \cos(\theta_1) + N_C \cos(\theta_2) - N_A - F = 0$

$$\begin{pmatrix} N_A \\ N_B \\ N_C \end{pmatrix} = \text{Find}(N_A, N_B, N_C) \qquad \begin{pmatrix} N_A \\ N_B \\ N_C \end{pmatrix} = \begin{pmatrix} 266.7 \\ 208.4 \\ 304.6 \end{pmatrix} \text{lb}$$

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Problem 5-14

The smooth rod of mass M rests inside the glass. Determine the reactions on the rod.



Given:

$M = 20\ \text{gm}$

$a = 75\ \text{mm}$

$b = 200\ \text{mm}$

$\theta = 40\ \text{deg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

Solution:

Initial Guesses:

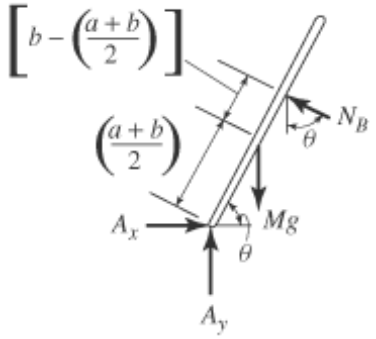
$A_x = 1\ \text{N} \quad A_y = 1\ \text{N} \quad N_B = 1\ \text{N}$

Given

$A_x - N_B \sin(\theta) = 0$

$A_y - M g + N_B \cos(\theta) = 0$

$-M g \frac{a+b}{2} \cos(\theta) + N_B b = 0$



$$\begin{pmatrix} A_x \\ A_y \\ N_B \end{pmatrix} = \text{Find}(A_x, A_y, N_B) \qquad \begin{pmatrix} A_x \\ A_y \\ N_B \end{pmatrix} = \begin{pmatrix} 0.066 \\ 0.117 \\ 0.103 \end{pmatrix} \text{N}$$

Problem 5-15

The "spanner wrench" is subjected to the force **F**. The support at A can be considered a pin, and the surface of contact at B is smooth. Determine the reactions on the spanner wrench.

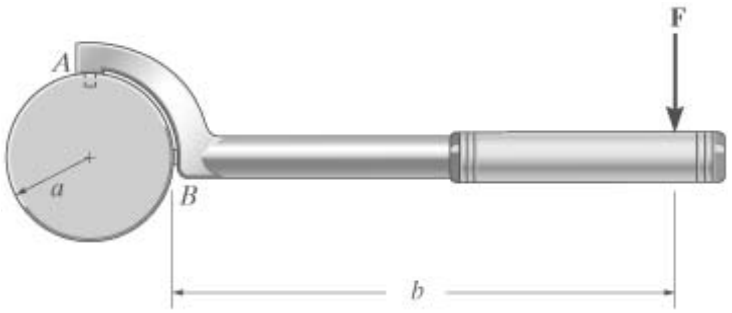
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Given:

$F = 20$ lb

$a = 1$ in

$b = 6$ in



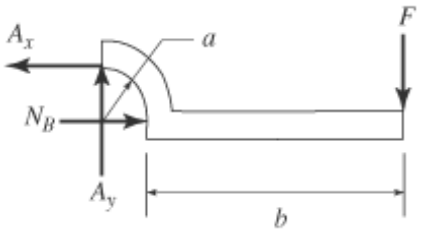
Solution:

Initial Guesses:

$A_x = 1$ lb

$A_y = 1$ lb

$N_B = 1$ lb



Given

$$-A_x + N_B = 0 \qquad A_y - F = 0 \qquad -F\,(a+b) + A_x\,a = 0$$

$$\begin{pmatrix} A_x \\ A_y \\ N_B \end{pmatrix} = \text{Find}\left(A_x, A_y, N_B\right) \qquad \begin{pmatrix} A_x \\ A_y \\ N_B \end{pmatrix} = \begin{pmatrix} 140 \\ 20 \\ 140 \end{pmatrix} \text{lb}$$

Problem 5-16

The automobile is being towed at constant velocity up the incline using the cable at *C*. The automobile has a mass *M* and center of mass at *G*. The tires are free to roll. Determine the reactions on both wheels at *A* and *B* and the tension in the cable at *C*.

Units Used:

$\text{Mg} = 10^3$ kg kN $= 10^3$ N

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Given:

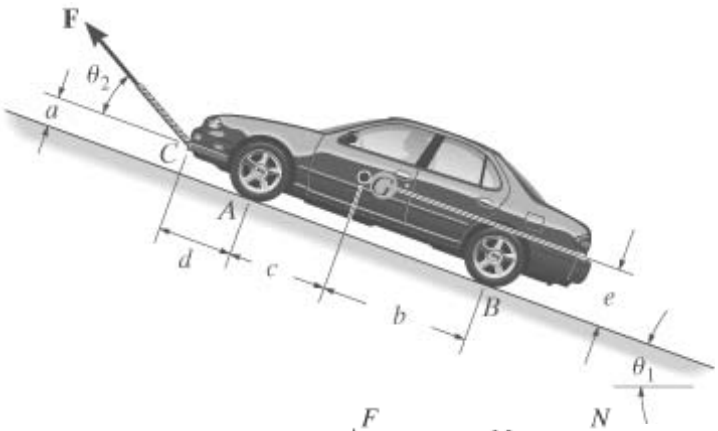
$M = 5\ \text{Mg}$ $d = 1.50\ \text{m}$

$a = 0.3\ \text{m}$ $e = 0.6\ \text{m}$

$b = 0.75\ \text{m}$ $\theta_1 = 20\ \text{deg}$

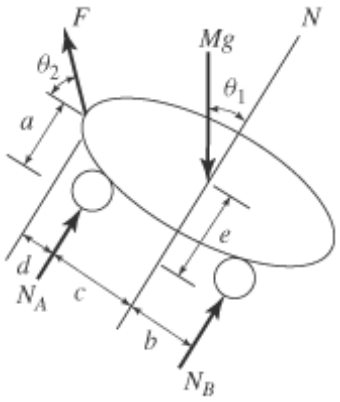
$c = 1\ \text{m}$ $\theta_2 = 30\ \text{deg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$



Solution:

Guesses $F = 1\ \text{kN}$ $N_A = 1\ \text{kN}$ $N_B = 1\ \text{kN}$



Given

$$N_A + N_B + F\ \sin(\theta_2) - M\ g\ \cos(\theta_1) = 0$$

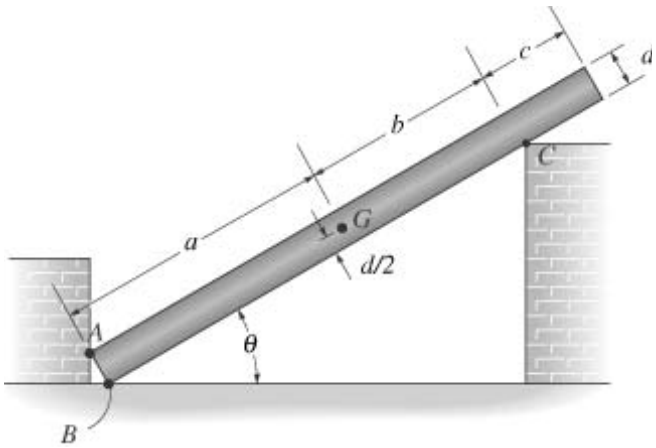
$$-F\ \cos(\theta_2) + M\ g\ \sin(\theta_1) = 0$$

$$F\ \cos(\theta_2)\ a - F\ \sin(\theta_2)\ b - M\ g\ \cos(\theta_1)\ c - M\ g\ \sin(\theta_1)\ e + N_B(c + d) = 0$$

$$\begin{pmatrix} F \\ N_A \\ N_B \end{pmatrix} = \text{Find}(F, N_A, N_B) \qquad \begin{pmatrix} F \\ N_A \\ N_B \end{pmatrix} = \begin{pmatrix} 19.37 \\ 13.05 \\ 23.36 \end{pmatrix} \text{kN}$$

Problem 5-17

The uniform bar has mass M and center of mass at G. The supports A, B, and C are smooth. Determine the reactions at the points of contact at A, B, and C.



Given:

$M = 100\ \text{kg}$

$a = 1.75\ \text{m}$

$b = 1.25\ \text{m}$

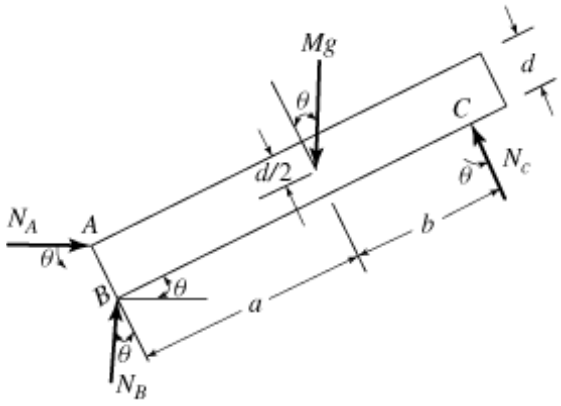
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$c = 0.5$ m

$d = 0.2$ m

$\theta = 30$ deg

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

The initial guesses:

$N_A = 20$ N

$N_B = 30$ N

$N_C = 40$ N

Given

$\circlearrowleft + \Sigma M_A = 0; \quad -Mg\cos(\theta)a - Mg\sin(\theta)\dfrac{d}{2} + N_B \sin(\theta)d + N_C(a+b) = 0$

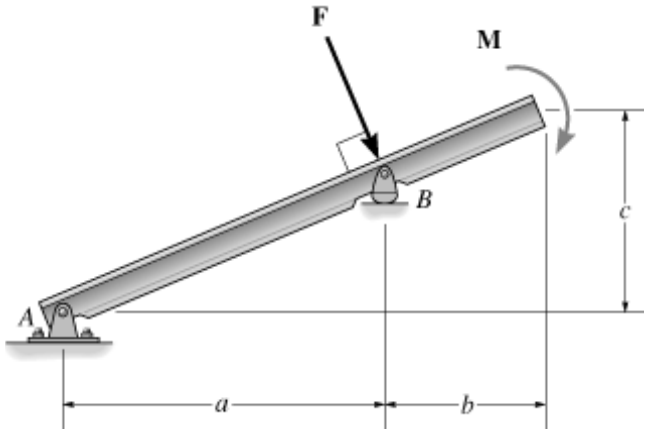
$+\uparrow \Sigma F_y = 0; \quad N_B - Mg + N_C \cos(\theta) = 0$

$+\uparrow \Sigma F_y = 0; \quad N_A - N_C \sin(\theta) = 0$

$\begin{pmatrix} N_C \\ N_B \\ N_A \end{pmatrix} = \text{Find}(N_C, N_B, N_A)$ $\quad \begin{pmatrix} N_C \\ N_B \\ N_A \end{pmatrix} = \begin{pmatrix} 493 \\ 554 \\ 247 \end{pmatrix}$ N

Problem 5-18

The beam is pin-connected at *A* and rocker-supported at *B*. Determine the reactions at the pin *A* and at the roller at *B*.



Given:

$F = 500$ N

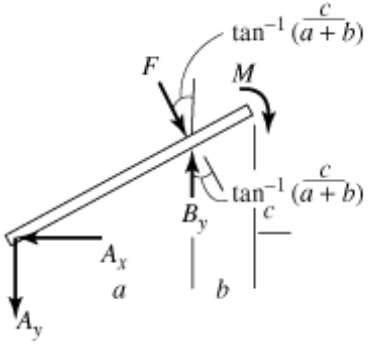
$M = 800$ N·m

$a = 8$ m

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$b = 4$ m

$c = 5$ m



Solution:

$$\alpha = \text{atan}\left(\frac{c}{a+b}\right)$$

$$\circlearrowleft + \Sigma M_A = 0; \qquad -F\frac{a}{\cos(\alpha)} - M + B_y a = 0$$

$$B_y = \frac{F\,a + M\cos(\alpha)}{\cos(\alpha)\,a}$$

$$B_y = 642 \text{ N}$$

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad -A_x + F\sin(\alpha) = 0 \qquad A_x = F\sin(\alpha) \qquad A_x = 192 \text{ N}$$

$$+\uparrow \Sigma F_y = 0; \qquad -A_y - F\cos(\alpha) + B_y = 0 \qquad A_y = -F\cos(\alpha) + B_y \qquad A_y = 180 \text{ N}$$

Problem 5-19

Determine the magnitude of the reactions on the beam at *A* and *B*. Neglect the thickness of the beam.

Given:

$F_1 = 600$ N

$F_2 = 400$ N

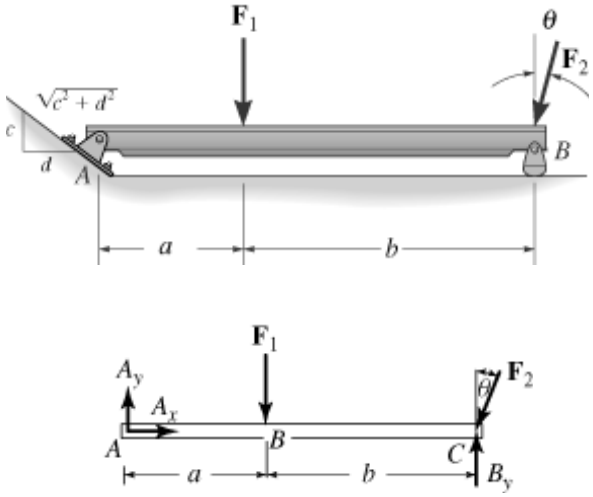
$\theta = 15$ deg

$a = 4$ m

$b = 8$ m

$c = 3$

$d = 4$



Solution:

$$\circlearrowleft + \Sigma M_A = 0; \qquad B_y(a+b) - F_2\cos(\theta)(a+b) - F_1 a = 0$$

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$$B_y = \frac{F_2 \cos(\theta)\,(a+b) + F_1\,a}{a+b} \qquad B_y = 586 \text{ N}$$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad A_x - F_2 \sin(\theta) = 0$

$$A_x = F_2 \sin(\theta) \qquad A_x = 104 \text{ N}$$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - F_2 \cos(\theta) + B_y - F_1 = 0$

$$A_y = F_2 \cos(\theta) - B_y + F_1 \qquad A_y = 400 \text{ N}$$

$$F_A = \sqrt{A_x^2 + A_y^2} \qquad F_A = 413 \text{ N}$$

Problem 5-20

Determine the reactions at the supports.

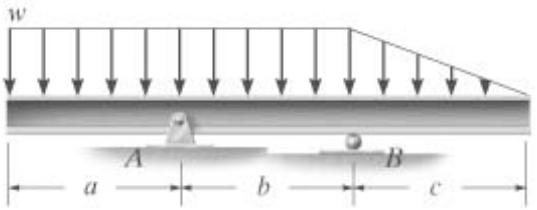
Given:

$w = 250 \dfrac{\text{lb}}{\text{ft}}$

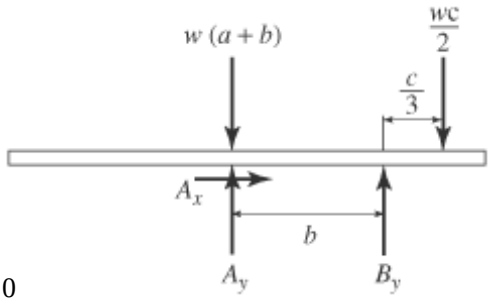
$a = 6$ ft

$b = 6$ ft

$c = 6$ ft



Solution:



Guesses

$A_x = 1$ lb $\quad A_y = 1$ lb $\quad B_y = 1$ lb

Given

$A_x = 0$ $\quad A_y + B_y - w(a+b) - \dfrac{1}{2}\,wc = 0$

$$wa\left(\frac{a}{2}\right) - wb\left(\frac{b}{2}\right) - \frac{1}{2}wc\left(b + \frac{c}{3}\right) + B_y\,b = 0$$

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$$\begin{pmatrix} A_x \\ A_y \\ B_y \end{pmatrix} = \text{Find}(A_x, A_y, B_y) \qquad \begin{pmatrix} A_x \\ A_y \\ B_y \end{pmatrix} = \begin{pmatrix} 0 \\ 2750 \\ 1000 \end{pmatrix} \text{lb}$$

Problem 5-21

When holding the stone of weight *W* in equilibrium, the humerus *H*, assumed to be smooth, exerts normal forces $\mathbf{F}_C$ and $\mathbf{F}_A$ on the radius *C* and ulna *A* as shown. Determine these forces and the force $\mathbf{F}_B$ that the biceps *B* exerts on the radius for equilibrium. The stone has a center of mass at *G*. Neglect the weight of the arm.

Given:

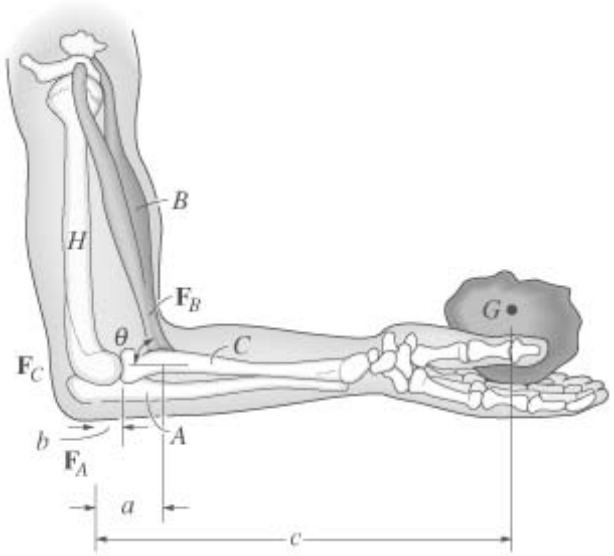
$W = 5$ lb

$\theta = 75$ deg

$a = 2$ in

$b = 0.8$ in

$c = 14$ in

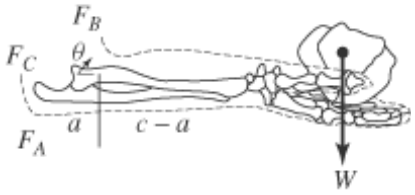


Solution:

$$\circlearrowleft + \Sigma M_B = 0; \quad -W(c-a) + F_A\, a = 0$$

$$F_A = W\left(\frac{c-a}{a}\right)$$

$$F_A = 30\,\text{lb}$$



$$+\uparrow \Sigma F_y = 0; \quad F_B \sin(\theta) - W - F_A = 0$$

$$F_B = \frac{W + F_A}{\sin(\theta)}$$

$$F_B = 36.2\,\text{lb}$$

$$\xrightarrow{+} \Sigma F_x = 0; \quad F_C - F_B \cos(\theta) = 0$$

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$$F_C = F_B \cos(\theta)$$

$$F_C = 9.378\,\text{lb}$$

Problem 5-22

The uniform door has a weight *W* and a center of gravity at *G*. Determine the reactions at the hinges if the hinge at *A* supports only a horizontal reaction on the door, whereas the hinge at *B* exerts both horizontal and vertical reactions.

Given:

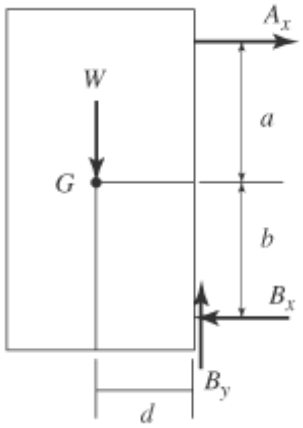
$W = 100$ lb

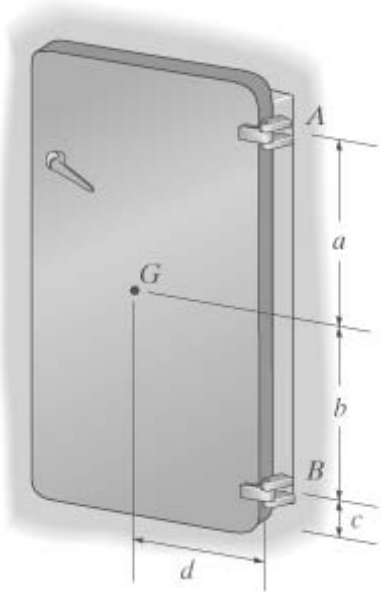
$a = 3$ ft

$b = 3$ ft

$c = 0.5$ ft

$d = 2$ ft





Solution:

$\Sigma M_B = 0; \quad W d - A_x (a + b) = 0$

$$A_x = W\left(\frac{d}{a+b}\right) \qquad A_x = 33.3\,\text{lb}$$

$\Sigma F_x = 0; \quad B_x = A_x \qquad B_x = 33.3\,\text{lb}$

$\Sigma F_y = 0; \quad B_y = W \qquad B_y = 100\,\text{lb}$

Problem 5-23

The ramp of a ship has weight W and center of gravity at *G*. Determine the cable force in *CD* needed to just start lifting the ramp, (i.e., so the reaction at *B* becomes zero). Also, determine the horizontal and vertical components of force at the hinge (pin) at *A*.

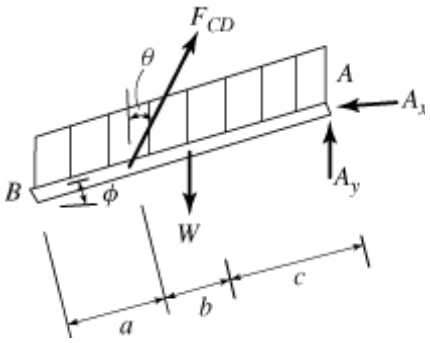
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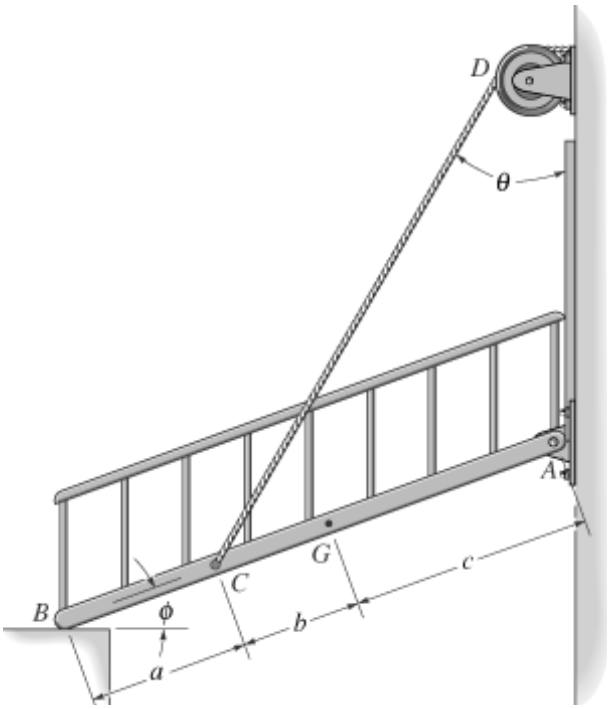
Given:

$W = 200$ lb $\quad a = 4$ ft

$\theta = 30$ deg $\quad b = 3$ ft

$\phi = 20$ deg $\quad c = 6$ ft





Solution:

$\Sigma M_A = 0;$

$$-F_{CD}\cos(\theta)(b+c)\cos(\phi) + F_{CD}\sin(\theta)(b+c)\sin(\phi) + W c\cos(\phi) = 0$$

$$F_{CD} = \frac{W\,c\,\cos(\phi)}{(b+c)(\cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi))} \qquad F_{CD} = 195\,\text{lb}$$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad F_{CD}\sin(\theta) - A_x = 0$

$$A_x = F_{CD}\sin(\theta) \qquad A_x = 97.5\,\text{lb}$$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - W + F_{CD}\cos(\theta) = 0$

$$A_y = W - F_{CD}\cos(\theta) \qquad A_y = 31.2\,\text{lb}$$

Problem 5-24

The drainpipe of mass M is held in the tines of the fork lift. Determine the normal forces at A and B as functions of the blade angle θ and plot the results of force (ordinate) versus θ (abscissa) for $0 \le \theta \le 90$ deg.

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Units used:

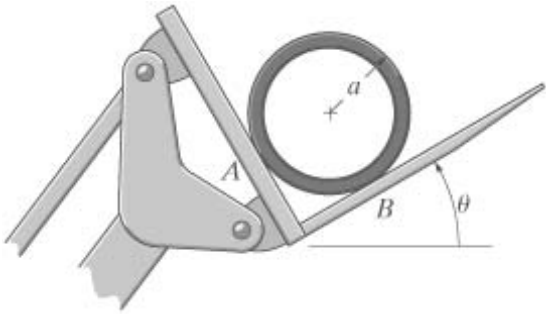
$$\text{Mg} = 10^3 \text{ kg}$$

Given:

$$M = 1.4 \text{ Mg}$$

$$a = 0.4 \text{ m}$$

$$g = 9.81 \frac{\text{m}}{\text{s}^2}$$

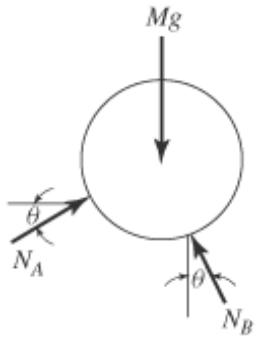


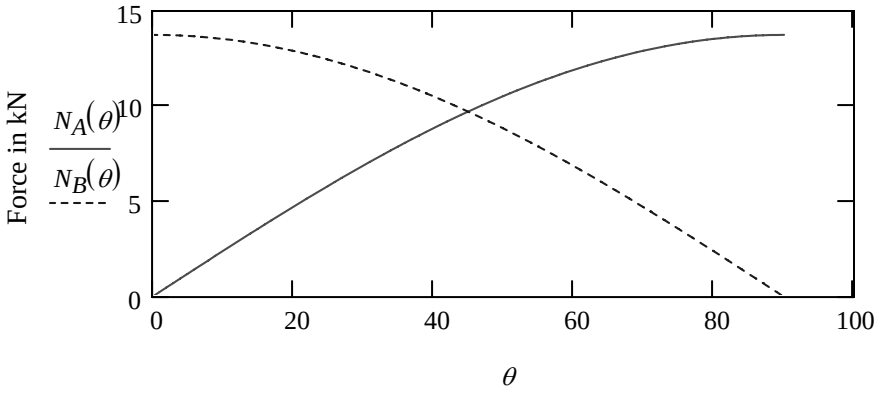
Solution:

$$\theta = 0 .. 90$$

$$N_A(\theta) = \frac{M g \sin(\theta \text{ deg})}{10^3}$$

$$N_B(\theta) = \frac{M g \cos(\theta \text{ deg})}{10^3}$$





Problem 5-25

While *slowly* walking, a man having a total mass *M* places all his weight on *one foo*t. Assuming that the normal force $\mathbf{N}_C$ of the ground acts on his foot at *C*, determine the resultant vertical compressive force $\mathbf{F}_B$ which the tibia *T* exerts on the astragalus *B*, and the vertical tension $\mathbf{F}_A$ in the achilles tendon *A* at the instant shown.

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Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

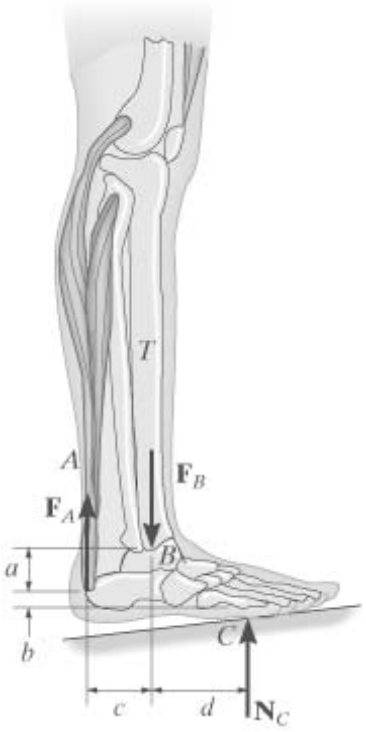
$M = 80\ \text{kg}$

$a = 15\ \text{mm}$

$b = 5\ \text{mm}$

$c = 20\ \text{mm}$

$d = 100\ \text{mm}$



Solution:

$N_C = Mg$

$N_C = 785\ \text{N}$

$\Sigma M_A = 0;\quad -F_B\,c + N_C\,(c + d) = 0$

$F_B = N_C\left(\frac{c + d}{c}\right)$

$F_B = 4.71\,\text{kN}$

$\Sigma F_y = 0;\quad F_A - F_B + N_C = 0$

$F_A = F_B - N_C$

$F_A = 3.92\,\text{kN}$

Problem 5-26

Determine the reactions at the roller *A* and pin *B*.

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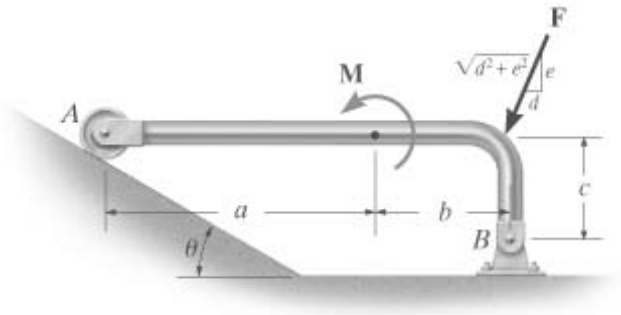
Given:

$M = 800\ \text{lb ft}$ $c = 3\ \text{ft}$

$F = 390\ \text{lb}$ $d = 5$

$a = 8\ \text{ft}$ $e = 12$

$b = 4\ \text{ft}$ $\theta = 30\ \text{deg}$



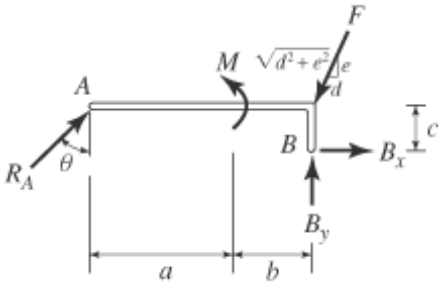
Solution:

Guesses $R_A = 1\ \text{lb}$ $B_x = 1\ \text{lb}$ $B_y = 1\ \text{lb}$

Given

$$R_A \sin(\theta) + B_x - \left(\frac{d}{\sqrt{e^2 + d^2}}\right) F = 0$$

$$R_A \cos(\theta) + B_y - \left(\frac{e}{\sqrt{e^2 + d^2}}\right) F = 0$$



$$M - R_A \cos(\theta)(a + b) + B_x c = 0$$

$$\begin{pmatrix} R_A \\ B_x \\ B_y \end{pmatrix} = \text{Find}(R_A, B_x, B_y) \qquad R_A = 105.1\ \text{lb} \qquad \begin{pmatrix} B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 97.4 \\ 269 \end{pmatrix} \text{lb}$$

Problem 5-27

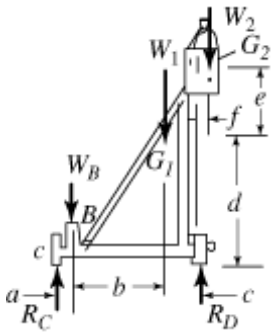
The platform assembly has weight W_1 and center of gravity at G_1. If it is intended to support a maximum load W_2 placed at point G_2, determine the smallest counterweight W that should be placed at B in order to prevent the platform from tipping over.

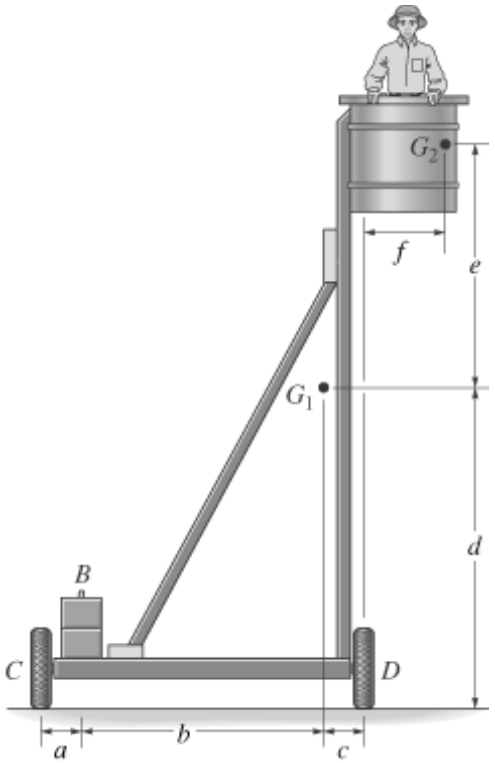
Given:

$W_1 = 250\ \text{lb}$ $a = 1\ \text{ft}$ $c = 1\ \text{ft}$ $e = 6\ \text{ft}$

$W_2 = 400\ \text{lb}$ $b = 6\ \text{ft}$ $d = 8\ \text{ft}$ $f = 2\ \text{ft}$

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Solution:

When tipping occurs, $R_c = 0$

$$\circlearrowleft + \Sigma M_D = 0; \qquad -W_2\ f + W_1\ c + W_B\ (b + c) = 0$$

$$W_B = \frac{W_2\ f - W_1\ c}{b + c}$$

$$W_B = 78.6\ \text{lb}$$

Problem 5-28

The articulated crane boom has a weight W and mass center at G. If it supports a load L, determine the force acting at the pin A and the compression in the hydraulic cylinder BC when the boom is in the position shown.

Units Used:

$$\text{kip} = 10^3\ \text{lb}$$

Given:

$$W = 125\ \text{lb}$$

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$L = 600$ lb

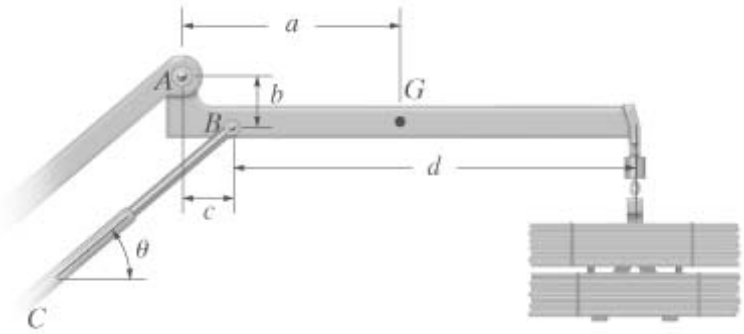
$a = 4$ ft

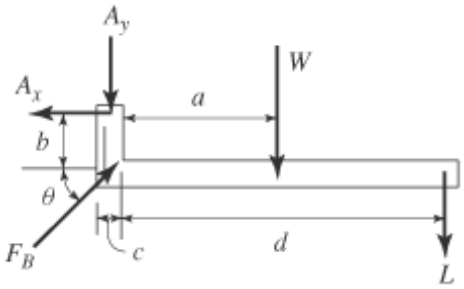
$b = 1$ ft

$c = 1$ ft

$d = 8$ ft

$\theta = 40$ deg





Solution:

Guesses $A_x = 1$ lb $A_y = 1$ lb $F_B = 1$ lb

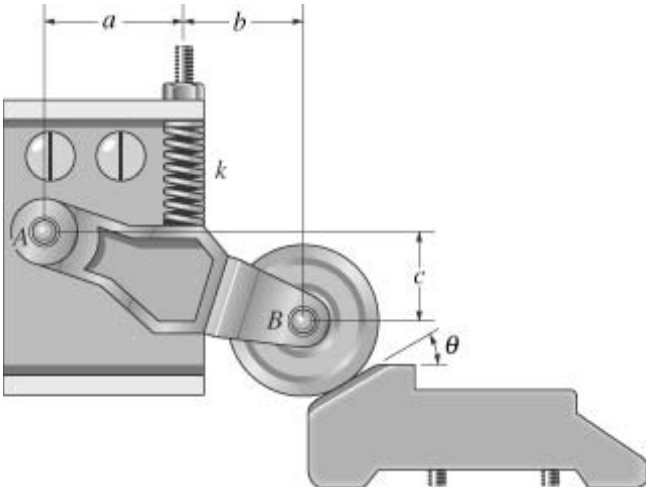
Given $-A_x + F_B\cos(\theta) = 0$ $-A_y + F_B\sin(\theta) - W - L = 0$

$F_B\cos(\theta)b + F_B\sin(\theta)c - W a - L(d + c) = 0$

$$\begin{pmatrix} A_x \\ A_y \\ F_B \end{pmatrix} = \text{Find}(A_x, A_y, F_B) \qquad F_B = 4.19\,\text{kip} \qquad \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 3.208 \\ 1.967 \end{pmatrix}\text{kip}$$

Problem 5-29

The device is used to hold an elevator door open. If the spring has stiffness k and it is compressed a distnace δ, determine the horizontal and vertical components of reaction at the pin A and the resultant force at the wheel bearing B.



Given:

$k = 40\ \frac{\text{N}}{\text{m}}$ $b = 125$ mm

$\delta = 0.2$ m $c = 100$ mm

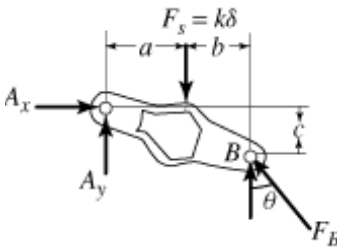
$a = 150$ mm $\theta = 30$ deg

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Solution:

$$F_s = k\delta$$

$\circlearrowleft + \Sigma M_A = 0;$ $\quad -F_s\, a + F_B \cos(\theta)(a+b) - F_B \sin(\theta) c = 0$



$$F_B = F_s \frac{a}{\cos(\theta)\,(a+b) - \sin(\theta)\,c}$$

$$F_B = 6.378 \text{ N}$$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad A_x - F_B \sin(\theta) = 0$

$$A_x = F_B \sin(\theta)$$

$$A_x = 3.189 \text{ N}$$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - F_s + F_B \cos(\theta) = 0$

$$A_y = F_s - F_B \cos(\theta)$$

$$A_y = 2.477 \text{ N}$$

Problem 5-30

Determine the reactions on the bent rod which is supported by a smooth surface at *B* and by a collar at *A*, which is fixed to the rod and is free to slide over the fixed inclined rod.

Given:

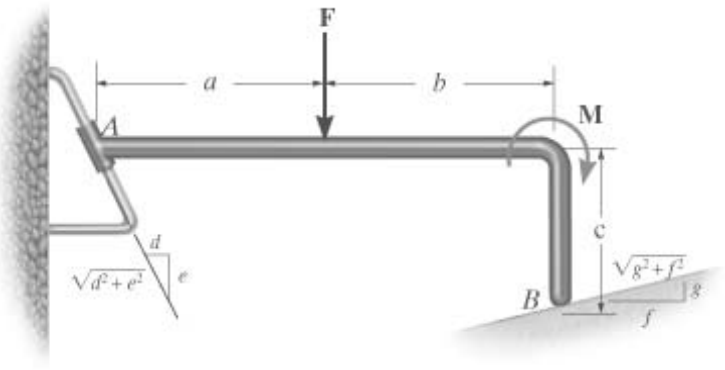
$F = 100$ lb

$M = 200$ lb ft

$a = 3$ ft

$b = 3$ ft

$c = 2$ft



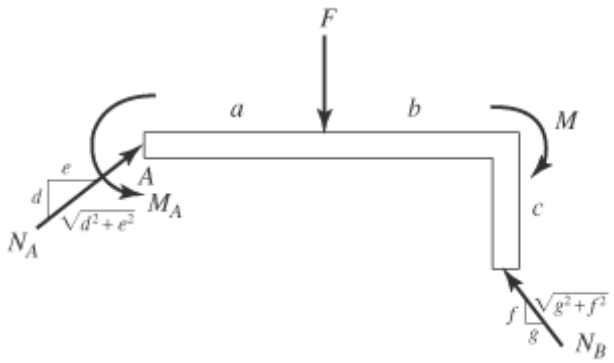
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$d = 3$

$e = 4$

$f = 12$

$g = 5$



Solution:

Initial Guesses:

$N_A = 20$ lb $\quad N_B = 10$ lb $\quad M_A = 30$ lb ft

Given

$$\Sigma M_A = 0; \quad M_A - F\,a - M + N_B\left(\frac{f}{\sqrt{f^2+g^2}}\right)(a+b) - N_B\left(\frac{g}{\sqrt{f^2+g^2}}\right)c = 0$$

$$\Sigma F_x = 0; \quad N_A\left(\frac{e}{\sqrt{e^2+d^2}}\right) - N_B\left(\frac{g}{\sqrt{f^2+g^2}}\right) = 0$$

$$\Sigma F_y = 0; \quad N_A\left(\frac{d}{\sqrt{e^2+d^2}}\right) + N_B\left(\frac{f}{\sqrt{f^2+g^2}}\right) - F = 0$$

$$\begin{pmatrix} N_A \\ N_B \\ M_A \end{pmatrix} = \text{Find}\left(N_A, N_B, M_A\right) \qquad \begin{pmatrix} N_A \\ N_B \end{pmatrix} = \begin{pmatrix} 39.7 \\ 82.5 \end{pmatrix} \text{lb} \qquad M_A = 106\,\text{lb}\cdot\text{ft}$$

Problem 5-31

The cantilevered jib crane is used to support the load F. If the trolley T can be placed anywhere in the range $x_1 \le x \le x_2$, determine the maximum magnitude of reaction at the supports A and B. Note that the supports are collars that allow the crane to rotate freely about the vertical axis. The collar at B supports a force in the vertical direction, whereas the one at A does not.

Units Used:

kip = 1000 lb

Given:

$F = 780$ lb

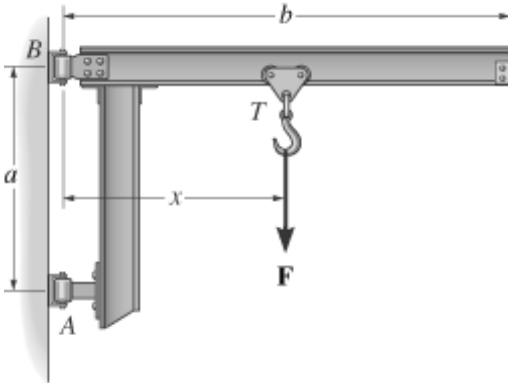
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$a = 4$ ft

$b = 8$ ft

$x_1 = 1.5$ ft

$x_2 = 7.5$ ft



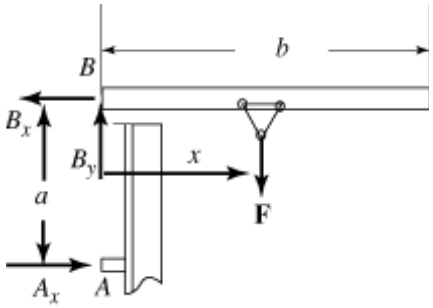
Solution:

The maximum occurs when $x = x_2$

$\curvearrowleft \Sigma M_A = 0; \quad -F x_2 + B_x a = 0$

$$B_x = F \frac{x_2}{a}$$

$$B_x = 1.462 \times 10^3 \text{ lb}$$



$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad A_x - B_x = 0 \quad A_x = B_x \quad A_x = 1.462 \times 10^3 \text{ lb}$

$+\uparrow \Sigma F_y = 0; \quad B_y - F = 0 \quad B_y = F \quad B_y = 780 \text{ lb}$

$$F_B = \sqrt{B_x^2 + B_y^2} \qquad F_B = 1.657 \text{ kip}$$

Problem 5-32

The uniform rod *AB* has weight *W*. Determine the force in the cable when the rod is in the position shown.

Given:

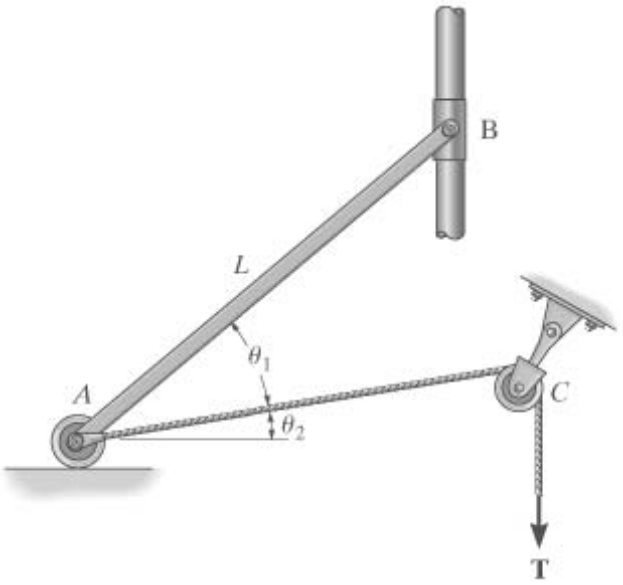
$W = 15$ lb

$L = 5$ ft

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$\theta_1 = 30 \text{ deg}$

$\theta_2 = 10 \text{ deg}$

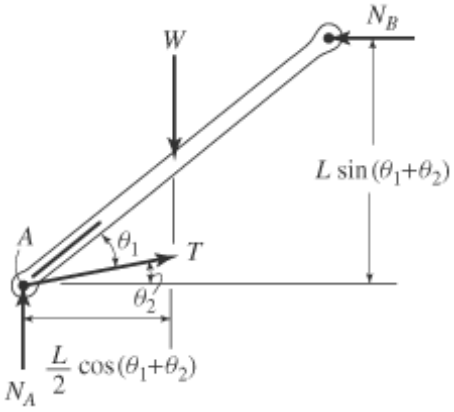


Solution:

$\Sigma M_A = 0; \quad N_B\, L \sin(\theta_1 + \theta_2) - W\left(\frac{L}{2}\right)\cos(\theta_1 + \theta_2) = 0$

$$N_B = \frac{W\cos(\theta_1 + \theta_2)}{2\,\sin(\theta_1 + \theta_2)}$$

$N_B = 8.938 \text{ lb}$



$\Sigma F_x = 0; \quad T\cos(\theta_2) - N_B$

$$T = \frac{N_B}{\cos(\theta_2)}$$

$T = 9.08 \text{ lb}$

Problem 5-33

The power pole supports the three lines, each line exerting a vertical force on the pole due to its weight as shown. Determine the reactions at the fixed support *D*. If it is possible for wind or ice to snap the lines, determine which line(s) when removed create(s) a condition for the greatest moment reaction at *D*.

Units Used:

kip $= 10^3$ lb

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Given:

$W_1 = 800$ lb

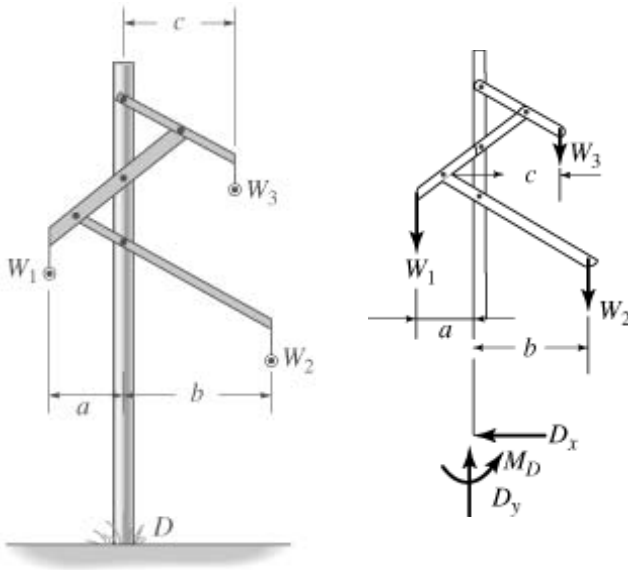
$W_2 = 450$ lb

$W_3 = 400$ lb

$a = 2$ft

$b = 4$ ft

$c = 3$ ft



Solution:

$\xrightarrow{+} \Sigma F_x = 0;$ $D_x = 0$

$+\uparrow \Sigma F_y = 0;$ $D_y - (W_1 + W_2 + W_3) = 0$

$D_y = W_1 + W_2 + W_3$ $D_y = 1.65\,\text{kip}$

$\circlearrowleft + \Sigma M_D = 0;$ $-W_2 b - W_3 c + W_1 a + M_D = 0$

$M_D = W_2\, b + W_3\, c - W_1\, a$ $M_D = 1.4\,\text{kip}\cdot\text{ft}$

Examine all cases. For these numbers we require line 1 to snap.

$M_{Dmax} = W_2\, b + W_3\, c$ $M_{Dmax} = 3\,\text{kip}\cdot\text{ft}$

Problem 5-34

The picnic table has a weight W_T and a center of gravity at G_T. If a man weighing W_M has a center of gravity at G_M and sits down in the centered position shown, determine the vertical reaction at each of the two legs at B.Neglect the thickness of the legs. What can you conclude from the results?

Given:

$W_T = 50$ lb

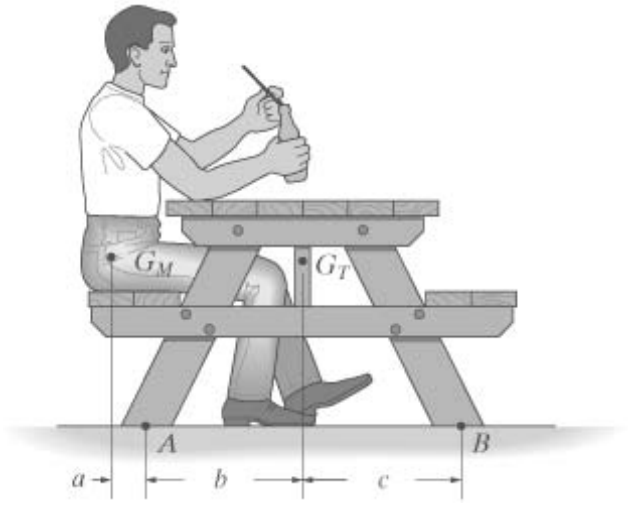
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$W_M = 225$ lb

$a = 6$ in

$b = 20$ in

$c = 20$ in



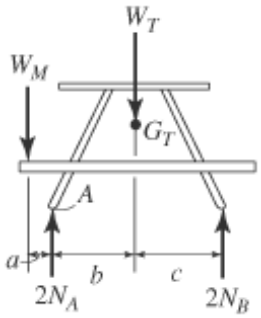
Solution:

$\Sigma M_A = 0; \quad 2\,N_B(b + c) + W_M\,a - W_T\,b = 0$

$$N_B = \frac{W_T\,b - W_M\,a}{2\,(b + c)}$$

$N_B = -4.37$ lb

Since N_B has a negative sign, the table will tip over.



Problem 5-35

If the wheelbarrow and its contents have a mass of M and center of mass at G, determine the magnitude of the resultant force which the man must exert on *each* of the two handles in order to hold the wheelbarrow in equilibrium.

Given:

$M = 60$ kg

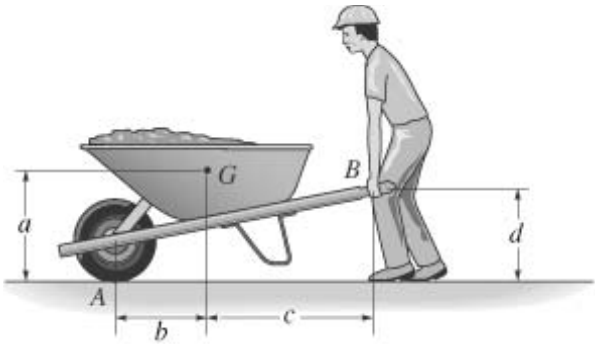
$a = 0.6$ m

$b = 0.5$ m

$c = 0.9$ m

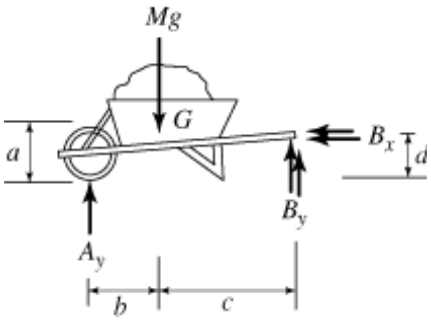
$d = 0.5$ m

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



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Solution:



$\circlearrowleft \Sigma M_B = 0;$ $\quad -A_y\,(b + c) + M g c = 0$

$$A_y = \frac{M g c}{b + c}$$

$$A_y = 378.386\,\text{N}$$

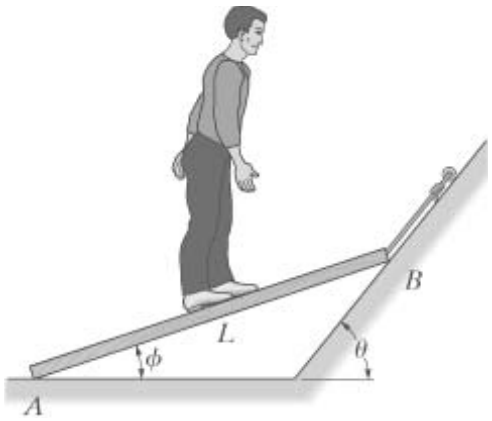
$\xrightarrow{+} \Sigma F_x = 0;$ $\quad B_x = 0\text{N}$ $\quad B_x = 0$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - M g + 2\,B_y = 0$

$$B_y = \frac{M g - A_y}{2} \qquad B_y = 105.107\ \text{N}$$

Problem 5-36

The man has weight W and stands at the center of the plank. If the planes at A and B are smooth, determine the tension in the cord in terms of W and θ.



Solution:

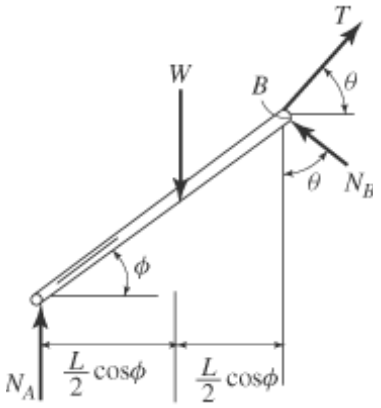
$\Sigma M_B = 0;$ $\quad W\,\frac{L}{2}\cos(\phi) - N_A\,L\cos(\phi) = 0$

$$N_A = \frac{W}{2}$$

$\Sigma F_x = 0;$ $\quad T\cos(\theta) - N_B\sin(\theta) = 0$ $\qquad (1)$

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$\Sigma F_y = 0;$ $\quad T\sin(\theta) + N_B\cos(\theta) + N_A - W = 0$ (2)



Solving Eqs. (1) and (2) yields:

$$T = \frac{W}{2}\sin(\theta)$$

$$N_B = \frac{W}{2}\cos(\theta)$$

Problem 5-37

When no force is applied to the brake pedal of the lightweight truck, the retainer spring AB keeps the pedal in contact with the smooth brake light switch at C. If the force on the switch is **F**, determine the unstretched length of the spring if the stiffness of the spring is k.

Given:

$F = 3$ N

$k = 80 \frac{\text{N}}{\text{m}}$

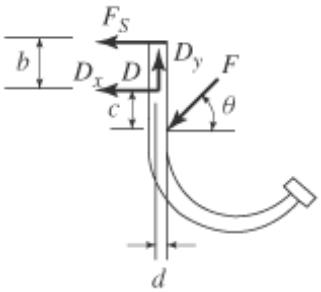
$a = 100$ mm

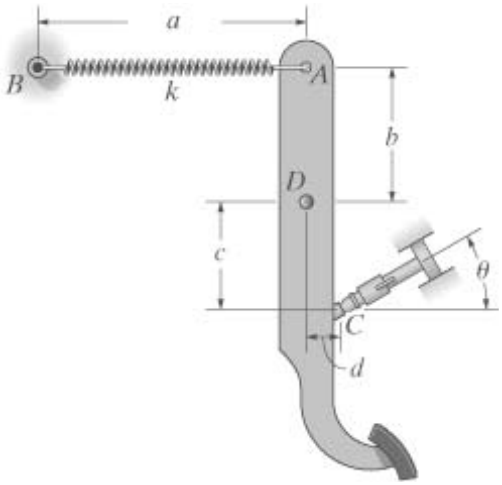
$b = 50$ mm

$c = 40$ mm

$d = 10$ mm

$\theta = 30$ deg





Solution:

$\Sigma M_D = 0;$ $\quad F_S\, b - F\cos(\theta)c - F\sin(\theta)d = 0$

$$F_S = F\,\frac{\cos(\theta)\,c + \sin(\theta)\,d}{b} \qquad F_S = 2.378 \text{ N}$$

$$F_S = kx \qquad x = \frac{F_S}{k} \qquad x = 29.73\,\text{mm}$$

$$L_0 = a - x \qquad L_0 = 70.3\,\text{mm}$$

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Problem 5-38

The telephone pole of negligible thickness is subjected to the force **F** directed as shown. It is supported by the cable *BCD* and can be assumed pinned at its base *A*. In order to provide clearance for a sidewalk right of way, where *D* is located, the strut *CE* is attached at *C*, as shown by the dashed lines (cable segment *CD* is removed). If the tension in *CD'* is to be twice the tension in *BCD*, determine the height *h* for placement of the strut *CE*.

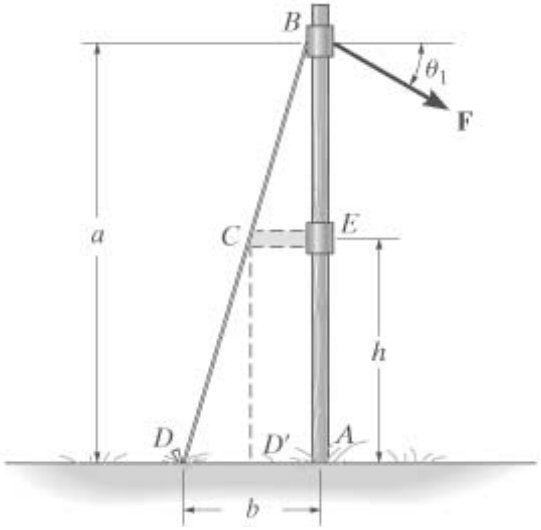
Given:

$F = 80$ lb

$\theta = 30$ deg

$a = 30$ ft

$b = 10$ ft



Solution:

$\circlearrowleft + \Sigma M_A = 0;$ $\quad -F\cos(\theta)\,a + \left(\frac{b}{\sqrt{a^2+b^2}}\right) T_{BCD}\,a = 0$

$$T_{BCD} = F\cos(\theta)\,\frac{\sqrt{a^2+b^2}}{b} \qquad T_{BCD} = 219.089\,\text{lb}$$

Require $T_{CD'} = 2\,T_{BCD}$ $\qquad T_{CD'} = 438.178\,\text{lb}$

$+\Sigma M_A = 0;$ $\quad T_{CD'}\,d - F\cos(\theta)\,a = 0$

$$d = F\,a\left(\frac{\cos(\theta)}{T_{CD'}}\right) \qquad d = 4.7434\,\text{ft}$$

Geometry:

$$\frac{a-h}{d} = \frac{a}{b} \qquad h = a - a\left(\frac{d}{b}\right) \qquad h = 15.8\,\text{ft}$$

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Problem 5-39

The worker uses the hand truck to move material down the ramp. If the truck and its contents are held in the position shown and have weight W with center of gravity at *G*, determine the resultant normal force of both wheels on the ground *A* and the magnitude of the force required at the grip *B*.

Given:

$W = 100$ lb $\quad e = 1.5$ ft

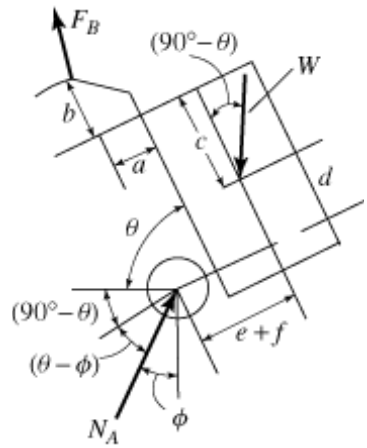
$a = 1$ ft $\quad f = 0.5$ ft

$b = 1.5$ ft $\quad \theta = 60$ deg

$c = 2$ ft $\quad \phi = 30$ deg

$d = 1.75$ ft





Solution:

$\curvearrowleft \Sigma M_B = 0;$

$$N_A \cos(\theta-\phi)(b+c+d) + N_A \sin(\theta-\phi)(a-f) - W\cos(\theta)(b+c) - W\sin(\theta)(e+a) = 0$$

$$N_A = \frac{W\cos(\theta)(b+c) + W\sin(\theta)(e+a)}{\cos(\theta-\phi)(b+c+d) + \sin(\theta-\phi)(a-f)} \qquad N_A = 81.621\,\text{lb}$$

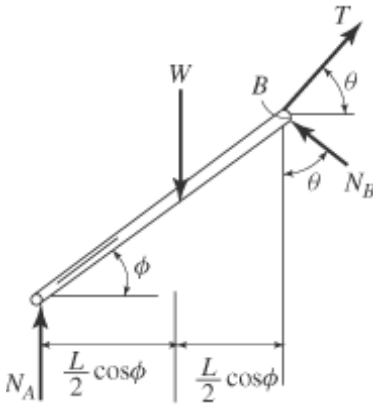
$\Sigma F_x = 0;$ $\quad -B_x + N_A \sin(\phi) = 0$ $\quad B_x = N_A \sin(\phi)$ $\quad B_x = 40.811\,\text{lb}$

$\Sigma F_y = 0;$ $\quad B_y + N_A(\cos(\phi) - W = 0)$ $\quad B_y = W - N_A \cos(\phi)$ $\quad B_y = 29.314\,\text{lb}$

$$F_B = \sqrt{B_x^2 + B_y^2} \qquad F_B = 50.2\,\text{lb}$$

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$\Sigma F_y = 0;$ $\quad T\sin(\theta) + N_B\cos(\theta) + N_A - W = 0$ (2)



Solving Eqs. (1) and (2) yields:

$$T = \frac{W}{2}\sin(\theta)$$

$$N_B = \frac{W}{2}\cos(\theta)$$

Problem 5-37

When no force is applied to the brake pedal of the lightweight truck, the retainer spring AB keeps the pedal in contact with the smooth brake light switch at C. If the force on the switch is **F**, determine the unstretched length of the spring if the stiffness of the spring is k.

Given:

$F = 3$ N

$k = 80 \frac{\text{N}}{\text{m}}$

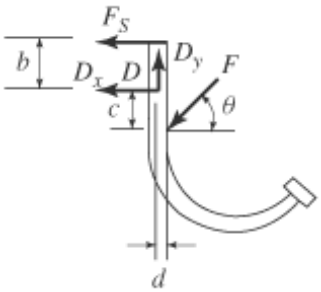
$a = 100$ mm

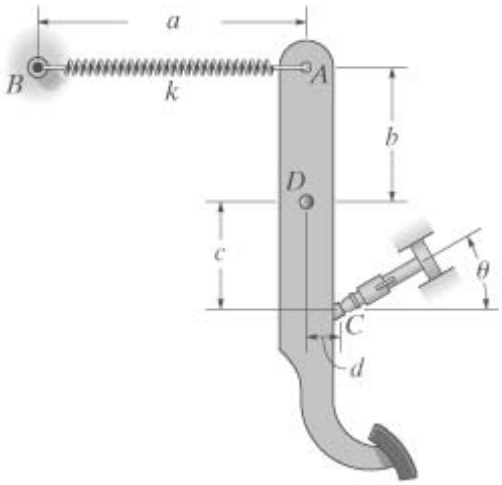
$b = 50$ mm

$c = 40$ mm

$d = 10$ mm

$\theta = 30$ deg





Solution:

$\Sigma M_D = 0;$ $\quad F_S\, b - F\cos(\theta)c - F\sin(\theta)d = 0$

$$F_S = F\,\frac{\cos(\theta)\,c + \sin(\theta)\,d}{b} \qquad F_S = 2.378 \text{ N}$$

$$F_S = kx \qquad x = \frac{F_S}{k} \qquad x = 29.73\,\text{mm}$$

$$L_0 = a - x \qquad L_0 = 70.3\,\text{mm}$$

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The shelf supports the electric motor which has mass m_1 and mass center at G_m. The platform upon which it rests has mass m_2 and mass center at G_p. Assuming that a single bolt B holds the shelf up and the bracket bears against the smooth wall at A, determine this normal force at A and the horizontal and vertical components of reaction of the bolt B on the bracket.

Given:

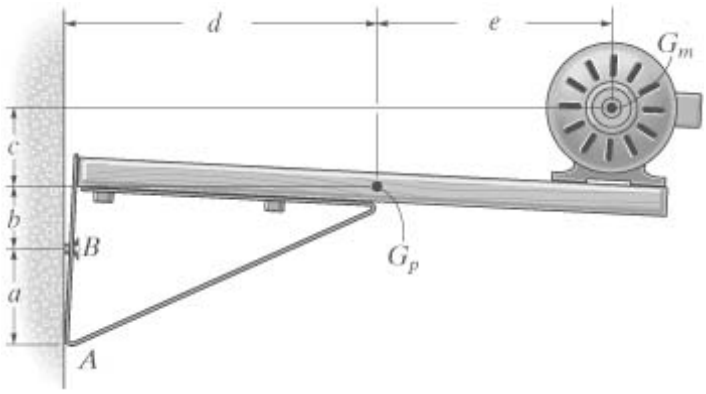
$m_1 = 15$ kg $\quad c = 50$ mm

$m_2 = 4$ kg $\quad d = 200$ mm

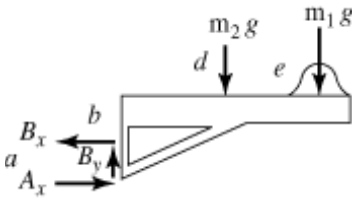
$a = 60$ mm $\quad e = 150$ mm

$b = 40$ mm

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:



$\circlearrowleft + \Sigma M_A = 0; \quad B_x a - m_2 g d - m_1 g(d + e) = 0$

$$B_x = g\,\frac{m_2\,d + m_1\,(d + e)}{a} \qquad B_x = 989 \text{ N}$$

$\xrightarrow{+} \Sigma F_x = 0; \quad A_x - B_x = 0$

$$A_x = B_x \qquad A_x = 989 \text{ N}$$

$+\uparrow \Sigma F_y = 0; \quad B_y - m_2\,g - m_1\,g = 0$

$$B_y = m_2\,g + m_1\,g \qquad B_y = 186 \text{ N}$$

Problem 5-42

A cantilever beam, having an extended length L, is subjected to a vertical force **F**. Assuming that the wall resists this load with linearly varying distributed loads over the length a of the beam portion inside the wall, determine the intensities w_1 and w_2 for equilibrium.

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Units Used:

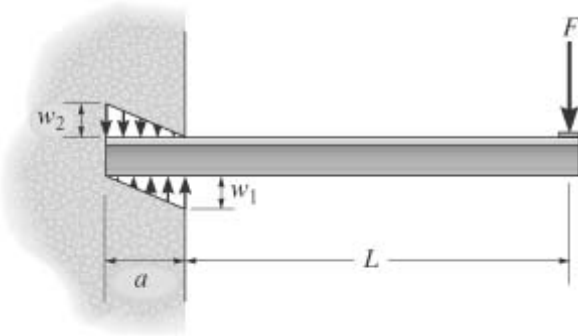
$$\text{kN} = 10^3 \text{ N}$$

Given:

$$F = 500 \text{ N}$$

$$a = 0.15 \text{ m}$$

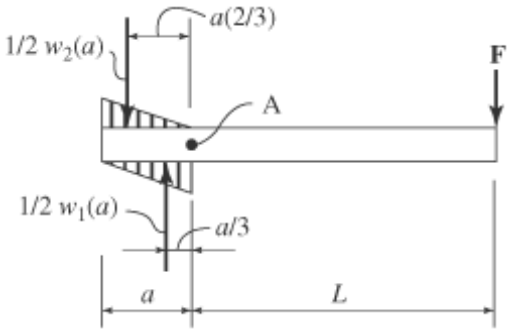
$$L = 3 \text{ m}$$



Solution:

The initial guesses

$$w_1 = 1 \frac{\text{kN}}{\text{m}} \qquad w_2 = 1 \frac{\text{kN}}{\text{m}}$$



Given

$$+\uparrow \Sigma F_y = 0; \qquad \frac{1}{2} w_1 \, a - \frac{1}{2} w_2 \, a - F = 0$$

$$\circlearrowleft + \Sigma M_A = 0; \qquad -F \, L - \frac{1}{2} w_1 \, a\left(\frac{a}{3}\right) + \frac{1}{2} w_2 \, a\left(\frac{2\,a}{3}\right) = 0$$

$$\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \text{Find}(w_1, w_2) \qquad \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 413 \\ 407 \end{pmatrix} \frac{\text{kN}}{\text{m}}$$

Problem 5-43

The upper portion of the crane boom consists of the jib *AB*, which is supported by the pin at *A*, the guy line *BC*, and the backstay *CD*, each cable being separately attached to the mast at *C*. If the load *F* is supported by the hoist line, which passes over the pulley at *B*, determine the magnitude of the resultant force the pin exerts on the jib at *A* for equilibrium, the tension in the guy line *BC*, and the tension *T* in the hoist line. Neglect the weight of the jib. The pulley at *B* has a radius of *r*.

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Units Used:

$$\text{kN} = 10^3 \text{ N}$$

Given:

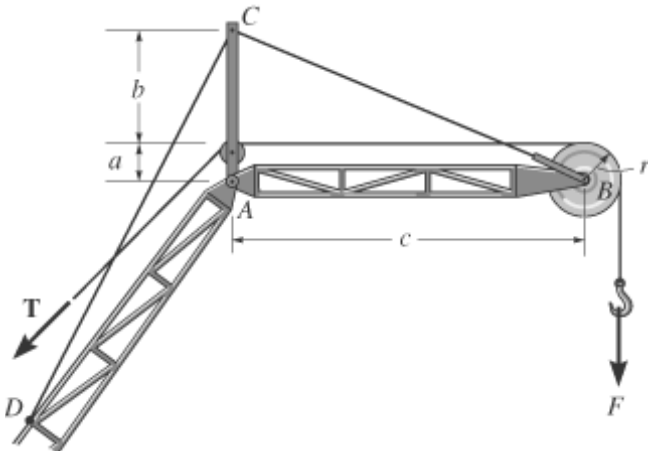
$F = 5$ kN

$r = 0.1$ m

$a = r$

$b = 1.5$ m

$c = 5$ m

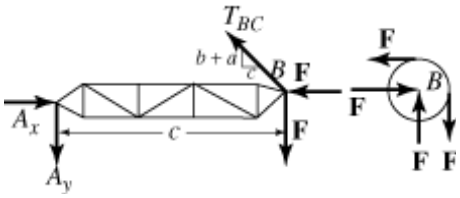


Solution:

From pulley, tension in the hoist line is

$$\circlearrowleft + \Sigma M_B = 0; \qquad T(a) - F(r) = 0$$

$$T = F\frac{r}{a} \qquad T = 5\,\text{kN}$$



From the jib,

$$\circlearrowleft + \Sigma M_A = 0; \qquad -F(c) + T_{BC}\frac{b+a}{\sqrt{c^2 + (b+a)^2}}c = 0$$

$$T_{BC} = F\frac{\sqrt{c^2 + (b+a)^2}}{b+a} \qquad T_{BC} = 16.406\,\text{kN}$$

$$+\uparrow \Sigma F_y = 0; \qquad -A_y + T_{BC}\left[\frac{b+a}{\sqrt{c^2 + (b+a)^2}}\right] - F = 0$$

$$A_y = T_{BC}\left[\frac{b+a}{\sqrt{c^2 + (b+a)^2}}\right] - F \qquad A_y = 0\,\text{kN}$$

$$\xrightarrow{+} \Sigma F_x = 0; \qquad A_x - T_{BC}\left[\frac{c}{\sqrt{c^2 + (b+a)^2}}\right] - F = 0$$

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$$A_x = T_{BC}\frac{c}{\sqrt{c^2 + (b+a)^2}} + F \qquad A_x = 20.6\,\text{kN}$$

$$F_A = \sqrt{A_x^2 + A_y^2} \qquad F_A = 20.6\,\text{kN}$$

Problem 5-44

The mobile crane has weight W_1 and center of gravity at G_1; the boom has weight W_2 and center of gravity at G_2. Determine the smallest angle of tilt θ of the boom, without causing the crane to overturn if the suspended load has weight W. Neglect the thickness of the tracks at A and B.

Given:

$W_1 = 120000$ lb

$W_2 = 30000$ lb

$W = 40000$ lb

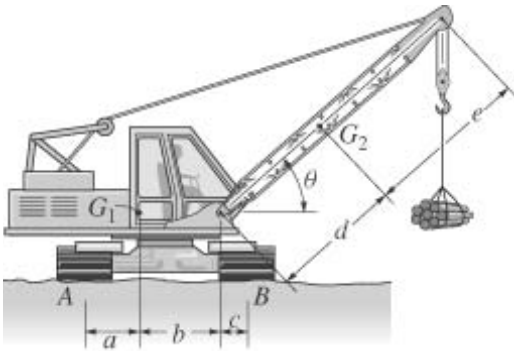
$a = 4$ ft

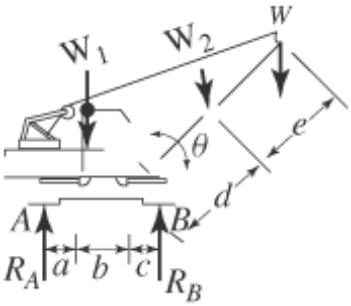
$b = 6$ ft

$c = 3$ ft

$d = 12$ ft

$e = 15$ ft





Solution:

When tipping occurs, $R_A = 0$

$$\curvearrowright \Sigma M_B = 0; \qquad -W_2\left(d\cos(\theta) - c\right) - W\left[(d+e)\cos(\theta) - c\right] + W_1(b+c) = 0$$

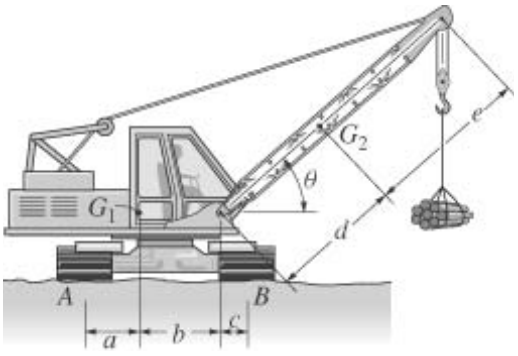
$$\theta = \text{acos}\left[\frac{W_2\,c + W\,c + W_1\,(b+c)}{W_2\,d + W\,(d+e)}\right]$$

$$\theta = 26.4\,\text{deg}$$

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Problem 5-45

The mobile crane has weight W_1 and center of gravity at G_1; the boom has weight W_2 and center of gravity at G_2. If the suspended load has weight W determine the normal reactions at the tracks A and B. For the calculation, neglect the thickness of the tracks.



Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

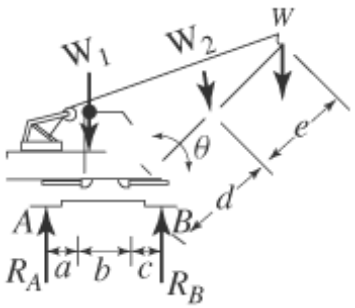
$W_1 = 120000$ lb $\quad a = 4$ ft

$W_2 = 30000$ lb $\quad b = 6$ ft

$W = 16000$ lb $\quad c = 3$ ft

$\theta = 30$ deg $\quad d = 12$ ft

$e = 15$ ft



Solution:

$\curvearrowleft + \Sigma M_B = 0;$

$$-W_2(d\cos(\theta) - c) - W[(d + e)\cos(\theta) - c] - R_A(a + b + c) + W_1(b + c) = 0$$

$$R_A = \frac{-W_2\,(d\cos(\theta) - c) - W[(d + e)\cos(\theta) - c] + W_1\,(b + c)}{a + b + c} \qquad R_A = 40.9\,\text{kip}$$

$+\uparrow \Sigma F_y = 0; \qquad R_A + R_B - W_1 - W_2 - W = 0$

$$R_B = -R_A + W_1 + W_2 + W \qquad R_B = 125\,\text{kip}$$

Problem 5-46

The man attempts to support the load of boards having a weight W and a center of gravity at G. If he is standing on a smooth floor, determine the smallest angle θ at which he can hold them up in the position shown. Neglect his weight.

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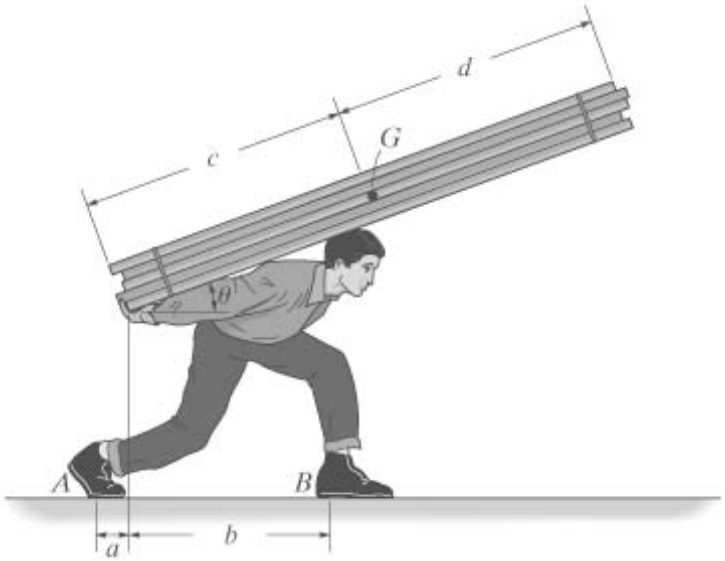
Given:

$a = 0.5$ ft

$b = 3$ ft

$c = 4$ ft

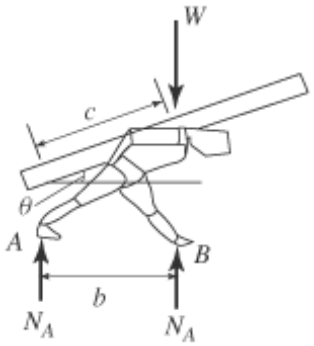
$d = 4$ ft



Solution:

$\Sigma M_B = 0; \quad -N_A(a + b) + W(b - c\cos(\theta)) = 0$

As θ becomes smaller, N_A goes to 0 so that,



$$\cos(\theta) = \frac{b}{c}$$

$$\theta = \text{acos}\left(\frac{b}{c}\right)$$

$$\theta = 41.4\,\text{deg}$$

Problem 5-47

The motor has a weight *W*. Determine the force that each of the chains exerts on the supporting hooks at *A*, *B*, and *C*. Neglect the size of the hooks and the thickness of the beam.

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Given:

$W = 850$ lb

$a = 0.5$ ft

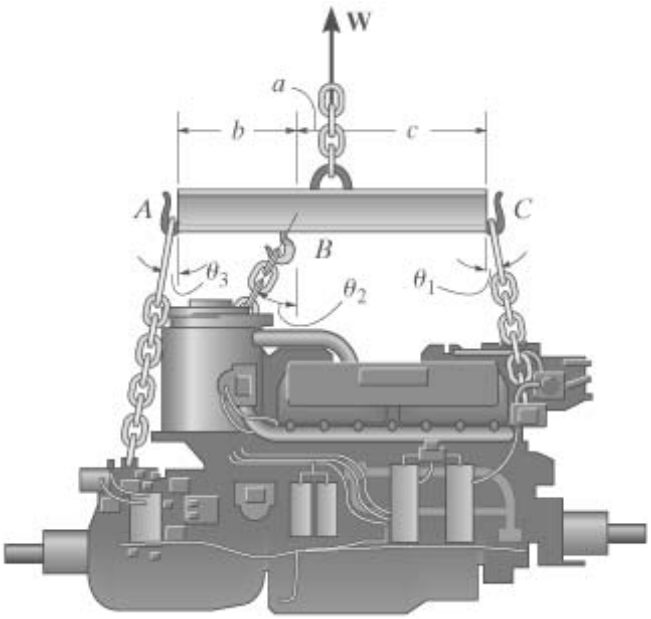
$b = 1$ ft

$c = 1.5$ ft

$\theta_1 = 10$ deg

$\theta_2 = 30$ deg

$\theta_3 = 10$ deg



Solution:

Guesses

$F_A = 1$ lb $\quad F_B = 1$ lb $\quad F_C = 1$ lb

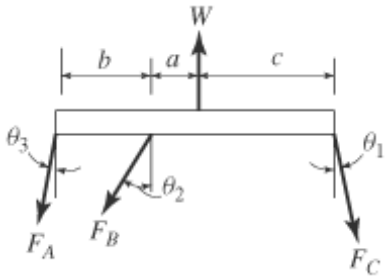
Given

$\Sigma M_B = 0; \quad F_A \cos(\theta_3) b + W\, a - F_C \cos(\theta_1)(a + c) = 0$

$\Sigma F_x = 0; \quad F_C \sin(\theta_1) - F_B \sin(\theta_2) - F_A \sin(\theta_3) = 0$

$\Sigma F_y = 0; \quad W - F_A \cos(\theta_3) - F_B \cos(\theta_2) - F_C \cos(\theta_1) = 0$

$$\begin{pmatrix} F_A \\ F_B \\ F_C \end{pmatrix} = \text{Find}(F_A, F_B, F_C)$$



$$\begin{pmatrix} F_A \\ F_B \\ F_C \end{pmatrix} = \begin{pmatrix} 432 \\ -0 \\ 432 \end{pmatrix} \text{lb}$$

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Problem 5-48

The boom supports the two vertical loads. Neglect the size of the collars at *D* and *B* and the thickness of the boom, and compute the horizontal and vertical components of force at the pin *A* and the force in cable *CB*.

Given:

$F_1 = 800$ N

$F_2 = 350$ N

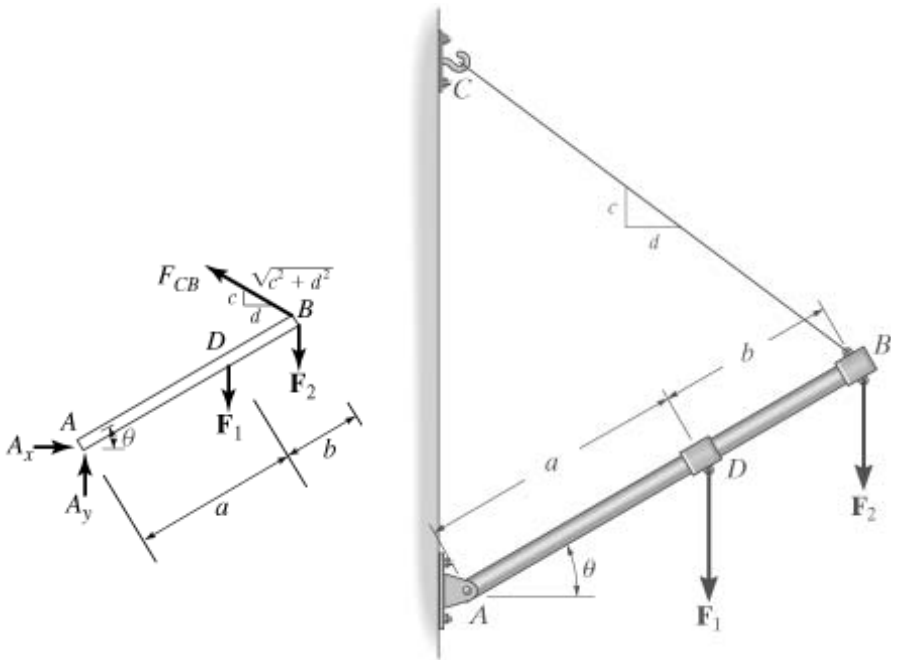
$a = 1.5$ m

$b = 1$ m

$c = 3$

$d = 4$

$\theta = 30$ deg



Solution:

$\circlearrowleft + \Sigma M_A = 0;$

$$-F_1 a\cos(\theta) - F_2(a+b)\cos(\theta) + \frac{d}{\sqrt{c^2+d^2}}F_{CB}\,(a+b)\sin(\theta) + \frac{c}{\sqrt{c^2+d^2}}F_{CB}(a+b)\cos(\theta) = 0$$

$$F_{CB} = \frac{\left[F_1 a + F_2(a+b)\right]\cos(\theta)\sqrt{c^2+d^2}}{d\sin(\theta)(a+b) + c\cos(\theta)(a+b)} \qquad F_{CB} = 782 \text{ N}$$

$\xrightarrow{+} \Sigma F_x = 0;$ $\quad A_x - \frac{d}{\sqrt{c^2+d^2}}F_{CB} = 0$

$$A_x = \frac{d}{\sqrt{c^2+d^2}}\,F_{CB} \qquad A_x = 625 \text{ N}$$

$+\uparrow \Sigma F_y = 0;$ $\quad A_y - F_1 - F_2 + \frac{c}{\sqrt{c^2+d^2}}F_{CB} = 0$

$$A_y = F_1 + F_2 - \frac{c}{\sqrt{c^2+d^2}}F_{CB} \qquad A_y = 681 \text{ N}$$

Problem 5-49

The boom is intended to support two vertical loads $\mathbf{F}_1$ and $\mathbf{F}_2$. If the cable *CB* can sustain a

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maximum load T_{max} before it fails, determine the critical loads if $F_1 = 2F_2$. Also, what is the magnitude of the maximum reaction at pin A?

Units Used:

$kN = 10^3 N$

Given:

$T_{max} = 1500\ N$

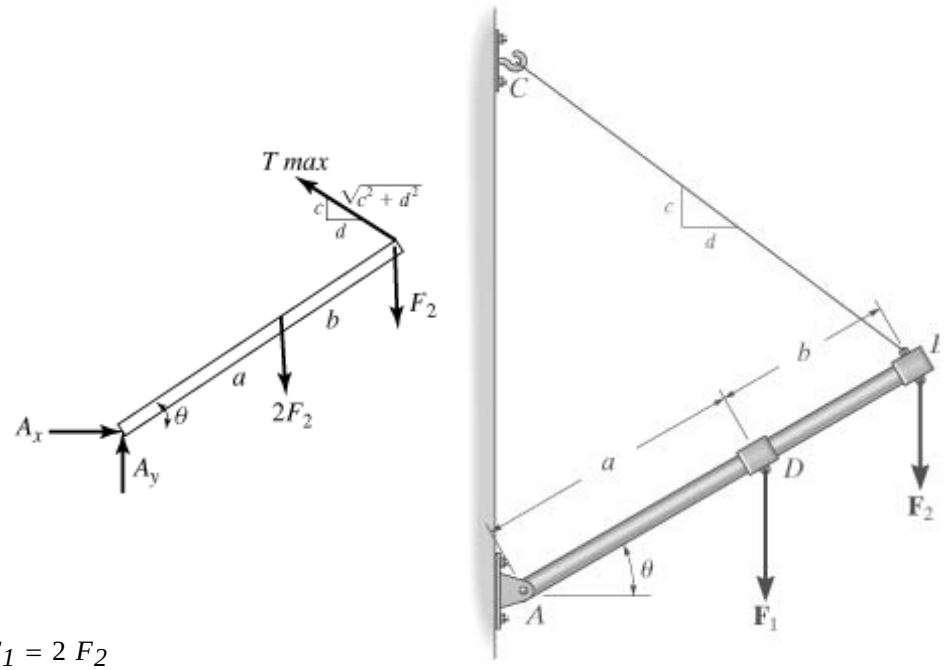
$a = 1.5\ m$

$b = 1\ m$

$c = 3$

$d = 4$

$\theta = 30 deg$



Solution:

$\curvearrowleft + \Sigma M_A = 0; \qquad F_1 = 2\ F_2$

$$-2\ F_2\, a\cos(\theta) - F_2(a+b)\cos(\theta) + \frac{d}{\sqrt{c^2+d^2}} T_{max}(a+b)\sin(\theta) + \frac{c}{\sqrt{c^2+d^2}} T_{max}(a+b)\cos(\theta) = 0$$

$$F_2 = \frac{(a+b)\,T_{max}\left(d\sin(\theta) + c\cos(\theta)\right)}{\sqrt{c^2+d^2}\cos(\theta)(3\ a+b)} \qquad F_2 = 724\,N$$

$$F_1 = 2\ F_2 \qquad F_1 = 1.448\,kN$$

$\xrightarrow{+} \Sigma F_x = 0; \qquad A_x - \frac{d}{\sqrt{c^2+d^2}} T_{max} = 0$

$$A_x = \frac{d}{\sqrt{c^2+d^2}} T_{max} \qquad A_x = 1.20\,kN$$

$+\uparrow \Sigma\ F_y = 0; \qquad A_y - F_2 - F_1 + \frac{c}{\sqrt{c^2+d^2}} T_{max} = 0$

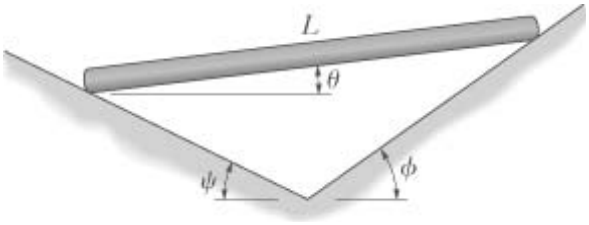
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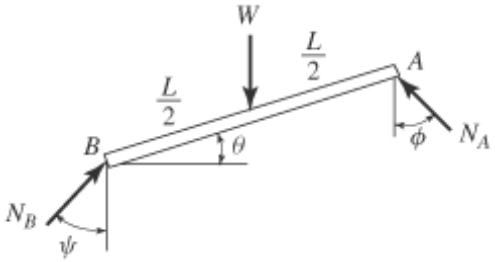
$$A_y = F_2 + F_1 - \frac{c}{\sqrt{c^2 + d^2}} T_{max} \qquad A_y = 1.27\,\text{kN}$$

$$F_A = \sqrt{A_x^2 + A_y^2} \qquad F_A = 1.749\,\text{kN}$$

Problem 5-50

The uniform rod of length L and weight W is supported on the smooth planes. Determine its position θ for equilibrium. Neglect the thickness of the rod.





Solution:

$\Sigma M_B = 0; \quad -W\frac{L}{2}\cos(\theta) + N_A\cos(\phi - \theta)L = 0 \qquad N_A = \frac{W\cos(\theta)}{2\cos(\phi - \theta)}$

$\Sigma M_A = 0; \quad W\frac{L}{2}\cos(\theta) - N_B\cos(\psi + \theta)L = 0 \qquad N_B = \frac{W\cos(\theta)}{2\cos(\psi + \theta)}$

$\Sigma F_x = 0; \quad N_B\sin(\psi) - N_A\sin(\phi) = 0$

$$\frac{W\cos(\theta)}{2\cos(\psi + \theta)}\sin(\psi) - \frac{W\cos(\theta)}{2\cos(\phi - \theta)}\sin(\phi) = 0$$

$$\sin(\psi)\cos(\phi - \theta) - \sin(\phi)\cos(\psi + \theta) = 0$$

$$\sin(\psi)(\cos(\phi)\cos(\theta) + \sin(\phi)\sin(\theta)) - \sin(\phi)(\cos(\psi)\cos(\theta) - \sin(\psi)\sin(\theta)) = 0$$

$$2\sin(\psi)\sin(\phi)\sin(\theta) = (\sin(\phi)\cos(\psi) - \sin(\psi)\cos(\phi))\cos(\theta)$$

$$\tan(\theta) = \frac{\sin(\phi)\cos(\psi) - \sin(\psi)\cos(\phi)}{2\sin(\psi)\sin(\phi)} = \frac{\cot(\psi) - \cot(\phi)}{2}$$

$$\theta = \text{atan}\left(\frac{\cot(\psi) - \cot(\phi)}{2}\right)$$

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Problem 5-51

The toggle switch consists of a cocking lever that is pinned to a fixed frame at A and held in place by the spring which has unstretched length δ. Determine the magnitude of the resultant force at A and the normal force on the peg at B when the lever is in the position shown.

Given:

$\delta = 200$ mm

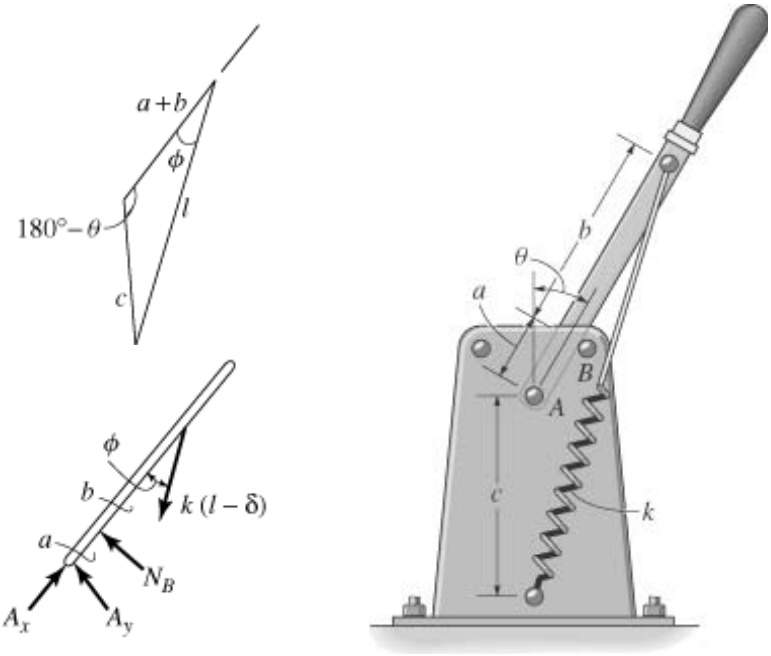
$k = 5 \frac{\text{N}}{\text{m}}$

$a = 100$ mm

$b = 300$ mm

$c = 300$ mm

$\theta = 30$ deg



Solution:

Using the law of cosines and the law of sines

$$l = \sqrt{c^2 + (a+b)^2 - 2\,c(a+b)\cos(180\ \text{deg} - \theta)}$$

$$\frac{\sin(\phi)}{c} = \frac{\sin(180\ \text{deg} - \theta)}{l} \qquad \phi = \operatorname{asin}\left(c\frac{\sin(180\ \text{deg} - \theta)}{l}\right) \qquad \phi = 12.808\,\text{deg}$$

$$F_s = k\,s = k(l - \delta) \qquad F_s = k(l - \delta) \qquad F_s = 2.3832\ \text{N}$$

$$\Sigma M_A = 0; \quad -F_s\sin(\phi)(a+b) + N_B a = 0 \qquad N_B = F_s\sin(\phi)\frac{a+b}{a} \qquad N_B = 2.11\ \text{N}$$

$$\Sigma F_x = 0; \quad A_x - F_s\cos(\phi) = 0 \qquad A_x = F_s\cos(\phi) \qquad A_x = 2.3239\ \text{N}$$

$$\Sigma F_y = 0; \quad A_y + N_B - F_s\sin(\phi) = 0 \qquad A_y = F_s\sin(\phi) - N_B \qquad A_y = -1.5850\ \text{N}$$

$$F_A = \sqrt{A_x^2 + A_y^2} \qquad F_A = 2.813\ \text{N}$$

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Problem 5-52

The rigid beam of negligible weight is supported horizontally by two springs and a pin. If the springs are uncompressed when the load is removed, determine the force in each spring when the load **P** is applied. Also, compute the vertical deflection of end *C*. Assume the spring stiffness *k* is large enough so that only small deflections occur. *Hint*: The beam rotates about *A* so the deflections in the springs can be related.

Solution:

$\Sigma M_A = 0;$

$$F_B L + F_C 2\ L - P\frac{3}{2}L = 0$$

$$F_B + 2\ F_C = 1.5\ P$$

$$\Delta_C = 2\ \Delta_B$$

$$\frac{F_C}{k} = \frac{2\ F_B}{k}$$

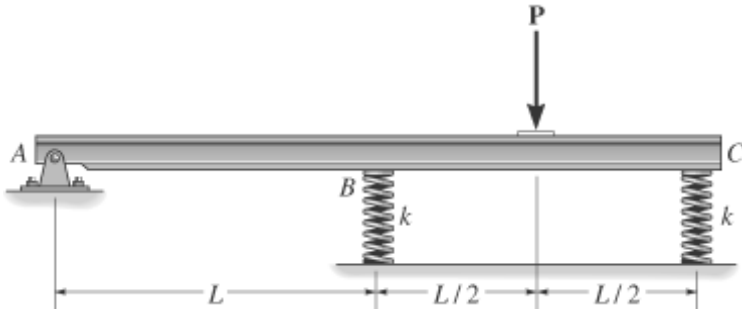
$$F_C = 2F_B$$

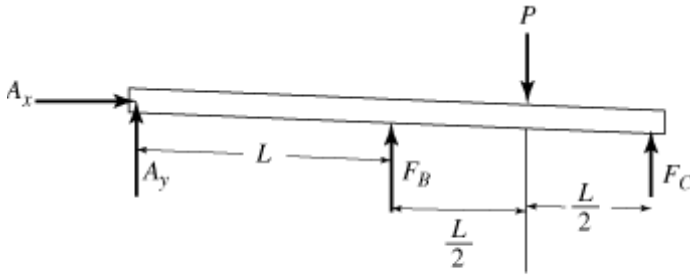
$$5\ F_B = 1.5\ P$$

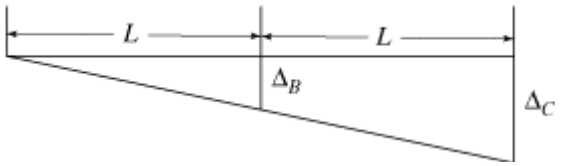
$$F_B = 0.3\ P$$

$$F_c = 0.6\ P$$

$$\Delta_C = \frac{0.6\ P}{k}$$







Problem 5-53

The rod supports a weight *W* and is pinned at its end *A*. If it is also subjected to a couple moment of *M*, determine the angle θ for equilibrium.The spring has an unstretched length δ and a stiffness *k*.

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Given:

$W = 200\ \text{lb}$

$M = 100\ \text{lb ft}$

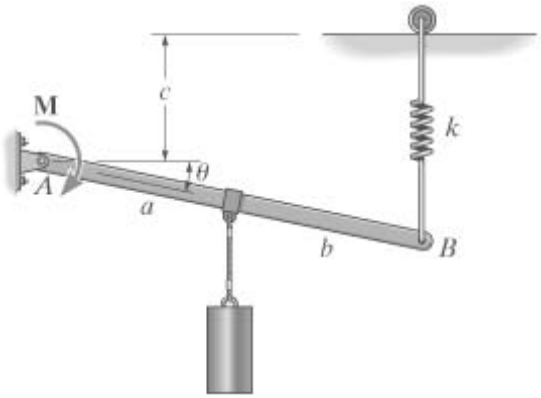
$\delta = 2\ \text{ft}$

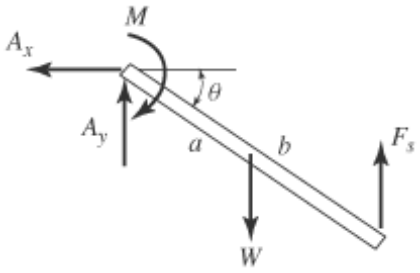
$k = 50\ \frac{\text{lb}}{\text{ft}}$

$a = 3\ \text{ft}$

$b = 3\ \text{ft}$

$c = 2\ \text{ft}$





Solution:

Initial Guess: $\theta = 10\ \text{deg}$

Given

$k\left[(a+b)\sin(\theta) + c - \delta\right](a+b)\cos(\theta) - W\,a\cos(\theta) - M = 0$

$\theta = \text{Find}(\theta)$ $\theta = 23.2\ \text{deg}$

Problem 5-54

The smooth pipe rests against the wall at the points of contact *A, B,* and *C.* Determine the reactions at these points needed to support the vertical force **F**. Neglect the pipe's thickness in the calculation.

Given:

$F = 45\ \text{lb}$

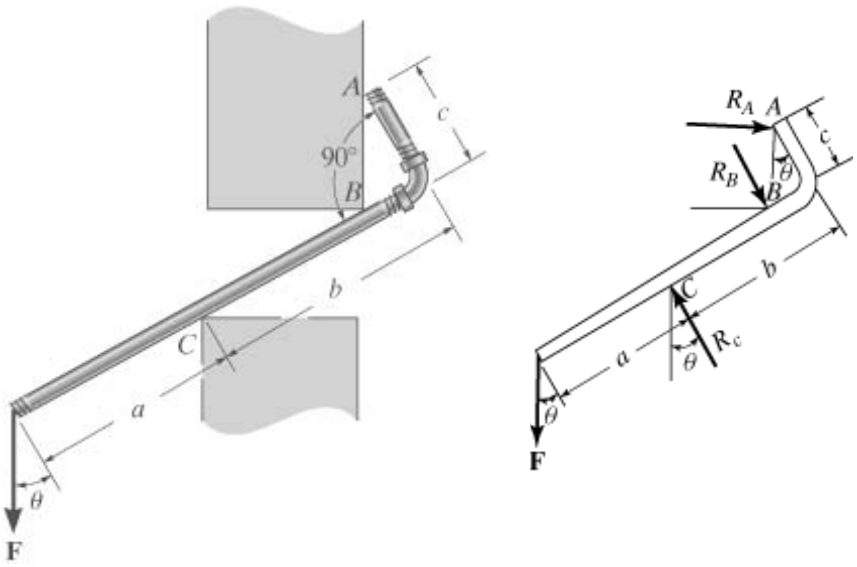
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$\theta = 30$ deg

$a = 16$ in

$b = 20$ in

$c = 8$ in



Solution:

Initial Guesses: $R_A = 1$ lb $R_B = 1$ lb $R_C = 1$ lb

Given

$$\circlearrowleft + \Sigma M_A = 0; \quad F\cos(\theta)(a+b) - F\sin(\theta)c - R_C b + R_B c\tan(\theta) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad R_C\cos(\theta) - R_B\cos(\theta) - F = 0$$

$$\xrightarrow{+} \Sigma F_x = 0; \quad R_A + R_B\sin(\theta) - R_C\sin(\theta) = 0$$

$$\begin{pmatrix} R_A \\ R_B \\ R_C \end{pmatrix} = \text{Find}(R_A, R_B, R_C) \qquad \begin{pmatrix} R_A \\ R_B \\ R_C \end{pmatrix} = \begin{pmatrix} 25.981 \\ 11.945 \\ 63.907 \end{pmatrix} \text{lb}$$

Problem 5-55

The rigid metal strip of negligible weight is used as part of an electromagnetic switch. If the stiffness of the springs at *A* and *B* is *k*, and the strip is originally horizontal when the springs are unstretched, determine the smallest force needed to close the contact gap at *C*.

Units Used:

$\text{mN} = 10^{-3}$ N

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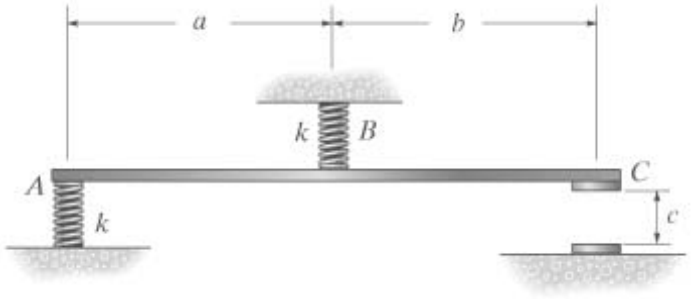
Given:

$a = 50\ \text{mm}$

$b = 50\ \text{mm}$

$c = 10\ \text{mm}$

$k = 5\ \frac{\text{N}}{\text{m}}$



Solution:

Initial Guesses: $F = 0.5\ \text{N}$ $y_A = 1\ \text{mm}$ $y_B = 1\ \text{mm}$

Given

$$\frac{c - y_A}{a + b} = \frac{y_B - y_A}{a} \qquad k\,y_A + k\,y_B - F = 0 \qquad k\,y_B a - F(a + b) = 0$$

$$\begin{pmatrix} y_A \\ y_B \\ F \end{pmatrix} = \text{Find}(y_A, y_B, F) \qquad \begin{pmatrix} y_A \\ y_B \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \text{mm} \qquad F = 10\,\text{mN}$$

Problem 5-56

The rigid metal strip of negligible weight is used as part of an electromagnetic switch. Determine the maximum stiffness k of the springs at A and B so that the contact at C closes when the vertical force developed there is **F**. Originally the strip is horizontal as shown.

Units Used:

$\text{mN} = 10^{-3}\ \text{N}$

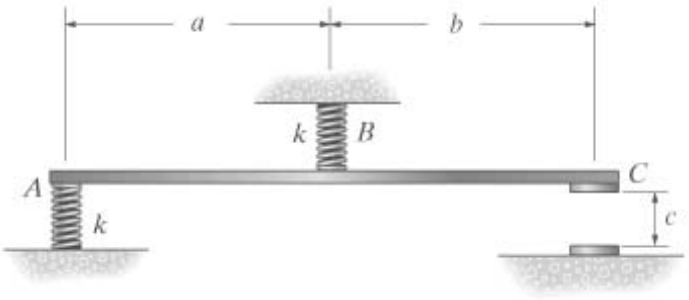
Given:

$a = 50\ \text{mm}$

$b = 50\ \text{mm}$

$c = 10\ \text{mm}$

$F = 0.5\ \text{N}$



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Solution:

Initial Guesses: $k = 1\,\frac{\text{N}}{\text{m}}$ $y_A = 1\ \text{mm}$ $y_B = 1\ \text{mm}$

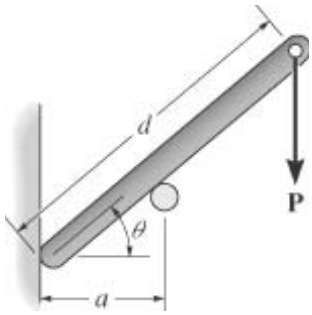
Given

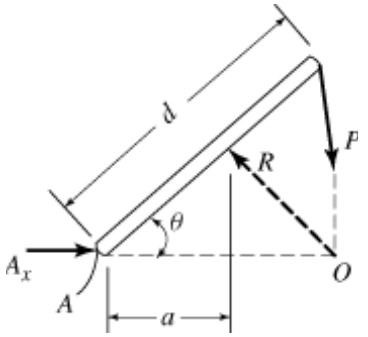
$$\frac{c - y_A}{a + b} = \frac{y_B - y_A}{a} \qquad k\,y_A + k\,y_B - F = 0 \qquad k\,y_B a - F(a + b) = 0$$

$$\begin{pmatrix} y_A \\ y_B \\ k \end{pmatrix} = \text{Find}(y_A, y_B, k) \qquad \begin{pmatrix} y_A \\ y_B \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}\text{mm} \qquad k = 250\,\frac{\text{N}}{\text{m}}$$

Problem 5-57

Determine the distance d for placement of the load **P** for equilibrium of the smooth bar in the position θ as shown. Neglect the weight of the bar.





Solution:

$+\uparrow \Sigma F_y = 0;$ $\quad R\cos(\theta) - P = 0$

$\circlearrowleft + \Sigma M_A = 0;$ $\quad -P\,d\cos(\theta) + R\,\frac{a}{\cos(\theta)} = 0$

$$R\,d\cos(\theta)^2 = R\,\frac{a}{\cos(\theta)}$$

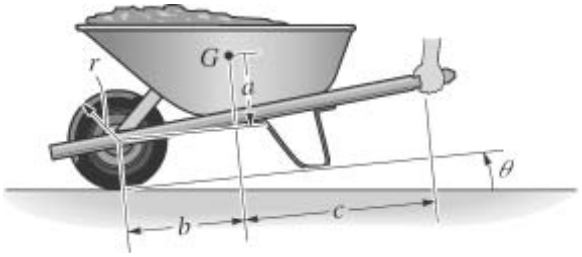
$$d = \frac{a}{\cos(\theta)^3}$$

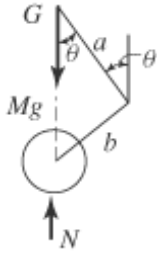
Problem 5-58

The wheelbarrow and its contents have mass m and center of mass at G. Determine the greatest

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angle of tilt θ without causing the wheelbarrow to tip over.





Solution:

Require point *G* to be over the wheel axle for tipping. Thus

$$b\cos(\theta) = a\sin(\theta)$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

Problem 5-59

Determine the force **P** needed to pull the roller of mass *M* over the smooth step.

Given:

$M = 50$ kg

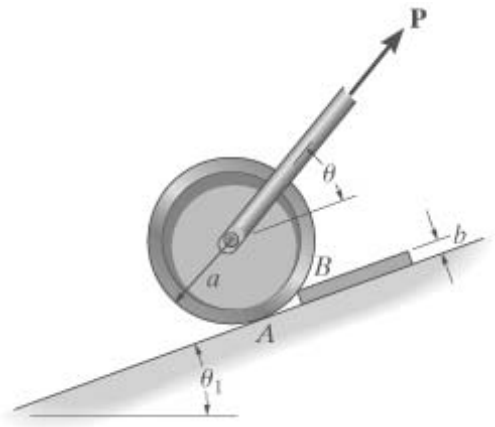
$a = 0.6$ m

$b = 0.1$ m

$\theta = 60$ deg

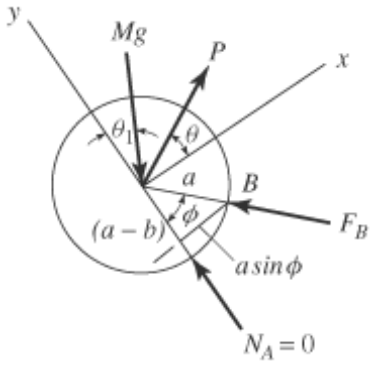
$\theta_1 = 20$ deg

$g = 9.81 \frac{\text{m}}{\text{s}^2}$



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Solution:



$$\phi = \text{acos}\left(\frac{a-b}{a}\right)$$

$$\phi = 33.56\,\text{deg}$$

$\Sigma M_B = 0,\quad M\,g\sin(\theta_1)(a-b) + M\,g\cos(\theta_1)a\sin(\phi)\ \ldots\ = 0$
$+\ P\cos(\theta)(a-b) - P\sin(\theta)a\sin(\phi)$

$$P = Mg\left[\frac{\sin(\theta_1)\ (a-b) + \cos(\theta_1)\ a\ \sin(\phi)}{\cos(\theta)\ (a-b) + \sin(\theta)\ a\ \sin(\phi)}\right]$$

$P = 441$ N

Problem 5-60

Determine the magnitude and direction θ of the minimum force **P** needed to pull the roller of mass M over the smooth step.

Given:

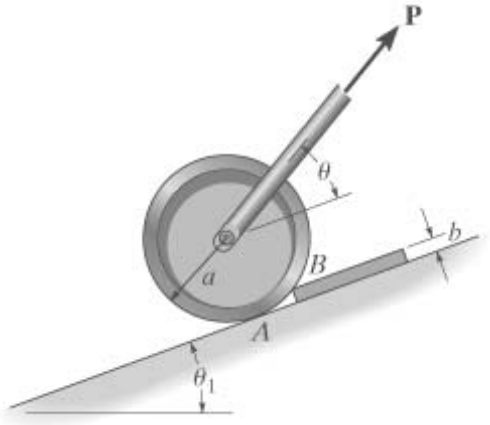
$a = 0.6$ m

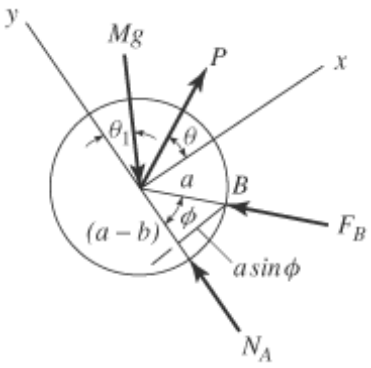
$b = 0.1$ m

$\theta_1 = 20$ deg

$M = 50$ kg

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$





Solution:

For P_{min}, N_A tends to 0 $\quad \phi = \text{acos}\left(\frac{a-b}{a}\right)$

$$\phi = 33.56\,\text{deg}$$

$\Sigma M_B = 0\quad M\,g\sin(\theta_1)(a-b) + M\,g\cos(\theta_1)a\sin(\phi)\ \ldots\ = 0$
$+\ \left[-P\cos(\theta)(a-b)\right] - P\sin(\theta)a\sin(\phi)$

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$$P = \frac{M\,g\left[\sin(\theta_1)(a-b) + \cos(\theta_1)\,a\sin(\phi)\right]}{\cos(\theta)(a-b) + a\sin(\phi)\sin(\theta)}$$

For P_{min}:

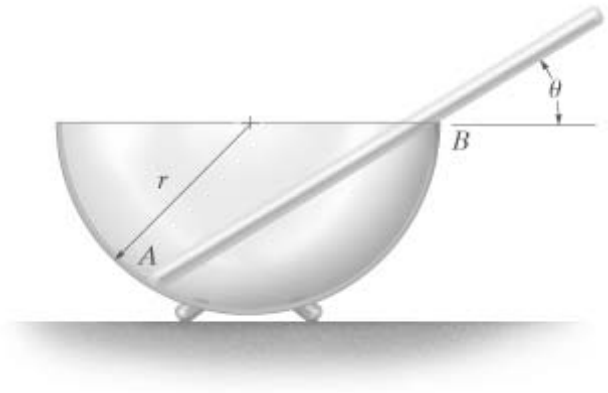
$$\frac{dP}{d\theta} = \frac{M\,g\left[\sin(\theta_1)(a-b) + \cos(\theta_1)\,a\sin(\phi)\right]}{\left[\cos(\theta)(a-b) + a\sin(\phi)\sin(\theta)\right]^2}\left[a\sin(\phi)\cos(\theta) - (a-b)\sin(\theta)\right] = 0$$

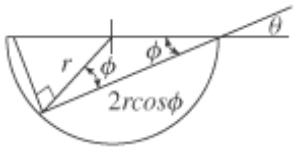
which gives, $\theta = \operatorname{atan}\left(\sin(\phi)\cdot\frac{a}{a-b}\right)$ $\theta = 33.6\,\text{deg}$

$$P = \frac{M\,g\left[\sin(\theta_1)(a-b) + \cos(\theta_1)\,a\sin(\phi)\right]}{\cos(\theta)(a-b) + a\sin(\phi)\sin(\theta)}$$ $P = 395\ \text{N}$

Problem 5-61

A uniform glass rod having a length L is placed in the smooth hemispherical bowl having a radius r. Determine the angle of inclination θ for equilibrium.

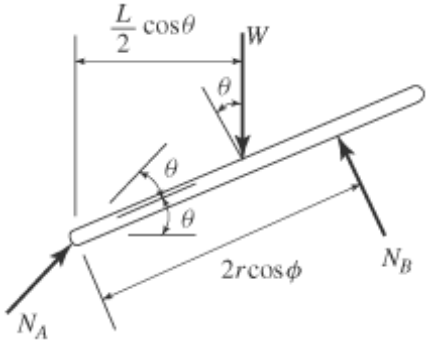




Solution:

By Observation $\phi = \theta$.

Equilibirium :



$\Sigma M_A = 0;\ N_B 2\,r\cos(\theta) - W\frac{L}{2}\cos(\theta) = 0$

$$N_B = \frac{W\,L}{4\,r}$$

$\Sigma F_x = 0;\quad N_A\cos(\theta) - W\sin(\theta) = 0$ $N_A = W\tan(\theta)$

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$$\Sigma F_y = 0; \quad W\tan(\theta)\sin(\theta) + \frac{W\,L}{4\,r} - W\cos(\theta) = 0$$

$$\sin(\theta)^2 - \cos(\theta)^2 = 1 - 2\cos(\theta)^2 = \frac{-L}{4\,r}\cos(\theta)$$

$$2\cos(\theta)^2 - \frac{L}{4\,r}\cos(\theta) - 1 = 0$$

$$\cos(\theta) = \frac{L + \sqrt{L^2 + 128\,r^2}}{16\,r} \qquad \theta = \operatorname{acos}\left(\frac{L + \sqrt{L^2 + 128\,r^2}}{16\,r}\right)$$

Problem 5-62

The disk has mass M and is supported on the smooth cylindrical surface by a spring having stiffness k and unstretched length l_0. The spring remains in the horizontal position since its end A is attached to the small roller guide which has negligible weight. Determine the angle θ to the nearest degree for equilibrium of the roller.

Given:

$M = 20$ kg

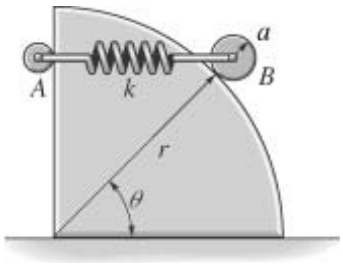
$k = 400\ \frac{\text{N}}{\text{m}}$

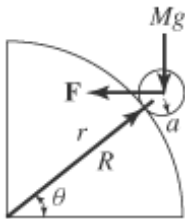
$l_0 = 1$ m

$r = 2$ m

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

$a = 0.2$ m





Guesses $F = 10$ N $R = 10$ N $\theta = 30$ deg

Solution: Given

$$\xrightarrow{+} \Sigma F_y = 0; \quad R\sin(\theta) - M\,g = 0$$

$$+\uparrow \Sigma F_x = 0; \quad R\cos(\theta) - F = 0$$

Spring $F = k\left[(r + a)\cos(\theta) - l_0\right]$

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$$\begin{pmatrix} F \\ R \\ \theta \end{pmatrix} = \text{Find}(F, R, \theta) \qquad \begin{pmatrix} F \\ R \end{pmatrix} = \begin{pmatrix} 163.633 \\ 255.481 \end{pmatrix} \text{N} \qquad \theta = 50.171\,\text{deg}$$

There is also another answer that we can find by choosing different starting guesses.

Guesses $F = 200$ N $R = 200$ N $\theta = 20$ deg

Solution: Given

$\overset{+}{\rightarrow} \Sigma F_y = 0; \qquad R\sin(\theta) - M\,g = 0$

$+\uparrow \Sigma F_x = 0; \qquad R\cos(\theta) - F = 0$

Spring $F = k\left[(r + a)\cos(\theta) - l_0\right]$

$$\begin{pmatrix} F \\ R \\ \theta \end{pmatrix} = \text{Find}(F, R, \theta) \qquad \begin{pmatrix} F \\ R \end{pmatrix} = \begin{pmatrix} 383.372 \\ 430.66 \end{pmatrix} \text{N} \qquad \theta = 27.102\,\text{deg}$$

Problem 5-63

Determine the *x*, *y*, *z* components of reaction at the fixed wall *A*.The force $\mathbf{F}_2$ is parallel to the *z* axis and the force $\mathbf{F}_1$ is parallel to the *y* axis.

Given:

$a = 2$ m $\qquad d = 2$ m

$b = 1$ m $\qquad F_1 = 200$ N

$c = 2.5$ m $\qquad F_2 = 150$ N

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Solution:

$\Sigma F_x = 0;\quad A_x = 0$

$\Sigma F_x = 0;\quad A_y = -F_1$

$A_y = -200\ \text{N}$

$\Sigma F_x = 0;\quad A_z = F_2$

$A_z = 150\ \text{N}$

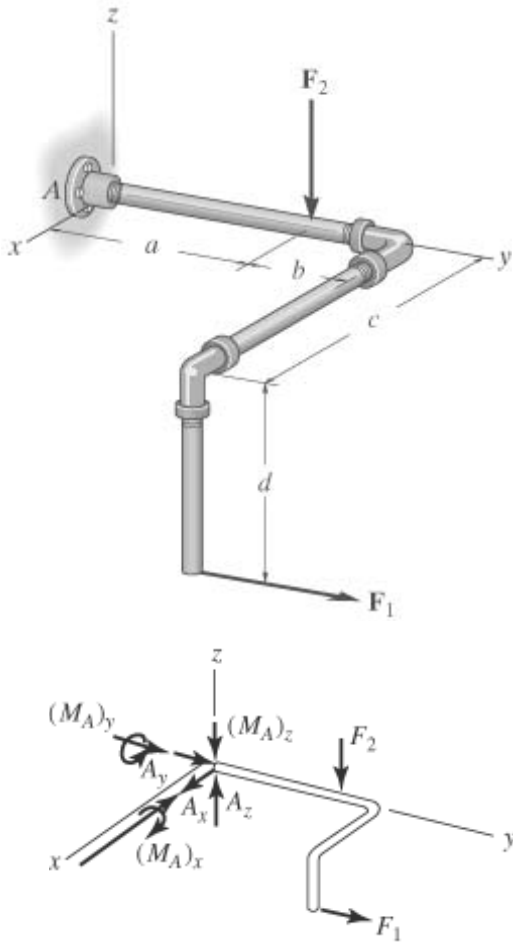
$\Sigma M_x = 0;\quad M_{Ax} = -F_2\,a + F_1\,d$

$M_{Ax} = 100\,\text{N}\cdot\text{m}$

$\Sigma M_y = 0;\quad M_{Ay} = 0$

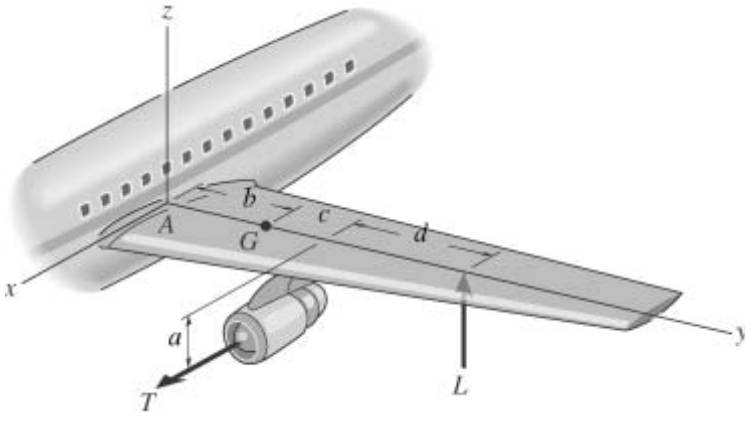
$\Sigma M_z = 0;\quad M_{Az} = F_1\,c$

$M_{Az} = 500\,\text{N}\cdot\text{m}$



Problem 5-64

The wing of the jet aircraft is subjected to thrust T from its engine and the resultant lift force L. If the mass of the wing is M and the mass center is at G, determine the x, y, z components of reaction where the wing is fixed to the fuselage at A.



Units Used:

$\text{Mg} = 10^3\ \text{kg}$

$\text{kN} = 10^3\ \text{N}$

$\text{g} = 9.81\,\dfrac{\text{m}}{\text{s}^2}$

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Given:

$T = 8$ kN

$L = 45$ kN

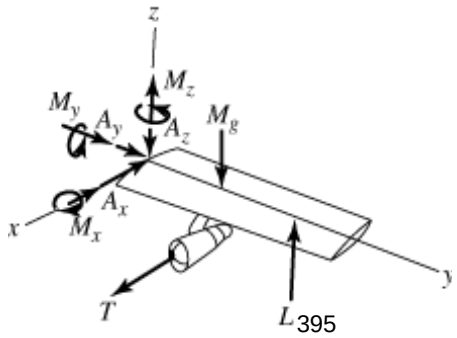
$M = 2.1$ Mg

$a = 2.5$ m

$b = 5$ m

$c = 3$ m

$d = 7$ m



Solution:

$\Sigma F_x = 0;$ $\quad -A_x + T = 0$

$A_x = T$ $\qquad A_x = 8\,\text{kN}$

$\Sigma F_y = 0;$ $\quad A_y = 0$ $\qquad A_y = 0$

$\Sigma F_z = 0;$ $\quad -A_z - M\,g + L = 0$

$A_z = L - M\,g$ $\qquad A_z = 24.4\,\text{kN}$

$\Sigma M_y = 0;$ $\quad M_y - T(a) = 0$

$M_y = T\,a$ $\qquad M_y = 20.0\,\text{kN}\cdot\text{m}$

$\Sigma M_x = 0;$ $\quad L(b + c + d) - M\,g\,b - M_x = 0$

$M_x = L(b + c + d) - M\,g\,b$ $\qquad M_x = 572\,\text{kN}\cdot\text{m}$

$\Sigma M_z = 0;$ $\quad M_z - T(b + c) = 0$

$M_z = T(b + c)$ $\qquad M_z = 64.0\,\text{kN}\cdot\text{m}$

Problem 5-65

The uniform concrete slab has weight *W*. Determine the tension in each of the three parallel supporting cables when the slab is held in the horizontal plane as shown.

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Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

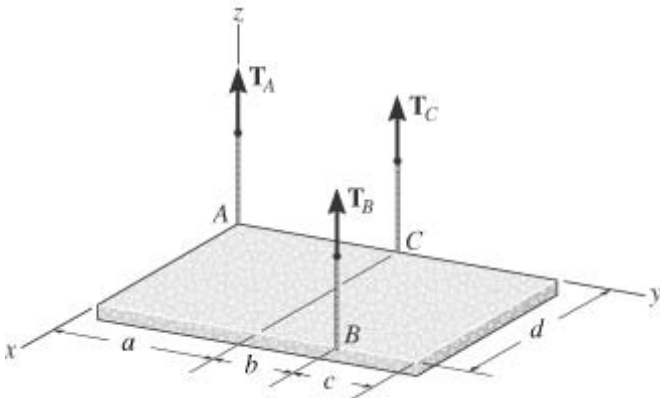
$W = 5500 \text{ lb}$

$a = 6 \text{ ft}$

$b = 3 \text{ ft}$

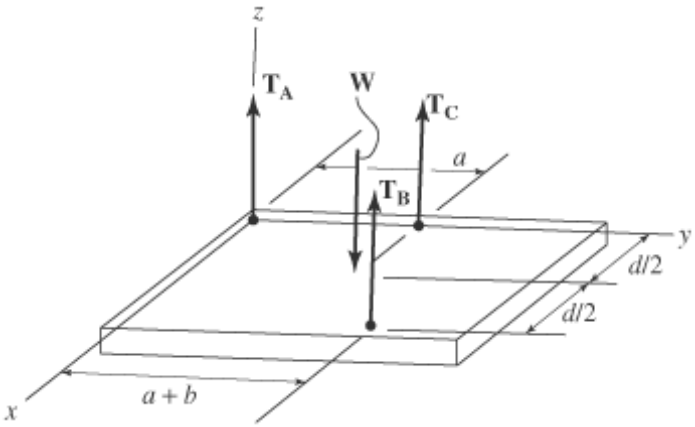
$c = 3 \text{ ft}$

$d = 6 \text{ ft}$



Solution:

Equations of Equilibrium : The cable tension T_B can be obtained directly by summing moments about the *y* axis.



$\Sigma M_y = 0; \qquad W\frac{d}{2} - T_B d = 0 \qquad T_B = \frac{W}{2} \qquad T_B = 2.75 \text{ kip}$

$\Sigma M_x = 0; \qquad T_C a + T_B(a+b) - W\left(\frac{a+b+c}{2}\right) = 0$

$T_C = \frac{1}{a}\left[W\frac{a+b+c}{2} - T_B(a+b)\right] \qquad T_C = 1.375 \text{ kip}$

$\Sigma F_z = 0; \qquad T_A + T_B + T_C - W = 0 \qquad T_A = -T_B - T_C + W \qquad T_A = 1.375 \text{ kip}$

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Problem 5-66

The air-conditioning unit is hoisted to the roof of a building using the three cables. If the tensions in the cables are $\mathbf{T_A}$, $\mathbf{T_B}$ and $\mathbf{T_C}$, determine the weight of the unit and the location (x, y) of its center of gravity G.

Given:

$T_A = 250$ lb

$T_B = 300$ lb

$T_C = 200$ lb

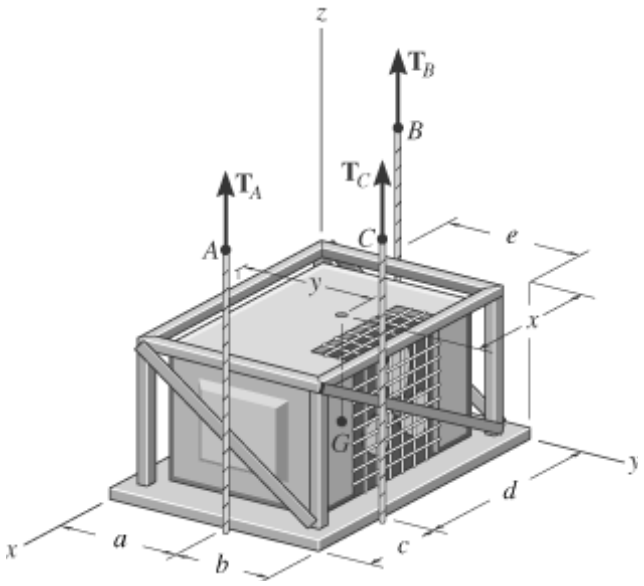
$a = 5$ ft

$b = 4$ ft

$c = 3$ ft

$d = 7$ ft

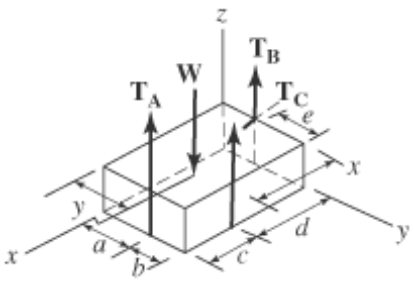
$e = 6$ ft



Solution:

$\Sigma F_z = 0; \quad T_A + T_B + T_C - W = 0$

$W = T_A + T_B + T_C \qquad W = 750\,\text{lb}$



$\Sigma M_y = 0; \quad W\,\text{x} - T_A(c + d) - T_C\,d = 0$

$$x = \frac{T_A(c + d) + T_C\,d}{W} \qquad x = 5.2\,\text{ft}$$

$\Sigma M_x = 0; \quad T_A\,a + T_B(a + b - e) + T_C(a + b) - W\,y = 0$

$$y = \frac{T_A\,a + T_B(a + b - e) + T_C(a + b)}{W} \qquad y = 5.267\,\text{ft}$$

Problem 5-67

The platform truck supports the three loadings shown. Determine the normal reactions on each of its three wheels.

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Given:

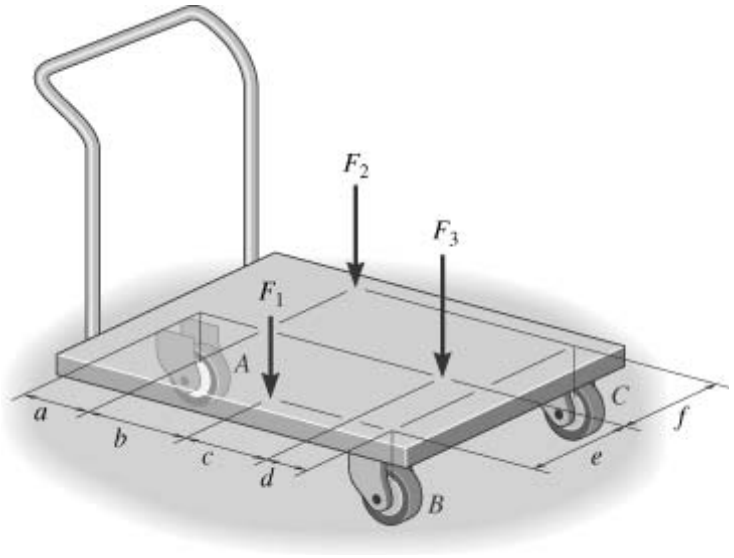
$F_1 = 380$ lb $\quad b = 12$ in

$F_2 = 500$ lb $\quad c = 10$ in

$F_3 = 800$ lb $\quad d = 5$ in

$e = 12$ in

$a = 8$ in $\quad f = 12$ in



Solution:

The initail guesses are $F_A = 1$ lb $\quad F_B = 1$ lb $\quad F_C = 1$ lb

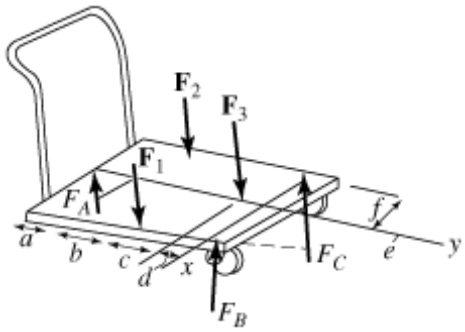
Given

$\Sigma M_x = 0; \quad F_1(c+d) + F_2(b+c+d) + F_3 d - F_A(a+b+c+d) = 0$

$\Sigma M_y = 0; \quad F_1 e - F_B e - F_2 f + F_C f = 0$

$\Sigma F_y = 0; \quad F_B + F_C - F_2 + F_A - F_1 - F_3 = 0$

$$\begin{pmatrix} F_A \\ F_B \\ F_C \end{pmatrix} = \text{Find}(F_A, F_B, F_C) \qquad \begin{pmatrix} F_A \\ F_B \\ F_C \end{pmatrix} = \begin{pmatrix} 663 \\ 449 \\ 569 \end{pmatrix} \text{lb}$$



Problem 5-68

Due to an unequal distribution of fuel in the wing tanks, the centers of gravity for the airplane fuselage *A* and wings *B* and *C* are located as shown. If these components have weights W_A, W_B and W_C, determine the normal reactions of the wheels *D*, *E*, and *F* on the ground.

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Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$W_A = 45000 \text{ lb}$

$W_B = 8000 \text{ lb}$

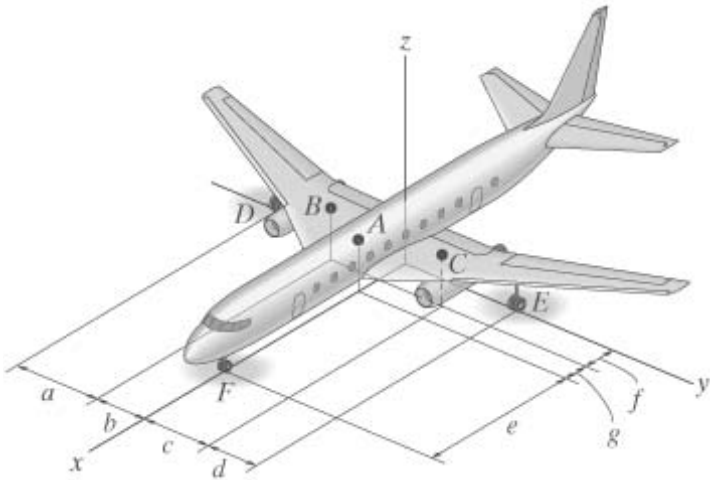
$W_C = 6000 \text{ lb}$

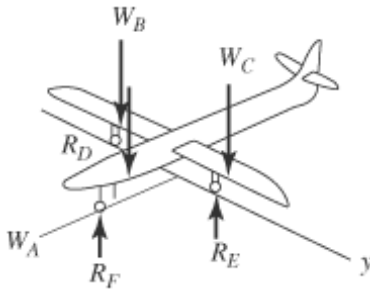
$a = 8 \text{ ft} \quad e = 20 \text{ ft}$

$b = 6 \text{ ft} \quad f = 4 \text{ ft}$

$c = 8 \text{ ft} \quad g = 3 \text{ ft}$

$d = 6 \text{ ft}$





Solution:

Initial guesses:

$R_D = 1 \text{ kip} \quad R_E = 1 \text{ kip} \quad R_F = 1 \text{ kip}$

Given

$\Sigma M_x = 0; \quad W_B b - R_D(a + b) - W_C c + R_E(c + d) = 0$

$\Sigma M_y = 0; \quad W_B f + W_A(g + f) + W_C f - R_F(e + g + f) = 0$

$\Sigma F_z = 0; \quad R_D + R_E + R_F - W_A - W_B - W_C = 0$

$$\begin{pmatrix} R_D \\ R_E \\ R_F \end{pmatrix} = \text{Find}(R_D, R_E, R_F) \qquad \begin{pmatrix} R_D \\ R_E \\ R_F \end{pmatrix} = \begin{pmatrix} 22.6 \\ 22.6 \\ 13.7 \end{pmatrix} \text{kip}$$

Problem 5-69

If the cable can be subjected to a maximum tension **T**, determine the maximum force **F** which may be applied to the plate. Compute the *x, y, z* components of reaction at the hinge *A* for this loading.

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Given:

$a = 3$ ft

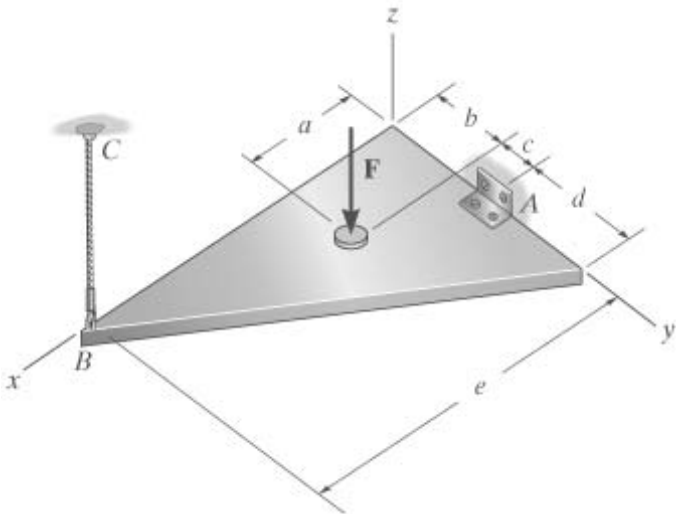
$b = 2$ ft

$c = 1$ ft

$d = 3$ ft

$e = 9$ ft

$T = 300$ lb

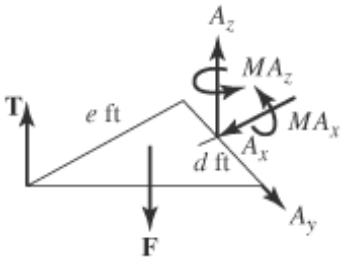


Solution:

Initial guesses:

$F = 10$ lb $\quad M_{Ax} = 10$ lb ft $\quad M_{Az} = 10$ lb ft

$A_x = 10$ lb $\quad A_y = 10$ lb $\quad A_z = 10$ lb



Given

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ T - F \end{pmatrix} = 0 \qquad \begin{pmatrix} M_{Ax} \\ 0 \\ M_{Az} \end{pmatrix} + \begin{pmatrix} a \\ -c \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -F \end{pmatrix} + \begin{pmatrix} e \\ -b - c \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ T \end{pmatrix} = 0$$

$$\begin{pmatrix} F \\ A_x \\ A_y \\ A_z \\ M_{Ax} \\ M_{Az} \end{pmatrix} = \text{Find}\left(F, A_x, A_y, A_z, M_{Ax}, M_{Az}\right) \qquad \begin{pmatrix} M_{Ax} \\ M_{Az} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{lb ft}$$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 600 \end{pmatrix} \text{lb} \qquad F = 900 \text{ lb}$$

Problem 5-70

The boom *AB* is held in equilibrium by a ball-and-socket joint *A* and a pulley and cord system as shown. Determine the *x, y, z* components of reaction at *A* and the tension in cable *DEC.*

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Given:

$$\mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ -1500 \end{pmatrix} \text{ lb}$$

$a = 5$ ft

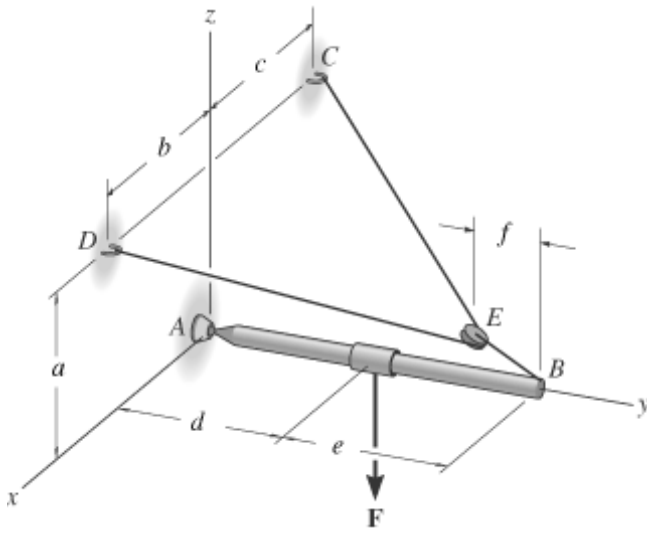
$b = 4$ ft

$c = b$

$d = 5$ ft

$e = 5$ ft

$f = 2$ ft

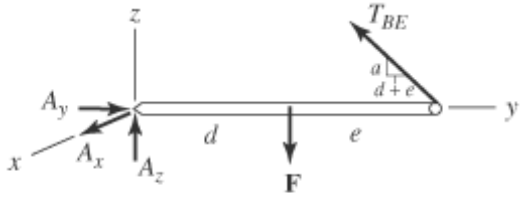


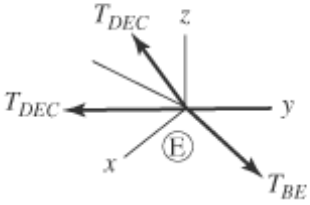
Solution:

$$\alpha = \text{atan}\left(\frac{a}{d+e}\right)$$

$$L = \sqrt{a^2 + (d+e)^2}$$

$$\beta = \text{atan}\left(\frac{b}{L - \frac{f\,L}{d+e}}\right)$$





Guesses $T_{BE} = 1$ lb $T_{DEC} = 1$ lb

$A_x = 1$ lb $A_y = 1$ lb $A_z = 1$ lb

Given $2\,T_{DEC}\cos(\beta) = T_{BE}$

$$\begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix} \times \mathbf{F} + \begin{pmatrix} 0 \\ d+e \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -T_{BE}\cos(\alpha) \\ T_{BE}\sin(\alpha) \end{pmatrix} = 0 \qquad \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} + \mathbf{F} + T_{BE}\begin{pmatrix} 0 \\ -\cos(\alpha) \\ \sin(\alpha) \end{pmatrix} = 0$$

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$$\begin{pmatrix} A_x \\ A_y \\ A_z \\ T_{BE} \\ T_{DEC} \end{pmatrix} = \text{Find}\left(A_x, A_y, A_z, T_{BE}, T_{DEC}\right) \quad T_{DEC} = 919\,\text{lb} \qquad \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} 0 \\ 1.5 \times 10^3 \\ 750 \end{pmatrix} \text{lb}$$

Problem 5-71

The cable *CED* can sustain a maximum tension T_{max} before it fails. Determine the greatest vertical force *F* that can be applied to the boom. Also, what are the *x, y, z* components of reaction at the ball-and-socket joint *A*?

Given:

$T_{max} = 800$ lb

$a = 5$ ft

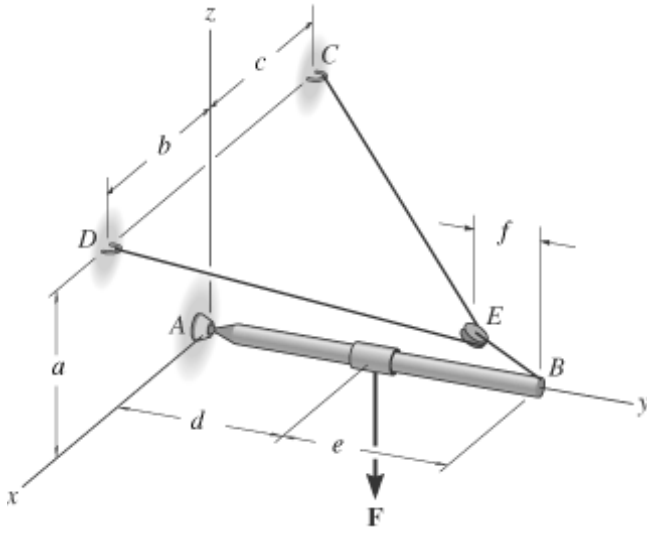
$b = 4$ ft

$c = b$

$d = 5$ ft

$e = 5$ ft

$f = 2$ ft

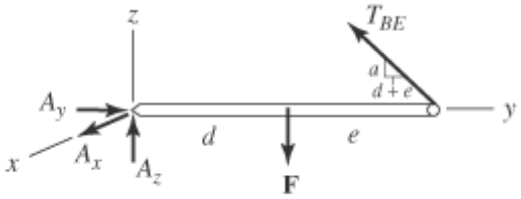


Solution:

$$\alpha = \text{atan}\left(\frac{a}{d+e}\right)$$

$$L = \sqrt{a^2 + (d+e)^2}$$

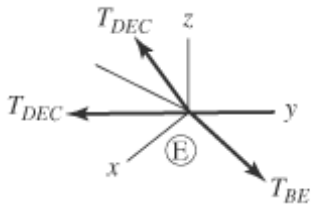
$$\beta = \text{atan}\left(\frac{b}{L - \frac{f\,L}{d+e}}\right) \qquad T_{DEC} = T_{max}$$



Guesses $T_{BE} = 1$ lb $F = 1$ lb

$A_x = 1$ lb $A_y = 1$ lb $A_z = 1$ lb

Given $2\,T_{DEC}\cos(\beta) = T_{BE}$



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$$\begin{pmatrix}0\\d\\0\end{pmatrix}\times\begin{pmatrix}0\\0\\-F\end{pmatrix}+\begin{pmatrix}0\\d+e\\0\end{pmatrix}\times\begin{pmatrix}0\\-T_{BE}\cos(\alpha)\\T_{BE}\sin(\alpha)\end{pmatrix}=0 \qquad \begin{pmatrix}A_x\\A_y\\A_z\end{pmatrix}+\begin{pmatrix}0\\0\\-F\end{pmatrix}+T_{BE}\begin{pmatrix}0\\-\cos(\alpha)\\\sin(\alpha)\end{pmatrix}=0$$

$$\begin{pmatrix}A_x\\A_y\\A_z\\T_{BE}\\F\end{pmatrix}=\text{Find}(A_x,A_y,A_z,T_{BE},F) \qquad F=1306\,\text{lb} \qquad \begin{pmatrix}A_x\\A_y\\A_z\end{pmatrix}=\begin{pmatrix}0\\1.306\times10^3\\653.197\end{pmatrix}\text{lb}$$

Problem 5-72

The uniform table has a weight W and is supported by the framework shown. Determine the smallest vertical force **P** that can be applied to its surface that will cause it to tip over. Where should this force be applied?

Given:

$W = 20$ lb

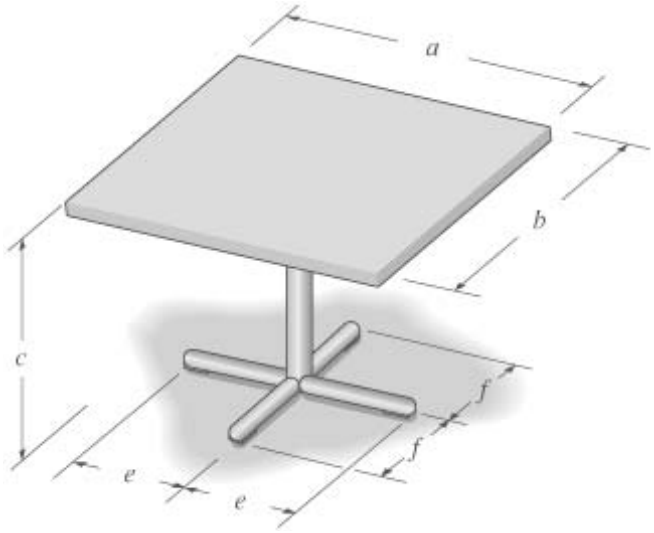
$a = 3.5$ ft

$b = 2.5$ ft

$c = 3$ ft

$e = 1.5$ ft

$f = 1$ ft

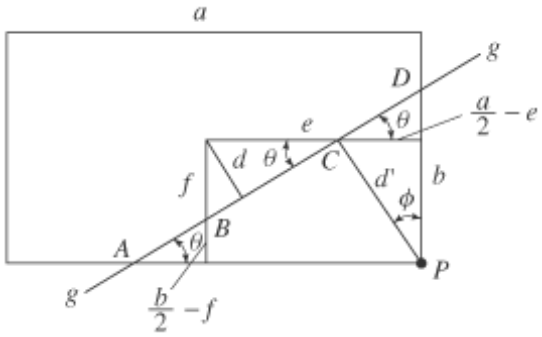


Solution:

$$\theta = \text{atan}\left(\frac{f}{e}\right) \qquad \theta = 33.69\,\text{deg}$$

$$d = e\sin(\theta) \qquad d = 0.832\,\text{ft}$$

$$\phi = \text{atan}\left(\frac{\frac{a}{2}-e}{\frac{b}{2}}\right) \qquad \phi = 11.31\,\text{deg}$$



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$$d' = \sqrt{\left(\frac{a}{2} - e\right)^2 + \left(\frac{b}{2}\right)^2} \qquad d' = 1.275\,\text{ft}$$

Tipping will occur about the g - g axis. Require P to be applied at the corner of the table for P_{min}.

$$W\,d = P\,d'\sin(90\ \text{deg} - \phi + \theta)$$

$$P = W\frac{d}{d'\sin(90\ \text{deg} - \phi + \theta)} \qquad P = 14.1\,\text{lb}$$

Problem 5-73

The windlass is subjected to load W. Determine the horizontal force P needed to hold the handle in the position shown, and the components of reaction at the ball-and-socket joint A and the smooth journal bearing B. The bearing at B is in proper alignment and exerts only force reactions perpendicular to the shaft on the windlass.

Given:

$W = 150$ lb

$a = 2$ ft

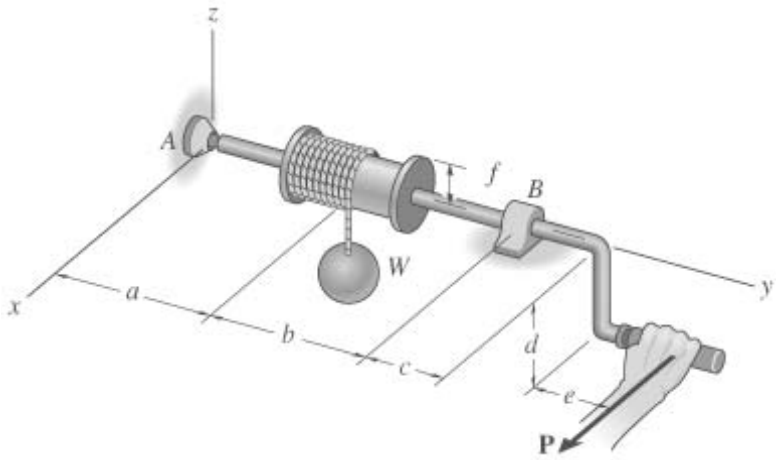
$b = 2$ ft

$c = 1$ ft

$d = 1$ ft

$e = 1$ ft

$f = 0.5$ ft



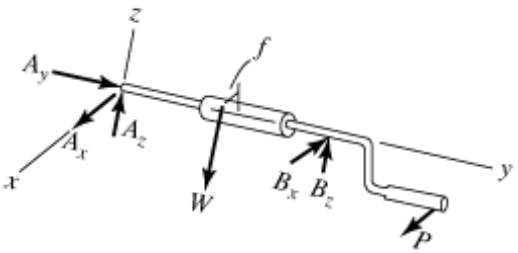
Solution:

$\Sigma M_y = 0;\quad W\,f - P\,d = 0$

$$P = \frac{W\,f}{d} \qquad P = 75\,\text{lb}$$

$\Sigma F_y = 0;\quad A_y = 0$ lb $\qquad A_y = 0$

$\Sigma M_x = 0;\quad -W\,a + B_z(a + b) = 0$



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$$B_z = \frac{W\,a}{a+b} \qquad B_z = 75\,\text{lb}$$

$\Sigma F_z = 0;$ $\quad A_z + B_z - W = 0$

$$A_z = W - B_z \qquad A_z = 75\,\text{lb}$$

$\Sigma M_z = 0;$ $\quad B_x(a+b) - (a+b+c+e)P = 0$

$$B_x = \frac{P(a+b+c+e)}{a+b} \qquad B_x = 112\,\text{lb}$$

$\Sigma F_x = 0;$ $\quad A_x - B_x + P = 0$

$$A_x = B_x - P \qquad A_x = 37.5\,\text{lb}$$

Problem 5-74

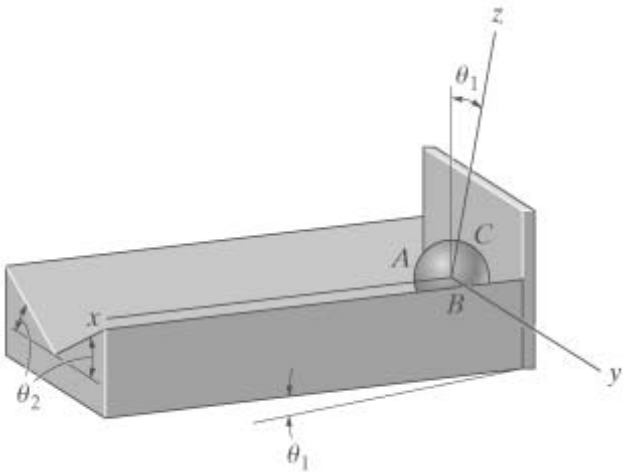
A ball of mass *M* rests between the grooves *A* and *B* of the incline and against a vertical wall at *C*. If all three surfaces of contact are smooth, determine the reactions of the surfaces on the ball. *Hint:* Use the *x, y, z* axes, with origin at the center of the ball, and the *z* axis inclined as shown.

Given:

$M = 2$ kg

$\theta_1 = 10$ deg

$\theta_2 = 45$ deg



Solution:

$\Sigma F_x = 0;$ $\quad F_C \cos(\theta_1) - M\,g \sin(\theta_1) = 0$

$$F_C = M\,g \cdot \tan(\theta_1)$$

$$F_C = 0.32\ \text{kg}\cdot\text{m}$$

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$\Sigma F_y = 0; \quad N_A \cos(\theta_2) - N_B \cos(\theta_2) = 0$

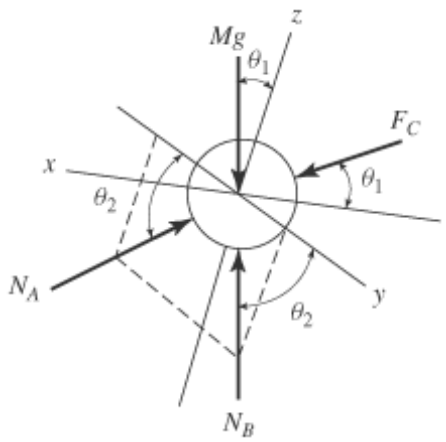
$N_A = N_B$

$\Sigma F_z = 0; \quad 2\, N_A \sin(\theta_2) - M\, g \cos(\theta_1) - F_c \sin(\theta_1) = 0$

$$N_A = \frac{1}{2} \cdot \frac{M\, g \cdot \cos(\theta_1) + F_c \cdot \sin(\theta_1)}{\sin(\theta_2)}$$

$N_A = 1.3 \text{ kg}\cdot\text{m}$

$N = N_A = N_B$



Problem 5-75

Member *AB* is supported by cable *BC* and at *A* by a *square* rod which fits loosely through the square hole at the end joint of the member as shown. Determine the components of reaction at *A* and the tension in the cable needed to hold the cylinder of weight *W* in equilibrium.

Units Used:

$\text{kip} = 10^3 \text{ lb}$

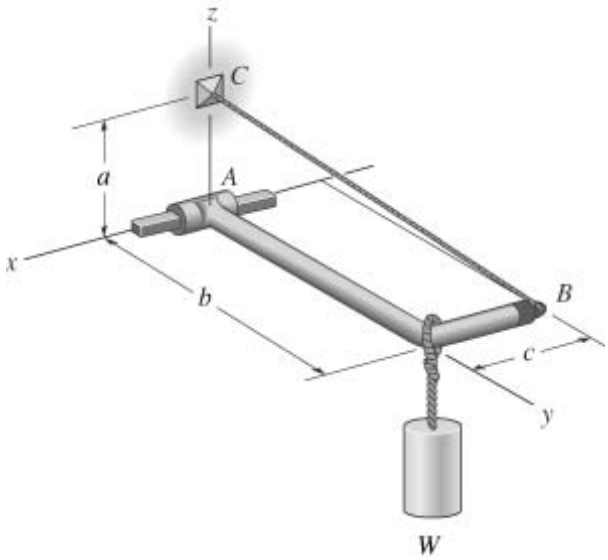
Given:

$W = 800 \text{ lb}$

$a = 2 \text{ ft}$

$b = 6 \text{ ft}$

$c = 3 \text{ ft}$



Solution:

$\Sigma F_x = 0 \qquad F_{BC}\left(\frac{c}{\sqrt{c^2 + b^2 + a^2}}\right) = 0 \qquad F_{BC} = 0 \text{ lb}$

$\Sigma F_y = 0 \qquad A_y = 0 \qquad A_y = 0\text{lb} \qquad A_y = 0\,\text{lb}$

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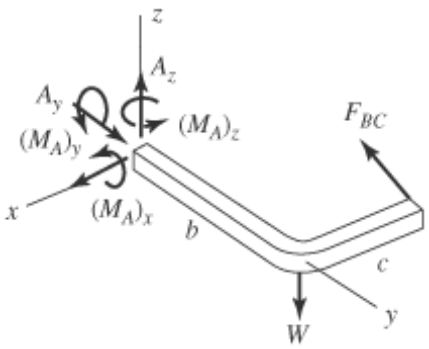
$\Sigma F_z = 0 \quad A_z - W = 0 \quad A_z = W \quad A_z = 800\,\text{lb}$

$\Sigma M_x = 0 \quad M_{Ax} - W\,b = 0$

$M_{Ax} = W\,b \quad M_{Ax} = 4.80\,\text{kip}\cdot\text{ft}$

$\Sigma M_y = 0 \quad M_{Ay} = 0\ \text{lb ft} \quad M_{Ay} = 0\,\text{lb}\cdot\text{ft}$

$\Sigma M_z = 0 \quad M_{Az} = 0\ \text{lb ft} \quad M_{Az} = 0\,\text{lb}\cdot\text{ft}$



Problem 5-76

The pipe assembly supports the vertical loads shown. Determine the components of reaction at the ball-and-socket joint *A* and the tension in the supporting cables *BC* and *BD*.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 3\ \text{kN} \quad d = 2\ \text{m}$

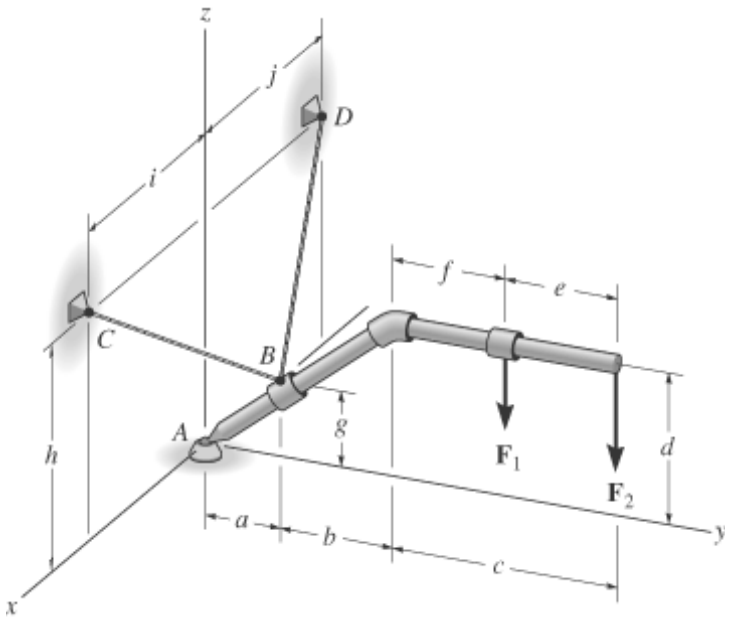
$F_2 = 4\ \text{kN} \quad e = 1.5\ \text{m}$

$a = 1\ \text{m} \quad g = 1\ \text{m}$

$b = 1.5\ \text{m} \quad h = 3\ \text{m}$

$c = 3\ \text{m} \quad i = 2\ \text{m}$

$f = c - e \quad j = 2\text{m}$

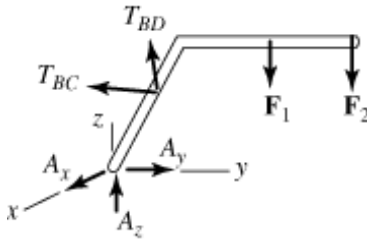


Solution:

The initial guesses are:

$T_{BD} = 1\ \text{kN} \quad T_{BC} = 1\ \text{kN}$

$A_x = 1\ \text{kN} \quad A_y = 1\ \text{kN} \quad A_z = 1\ \text{kN}$



The vectors

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$$\mathbf{r_1} = \begin{pmatrix} 0 \\ a+b+f \\ d \end{pmatrix} \qquad \mathbf{r_2} = \begin{pmatrix} 0 \\ a+b+c \\ d \end{pmatrix}$$

$$\mathbf{r_{BC}} = \begin{pmatrix} i \\ -a \\ h-g \end{pmatrix} \qquad \mathbf{r_{BD}} = \begin{pmatrix} -j \\ -a \\ h-g \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} 0 \\ a \\ g \end{pmatrix}$$

$$\mathbf{u_{BC}} = \frac{\mathbf{r_{BC}}}{\left|\mathbf{r_{BC}}\right|} \qquad \mathbf{u_{BD}} = \frac{\mathbf{r_{BD}}}{\left|\mathbf{r_{BD}}\right|} \qquad i = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad j = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad k = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Given $\quad A_x i + A_y j + A_z k - F_1 k - F_2 k + T_{BD}\mathbf{u_{BD}} + T_{BC}\mathbf{u_{BC}} = 0$

$$\mathbf{r_{AB}} \times \left(T_{BD}\mathbf{u_{BD}} + T_{BC}\mathbf{u_{BC}}\right) + \mathbf{r_1} \times \left(-F_1 k\right) + \mathbf{r_2} \times \left(-F_2 k\right) = 0$$

$$\begin{pmatrix} T_{BD} \\ T_{BC} \\ A_x \\ A_y \\ A_z \end{pmatrix} = \text{Find}\left(T_{BD}, T_{BC}, A_x, A_y, A_z\right) \qquad \begin{pmatrix} T_{BD} \\ T_{BC} \end{pmatrix} = \begin{pmatrix} 17 \\ 17 \end{pmatrix} \text{kN}$$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} 0 \\ 11.333 \\ -15.667 \end{pmatrix} \text{kN}$$

Problem 5-77

The hatch door has a weight W and center of gravity at *G*. If the force **F** applied to the handle at *C* has coordinate direction angles of α, β and γ, determine the magnitude of **F** needed to hold the door slightly open as shown. The hinges are in proper alignment and exert only force reactions on the door. Determine the components of these reactions if *A* exerts only *x* and *z* components of force and *B* exerts *x*, *y*, *z* force components.

Given:

$W = 80$ lb

$\alpha = 60$ deg

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$\beta = 45$ deg

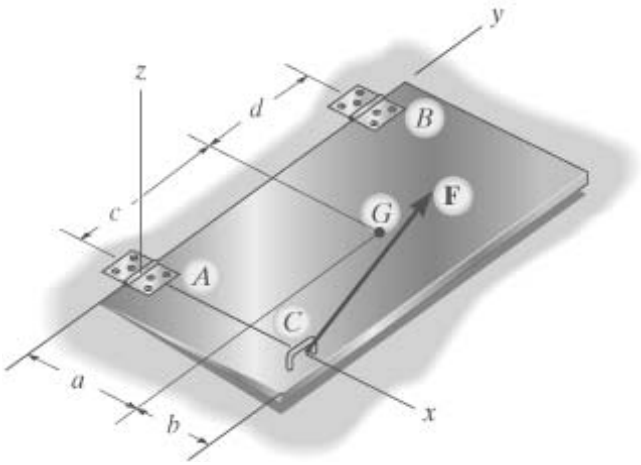
$\gamma = 60$ deg

$a = 3$ ft

$b = 2$ ft

$c = 4$ ft

$d = 3$ ft

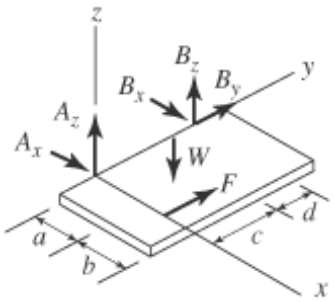


Solution:

Initial Guesses:

$A_x = 1$ lb $\quad A_z = 1$ lb $\quad F = 1$ lb

$B_x = 1$ lb $\quad B_y = 1$ lb $\quad B_z = 1$ lb



Given

$$\begin{pmatrix} A_x \\ 0 \\ A_z \end{pmatrix} + \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} + F\begin{pmatrix} \cos(\alpha) \\ \cos(\beta) \\ \cos(\gamma) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -W \end{pmatrix} = 0$$

$$\begin{pmatrix} a+b \\ 0 \\ 0 \end{pmatrix} \times \left[F\begin{pmatrix} \cos(\alpha) \\ \cos(\beta) \\ \cos(\gamma) \end{pmatrix} \right] + \begin{pmatrix} a \\ c \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -W \end{pmatrix} + \begin{pmatrix} 0 \\ c+d \\ 0 \end{pmatrix} \times \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = 0$$

$$\begin{pmatrix} A_x \\ A_z \\ B_x \\ B_y \\ B_z \\ F \end{pmatrix} = \text{Find}(A_x, A_z, B_x, B_y, B_z, F)$$

$$\begin{pmatrix} A_x \\ A_z \end{pmatrix} = \begin{pmatrix} -96.5 \\ -13.7 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{pmatrix} 48.5 \\ -67.9 \\ 45.7 \end{pmatrix} \text{lb}$$

$$F = 96\,\text{lb}$$

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Problem 5-78

The hatch door has a weight W and center of gravity at G. If the force **F** applied to the handle at C has coordinate direction angles α, β, γ determine the magnitude of **F** needed to hold the door slightly open as shown. If the hinge at A becomes loose from its attachment and is ineffective, what are the x, y, z components of reaction at hinge B?

Given:

$W = 80$ lb

$\alpha = 60$ deg

$\beta = 45$ deg

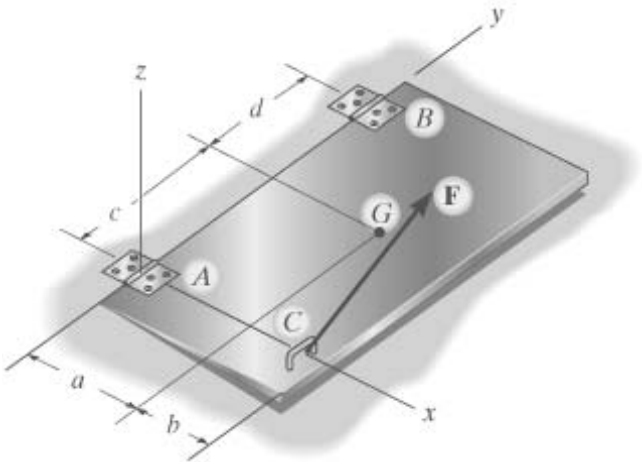
$\gamma = 60$ deg

$a = 3$ ft

$b = 2$ ft

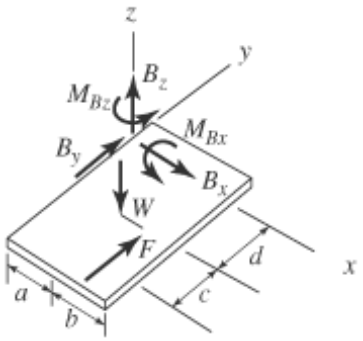
$c = 4$ ft

$d = 3$ ft



Solution:

$\Sigma M_y = 0;$ $\quad F = W \dfrac{a}{\cos(\gamma)\,(a+b)}$ $\quad F = 96\,\text{lb}$



$\Sigma F_x = 0;$ $\quad B_x + F\cos(\alpha) = 0$

$B_x = -F\cos(\alpha)$ $\quad B_x = -48\,\text{lb}$

$\Sigma F_y = 0;$ $\quad B_y + F\cos(\beta) = 0$

$B_y = -F\cos(\beta)$ $\quad B_y = -67.9\,\text{lb}$

$\Sigma F_z = 0;$ $\quad B_z - W + F\cos(\gamma) = 0$

$B_z = W - F\cos(\gamma)$ $\quad B_z = 32\,\text{lb}$

$\Sigma M_x = 0;$ $\quad M_{Bx} + W\,d - F\cos(\gamma)(c+d) = 0$

$M_{Bx} = -W\,d + F\cos(\gamma)\,(c+d)$ $\quad M_{Bx} = 96\,\text{lb}\cdot\text{ft}$

$\Sigma M_z = 0;$ $\quad M_{Bz} + F\cos(\alpha)(c+d) + F\cos(\beta)(a+b) = 0$

$M_{Bz} = -F\cos(\alpha)\,(c+d) - F\cos(\beta)\,(a+b)$ $\quad M_{Bz} = -675\,\text{lb}\cdot\text{ft}$

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Problem 5-79

The bent rod is supported at *A*, *B*, and *C* by smooth journal bearings. Compute the *x*, *y*, *z* components of reaction at the bearings if the rod is subjected to forces $\mathbf{F_1}$ and $\mathbf{F_2}$. $\mathbf{F_1}$ lies in the *y*-*z* plane. The bearings are in proper alignment and exert only force reactions on the rod.

Given:

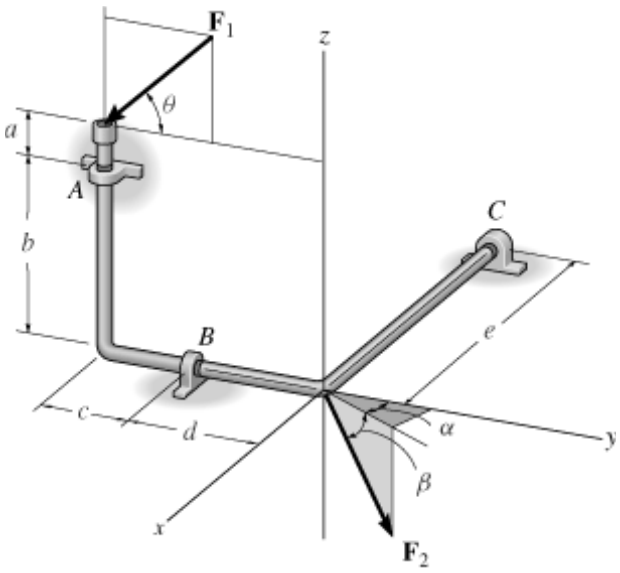
$F_1 = 300$ lb $\quad d = 3$ ft

$F_2 = 250$ lb $\quad e = 5$ ft

$a = 1$ ft $\quad \alpha = 30$ deg

$b = 4$ ft $\quad \beta = 45$ deg

$c = 2$ ft $\quad \theta = 45$ deg



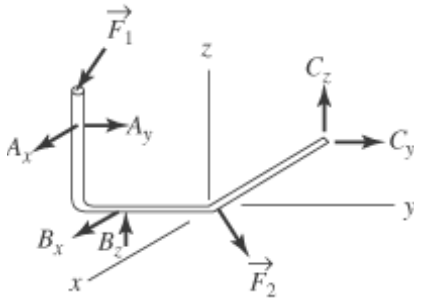
Solution:

The initial guesses:

$A_x = 100$ lb $\quad A_y = 200$ lb

$B_x = 300$ lb $\quad B_z = 400$ lb

$C_y = 500$ lb $\quad C_z = 600$ lb



Given

$$A_x + B_x + F_2 \cos(\beta)\sin(\alpha) = 0$$

$$A_y + C_y - F_1 \cos(\theta) + F_2 \cos(\beta)\cos(\alpha) = 0$$

$$B_z + C_z - F_1 \sin(\theta) - F_2 \sin(\beta) = 0$$

$$F_1 \cos(\theta)(a + b) + F_1 \sin(\theta)(c + d) - B_z d - A_y b = 0$$

$$A_x b + C_z e = 0$$

$$A_x(c + d) + B_x d - C_y e = 0$$

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$$\begin{pmatrix} A_x \\ A_y \\ B_x \\ B_z \\ C_y \\ C_z \end{pmatrix} = \text{Find}\left(A_x, A_y, B_x, B_z, C_y, C_z\right)$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 632.883 \\ -141.081 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} B_x \\ B_z \end{pmatrix} = \begin{pmatrix} -721.271 \\ 895.215 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} C_y \\ C_z \end{pmatrix} = \begin{pmatrix} 200.12 \\ -506.306 \end{pmatrix} \text{lb}$$

Problem 5-80

The bent rod is supported at *A*, *B*, and *C* by smooth journal bearings. Determine the magnitude of $\mathbf{F}_2$ which will cause the reaction $\mathbf{C}_y$ at the bearing *C* to be equal to zero. The bearings are in proper alignment and exert only force reactions on the rod.

Given:

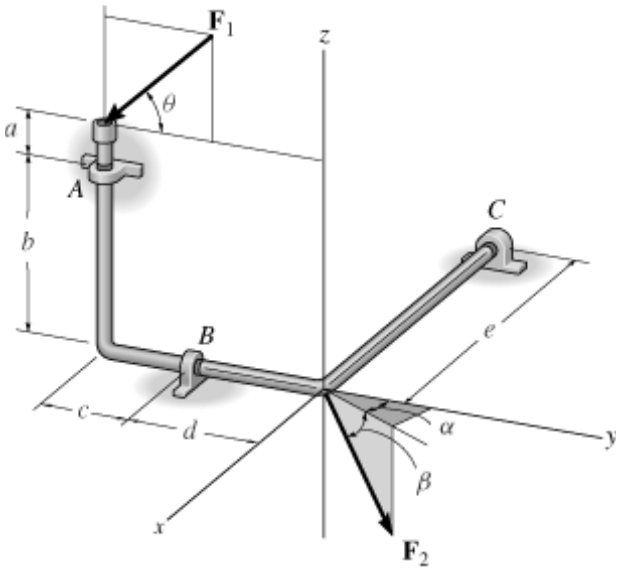
$F_1 = 300$ lb $\quad d = 3$ ft

$C_y = 0$ lb $\quad e = 5$ ft

$a = 1$ ft $\quad \alpha = 30$ deg

$b = 4$ ft $\quad \beta = 45$ deg

$c = 2$ ft $\quad \theta = 45$ deg



Solution:

The initial guesses:

$A_x = 100$ lb $\quad A_y = 200$ lb

$B_x = 300$ lb $\quad B_z = 400$ lb

$F_2 = 500$ lb $\quad C_z = 600$ lb

Given

$$A_x + B_x + F_2 \cos(\beta)\sin(\alpha) = 0$$

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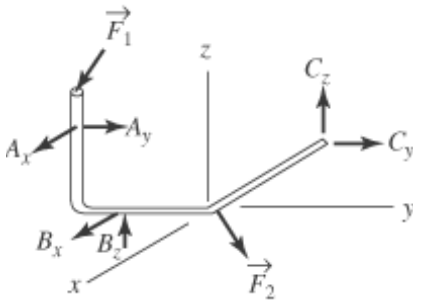
$$A_y + C_y - F_1\cos(\theta) + F_2\cos(\beta)\cos(\alpha) = 0$$

$$B_z + C_z - F_1\sin(\theta) - F_2\sin(\beta) = 0$$

$$F_1\cos(\theta)(a+b) + F_1\sin(\theta)(c+d) - B_z d - A_y b = 0$$

$$A_x b + C_z e = 0$$

$$A_x(c+d) + B_x d - C_y e = 0$$



$$\begin{pmatrix} A_x \\ A_y \\ B_x \\ B_z \\ C_z \\ F_2 \end{pmatrix} = \text{Find}(A_x, A_y, B_x, B_z, C_z, F_2) \qquad F_2 = 673.704\,\text{lb}$$

Problem 5-81

Determine the tension in cables *BD* and *CD* and the *x, y, z* components of reaction at the ball-and-socket joint at *A*.

Given:

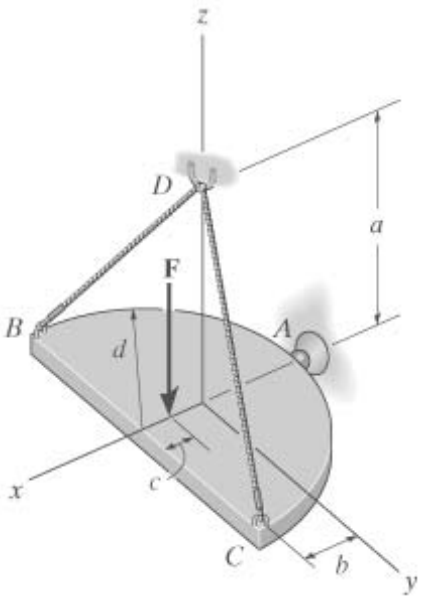
$F = 300$ N

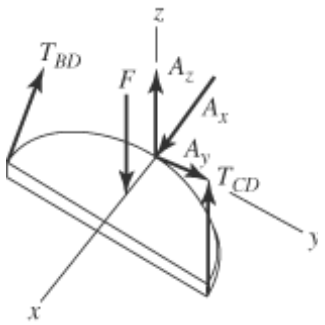
$a = 3$ m

$b = 1$ m

$c = 0.5$ m

$d = 1.5$ m





Solution:

$$r_{BD} = \begin{pmatrix} -b \\ d \\ a \end{pmatrix}$$

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$$r_{CD} = \begin{pmatrix} -b \\ -d \\ a \end{pmatrix}$$

Initial Guesses: $T_{BD} = 1\ \text{N}$ $T_{CD} = 1\ \text{N}$ $A_x = 1\ \text{N}$ $A_y = 1\ \text{N}$ $A_z = 1\ \text{N}$

Given

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} + T_{BD}\frac{r_{BD}}{|r_{BD}|} + T_{CD}\frac{r_{CD}}{|r_{CD}|} + \begin{pmatrix} 0 \\ 0 \\ -F \end{pmatrix} = 0$$

$$\begin{pmatrix} d \\ -d \\ 0 \end{pmatrix} \times \left(T_{BD}\frac{r_{BD}}{|r_{BD}|} \right) + \begin{pmatrix} d \\ d \\ 0 \end{pmatrix} \times \left(T_{CD}\frac{r_{CD}}{|r_{CD}|} \right) + \begin{pmatrix} d-c \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -F \end{pmatrix} = 0$$

$$\begin{pmatrix} T_{BD} \\ T_{CD} \\ A_x \\ A_y \\ A_z \end{pmatrix} = \text{Find}\left(T_{BD}, T_{CD}, A_x, A_y, A_z\right)$$

$$\begin{pmatrix} T_{BD} \\ T_{CD} \end{pmatrix} = \begin{pmatrix} 116.7 \\ 116.7 \end{pmatrix}\ \text{N}$$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} 66.7 \\ 0 \\ 100 \end{pmatrix}\ \text{N}$$

Problem 5-82

Determine the tensions in the cables and the components of reaction acting on the smooth collar at *A* necessary to hold the sign of weight *W* in equilibrium. The center of gravity for the sign is at *G*.

Given:

$W = 50\ \text{lb}$ $f = 2.5\ \text{ft}$

$a = 4\ \text{ft}$ $g = 1\ \text{ft}$

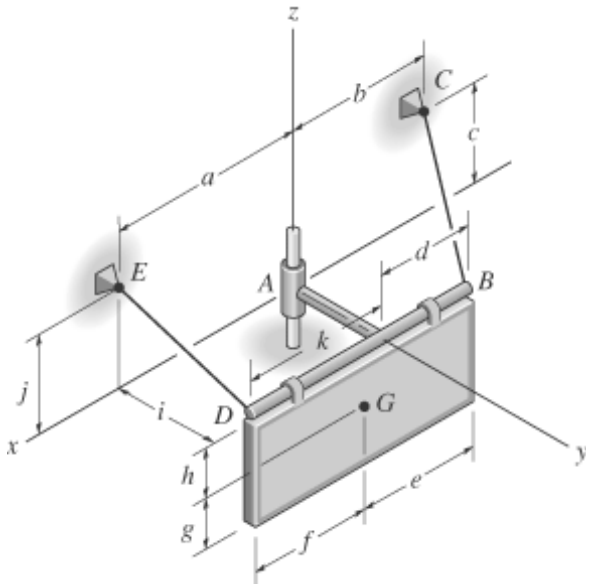
$b = 3\ \text{ft}$ $h = 1\ \text{ft}$

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$c = 2$ ft $\quad i = 2$ ft

$d = 2$ ft $\quad j = 2$ ft

$e = 2.5$ ft $\quad k = 3$ ft



Solution:

The initial guesses are:

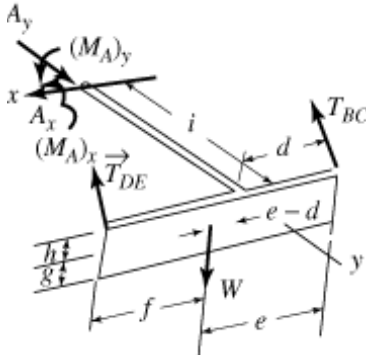
$T_{BC} = 10$ lb $\quad A_x = 10$ lb $\quad M_{Ax} = 10$ lb·ft

$T_{DE} = 10$ lb $\quad A_y = 10$ lb $\quad M_{Ay} = 10$ lb·ft

Given

$$(a-k)\frac{T_{DE}}{\sqrt{(a-k)^2+i^2+j^2}} + (-b+d)\frac{T_{BC}}{\sqrt{(d-b)^2+i^2+c^2}} + A_x = 0$$

$$-i\frac{T_{DE}}{\sqrt{(a-k)^2+i^2+j^2}} - i\frac{T_{BC}}{\sqrt{(d-b)^2+i^2+c^2}} + A_y = 0$$



$$j\frac{T_{DE}}{\sqrt{(a-k)^2+i^2+j^2}} + c\frac{T_{BC}}{\sqrt{(d-b)^2+i^2+c^2}} - W = 0$$

$$M_{Ax} + T_{DE}\,j\frac{i}{\sqrt{(a-k)^2+i^2+j^2}} + c\,T_{BC}\frac{i}{\sqrt{(d-b)^2+i^2+c^2}} - W\,i = 0$$

$$M_{Ay} - T_{DE}\,k\frac{j}{\sqrt{(a-k)^2+i^2+j^2}} + T_{BC}\,c\frac{d}{\sqrt{(d-b)^2+i^2+c^2}} + W(k-f) = 0$$

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$$-T_{DE}\,a\frac{i}{\sqrt{(a-k)^2+i^2+j^2}}+T_{BC}\,b\frac{i}{\sqrt{(d-b)^2+i^2+c^2}}=0$$

$$\begin{pmatrix} M_{Ax}\\ M_{Ay}\\ T_{BC}\\ T_{DE}\\ A_x\\ A_y \end{pmatrix} = \text{Find}\left(M_{Ax}, M_{Ay}, T_{BC}, T_{DE}, A_x, A_y\right)$$

$$\begin{pmatrix} T_{BC}\\ T_{DE} \end{pmatrix} = \begin{pmatrix} 42.857\\ 32.143 \end{pmatrix}\text{lb}$$

$$\begin{pmatrix} A_x\\ A_y \end{pmatrix} = \begin{pmatrix} 3.571\\ 50 \end{pmatrix}\text{lb}$$

$$\begin{pmatrix} M_{Ax}\\ M_{Ay} \end{pmatrix} = \begin{pmatrix} 2.698\times 10^{-13}\\ -17.857 \end{pmatrix}\text{lb}\cdot\text{ft}$$

Problem 5-83

The member is supported by a pin at *A* and a cable *BC*. If the load at *D* is *W*, determine the *x*, *y*, *z* components of reaction at these supports.

Units Used:

kip = 10^3 lb

Given:

$W = 300$ lb

$a = 1$ ft

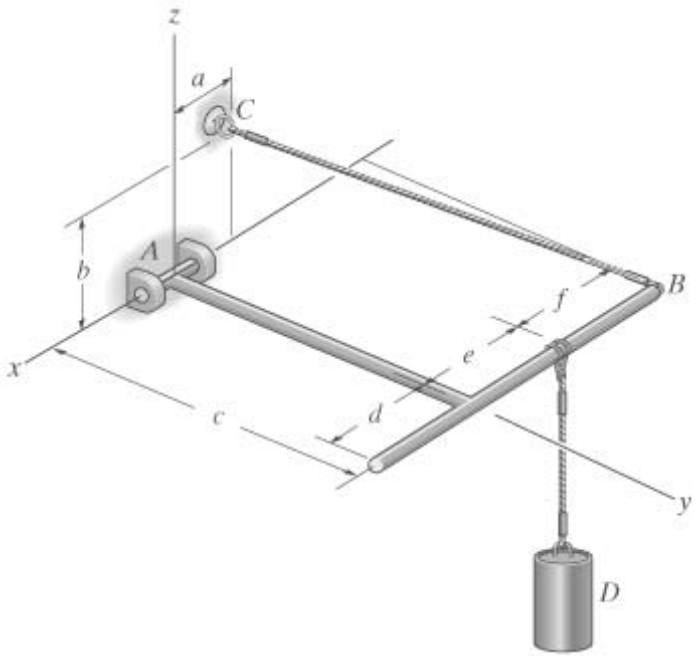
$b = 2$ ft

$c = 6$ ft

$d = 2$ ft

$e = 2$ ft

$f = 2$ ft



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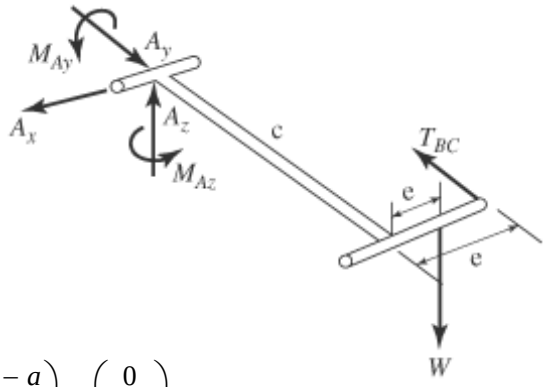
Solution:

Initial Guesses:

$T_{BC} = 1\ \text{lb}$ $A_x = 1\ \text{lb}$

$A_y = 1\ \text{lb}$ $A_z = 1\ \text{lb}$

$M_{Ay} = 1\ \text{lb ft}$ $M_{Az} = 1\ \text{lb ft}$



Given

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} + \frac{T_{BC}}{\sqrt{b^2 + c^2 + (e + f - a)^2}} \begin{pmatrix} e + f - a \\ -c \\ b \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -W \end{pmatrix} = 0$$

$$\begin{pmatrix} 0 \\ M_{Ay} \\ M_{Az} \end{pmatrix} + \begin{pmatrix} -e \\ c \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -W \end{pmatrix} + \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix} \times \left[\frac{T_{BC}}{\sqrt{b^2 + c^2 + (e + f - a)^2}} \begin{pmatrix} e + f - a \\ -c \\ b \end{pmatrix} \right] = 0$$

$$\begin{pmatrix} T_{BC} \\ A_x \\ A_y \\ A_z \\ M_{Ay} \\ M_{Az} \end{pmatrix} = \text{Find}\left(T_{BC}, A_x, A_y, A_z, M_{Ay}, M_{Az}\right)$$

$T_{BC} = 1.05\ \text{kip}$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} -450 \\ 900 \\ 0 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} M_{Ay} \\ M_{Az} \end{pmatrix} = \begin{pmatrix} -600 \\ -900 \end{pmatrix} \text{lb} \cdot \text{ft}$$

Problem 5-84

Determine the *x*, *y*, *z* components of reaction at the pin *A* and the tension in the cable *BC* necessary for equilibrium of the rod.

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Given:

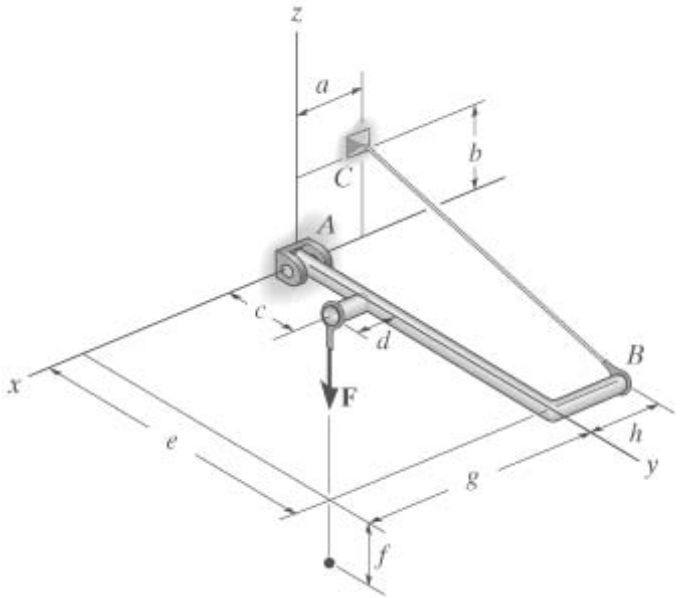
$F = 350$ lb $\quad e = 12$ ft

$a = 4$ ft $\quad f = 4$ ft

$b = 5$ ft $\quad g = 10$ ft

$c = 4$ ft $\quad h = 4$ ft

$d = 2$ ft $\quad i = 10$ ft



Solution:

Initial Guesses:

$F_{BC} = 1$ lb

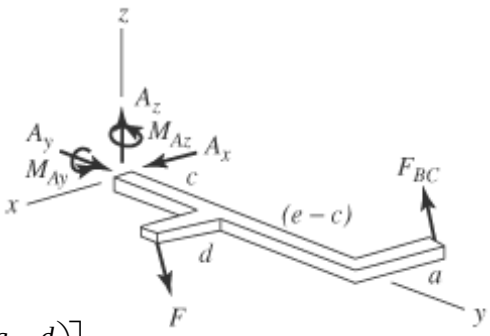
$A_y = 1$ lb

$M_{Ay} = 1$ lb·ft

$A_x = 1$ lb

$A_z = 1$ lb

$M_{Az} = 1$ lb·ft



Given

$$\begin{pmatrix} 0 \\ M_{Ay} \\ M_{Az} \end{pmatrix} + \begin{pmatrix} d \\ c \\ 0 \end{pmatrix} \times \left[\frac{F}{\sqrt{(g-d)^2 + (e-c)^2 + f^2}} \begin{pmatrix} g-d \\ e-c \\ -f \end{pmatrix} \right] \ldots = 0$$

$$+ \begin{pmatrix} -a \\ 0 \\ b \end{pmatrix} \times \left[\frac{F_{BC}}{\sqrt{(a-h)^2 + e^2 + b^2}} \begin{pmatrix} a-h \\ -e \\ b \end{pmatrix} \right]$$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} + \frac{F}{\sqrt{(g-d)^2 + (e-c)^2 + f^2}} \begin{pmatrix} g-d \\ e-c \\ -f \end{pmatrix} + \frac{F_{BC}}{\sqrt{(a-h)^2 + e^2 + b^2}} \begin{pmatrix} a-h \\ -e \\ b \end{pmatrix} = 0$$

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$$\begin{pmatrix} F_{BC} \\ A_x \\ A_y \\ A_z \\ M_{Ay} \\ M_{Az} \end{pmatrix} = \text{Find}\left(F_{BC}, A_x, A_y, A_z, M_{Ay}, M_{Az}\right) \qquad F_{BC} = 101\,\text{lb}$$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} -233.3 \\ -140 \\ 77.8 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} M_{Ay} \\ M_{Az} \end{pmatrix} = \begin{pmatrix} -388.9 \\ 93.3 \end{pmatrix} \text{lb}\cdot\text{ft}$$

Problem 5-85

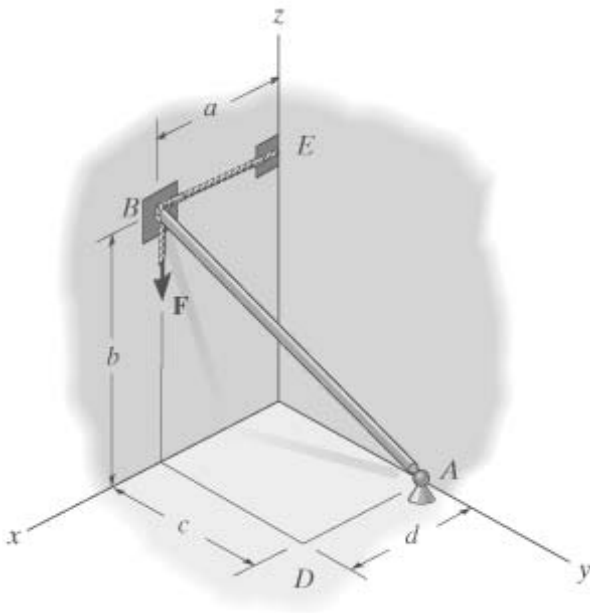
Rod *AB* is supported by a ball-and-socket joint at *A* and a cable at *B*. Determine the *x*, *y*, *z* components of reaction at these supports if the rod is subjected to a vertical force **F** as shown.

Given:

$F = 50$ lb

$a = 2$ ft $\quad c = 2$ ft

$b = 4$ ft $\quad d = 2$ ft



Solution:

$T_B = 10$ lb $\quad A_x = 10$ lb

$A_y = 10$ lb $\quad A_z = 10$ lb

$B_y = 10$ lb

Given

$\Sigma F_x = 0; \quad -T_B + A_x = 0$

$\Sigma F_y = 0; \quad A_y + B_y = 0$

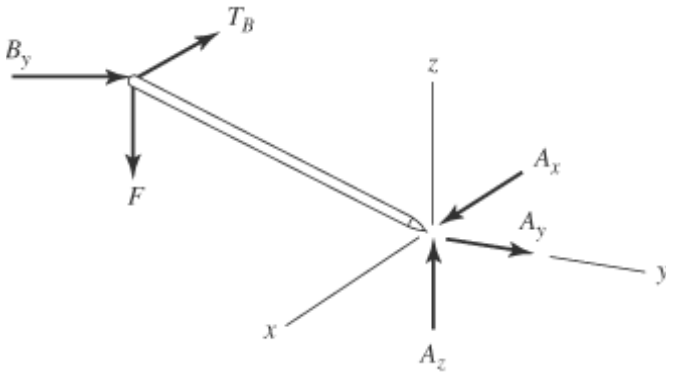
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$\Sigma F_z = 0; \quad -F + A_z = 0$

$\Sigma M_{Ax} = 0; \quad F(c) - B_y(b) = 0$

$\Sigma M_{Ay} = 0; \quad F(a) - T_B(b) = 0$

$\Sigma M_{Az} = 0; \quad B_y(a) - T_B(c) = 0$



Solving,

$$\begin{pmatrix} T_B \\ A_x \\ A_y \\ A_z \\ B_y \end{pmatrix} = \text{Find}\left(T_B, A_x, A_y, A_z, B_y\right) \qquad \begin{pmatrix} T_B \\ A_x \\ A_y \\ A_z \\ B_y \end{pmatrix} = \begin{pmatrix} 25 \\ 25 \\ -25 \\ 50 \\ 25 \end{pmatrix} \text{lb}$$

Problem 5-86

The member is supported by a square rod which fits loosely through a smooth square hole of the attached collar at *A* and by a roller at *B*. Determine the *x*, *y*, *z* components of reaction at these supports when the member is subjected to the loading shown.

Given:

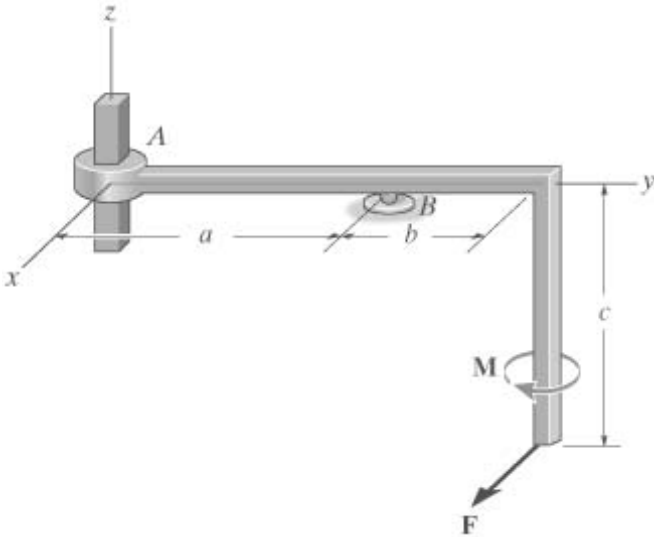
$M = 50 \text{ lb}\cdot\text{ft}$

$F = \begin{pmatrix} 20 \\ -40 \\ -30 \end{pmatrix} \text{lb}$

$a = 2 \text{ ft}$

$b = 1 \text{ ft}$

$c = 2 \text{ ft}$



Solution:

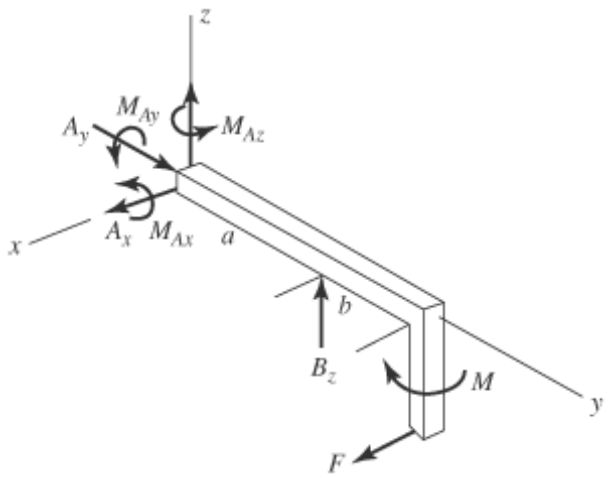
Initial Guesses

$A_x = 1 \text{ lb} \qquad A_y = 1 \text{ lb}$

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$M_{Ax} = 1\ \text{lb ft}$ $M_{Ay} = 1\ \text{lb ft}$

$M_{Az} = 1\ \text{lb ft}$ $B_z = 1\ \text{lb}$



Given

$$\begin{pmatrix} A_x \\ A_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ B_z \end{pmatrix} + F = 0$$

$$\begin{pmatrix} M_{Ax} \\ M_{Ay} \\ M_{Az} \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ B_z \end{pmatrix} + \left[\begin{pmatrix} 0 \\ a+b \\ -c \end{pmatrix} \times F + \begin{pmatrix} 0 \\ 0 \\ -M \end{pmatrix} \right] = 0$$

$$\begin{pmatrix} A_x \\ A_y \\ M_{Ax} \\ M_{Ay} \\ M_{Az} \\ B_z \end{pmatrix} = \text{Find}\left(A_x, A_y, M_{Ax}, M_{Ay}, M_{Az}, B_z\right)$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} -20 \\ 40 \end{pmatrix} \text{lb}$$

$$B_z = 30\,\text{lb}$$

$$\begin{pmatrix} M_{Ax} \\ M_{Ay} \\ M_{Az} \end{pmatrix} = \begin{pmatrix} 110 \\ 40 \\ 110 \end{pmatrix} \text{lb}\cdot\text{ft}$$

Problem 5-87

The platform has mass *M* and center of mass located at *G*. If it is lifted using the three cables, determine the force in each of these cables.

Units Used:

$\text{Mg} = 10^3\ \text{kg}$ $\text{kN} = 10^3\ \text{N}$ $g = 9.81\ \frac{\text{m}}{\text{s}^2}$

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Given:

$M = 3$ Mg

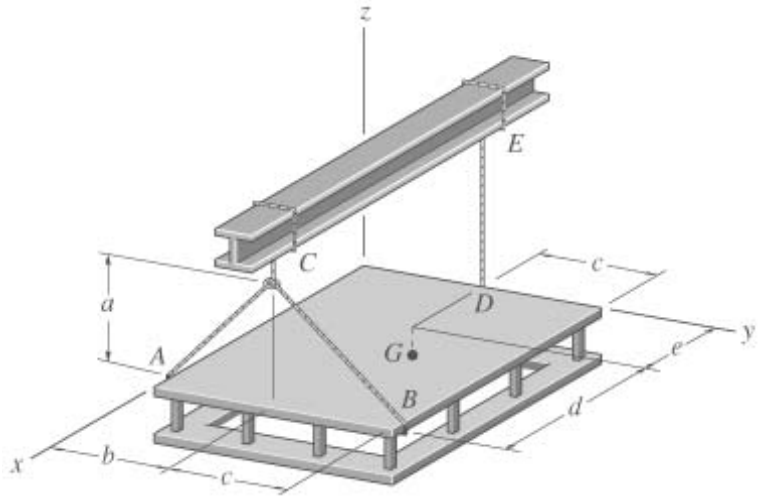
$a = 4$ m

$b = 3$ m

$c = 3$ m

$d = 4$ m

$e = 2$ m

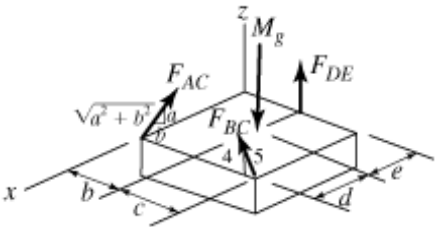


Solution:

The initial guesses are:

$F_{AC} = 10$ N $\quad F_{BC} = 10$ N $\quad F_{DE} = 10$ N

Given



$$\frac{b\,(F_{AC})}{\sqrt{a^2+b^2}} - \frac{c\,(F_{BC})}{\sqrt{a^2+c^2}} = 0$$

$$M\,g\,e - (F_{AC})a\frac{d+e}{\sqrt{a^2+b^2}} - F_{BC}\frac{a(d+e)}{\sqrt{a^2+c^2}} = 0$$

$$\frac{a}{\sqrt{a^2+c^2}}F_{BC}(b+c) - M\,g\,b + F_{DE}\,b = 0$$

$$\begin{pmatrix} F_{AC} \\ F_{BC} \\ F_{DE} \end{pmatrix} = \text{Find}(F_{AC}, F_{BC}, F_{DE}) \qquad \begin{pmatrix} F_{AC} \\ F_{BC} \\ F_{DE} \end{pmatrix} = \text{kN}$$

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Problem 5-88

The platform has a mass of M and center of mass located at G. If it is lifted using the three cables, determine the force in each of the cables. Solve for each force by using a single moment equation of equilibrium.

Units Used:

$Mg = 1000\ kg$

$kN = 10^3\ N$

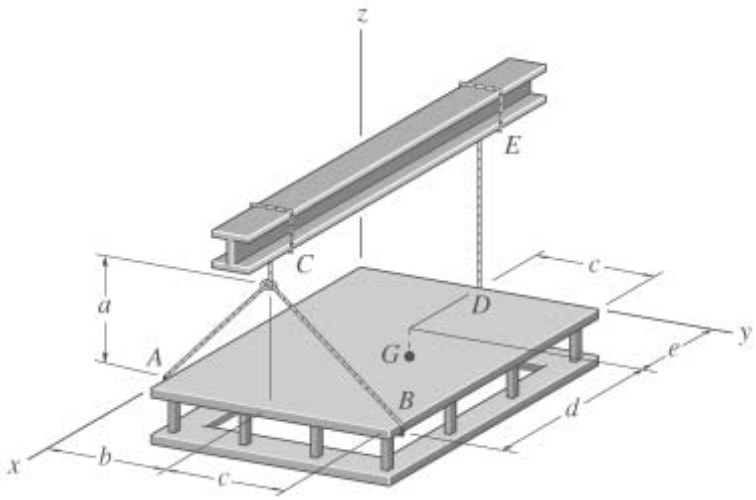
$g = 9.81\ \frac{m}{s^2}$

Given:

$M = 2\ Mg \qquad c = 3\ m$

$a = 4\ m \qquad d = 4\ m$

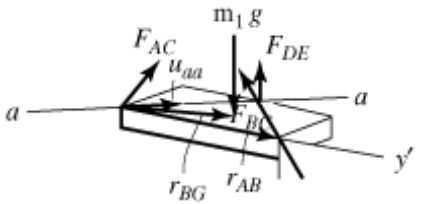
$b = 3\ m \qquad e = 2\ m$



Solution:

$$r_{BC} = \begin{pmatrix} 0 \\ -c \\ a \end{pmatrix} \qquad r_{AC} = \begin{pmatrix} 0 \\ b \\ a \end{pmatrix}$$

$$r_{AD} = \begin{pmatrix} -e-d \\ b \\ 0 \end{pmatrix} \qquad r_{BD} = \begin{pmatrix} -d-e \\ -c \\ 0 \end{pmatrix}$$



First find F_{DE}.

$\Sigma M_y' = 0; \qquad F_{DE}(d+e) - M\,g\,d = 0 \qquad F_{DE} = \frac{M\,g\,d}{d+e} \qquad F_{DE} = 4.1\ s^2\ kN$

Next find F_{BC}. Guess $F_{BC} = 1\ kN$

$$\text{Given} \quad \left[\begin{pmatrix} e \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -M\,g \end{pmatrix} + \begin{pmatrix} e+d \\ c \\ 0 \end{pmatrix} \times \left(F_{BC} \frac{r_{BC}}{|r_{BC}|} \right) \right] r_{AD} = 0 \qquad F_{BC} = \text{Find}(F_{BC})$$

$F_{BC} =$ kN

Now find F_{AC}. Guess $F_{AC} = 1\ kN$

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Given $\left[\begin{pmatrix} e \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -M\,g \end{pmatrix} + \begin{pmatrix} e+d \\ -b \\ 0 \end{pmatrix} \times \left(F_{AC} \frac{r_{AC}}{|r_{AC}|} \right)\right] r_{BD} = 0$ $F_{AC} = \text{Find}(F_{AC})$

$F_{AC} =$ kN

Problem 5-89

The cables exert the forces shown on the pole. Assuming the pole is supported by a ball-and-socket joint at its base, determine the components of reaction at A. The forces $\mathbf{F_1}$ and $\mathbf{F_2}$ lie in a horizontal plane.

Given:

$F_1 = 140$ lb

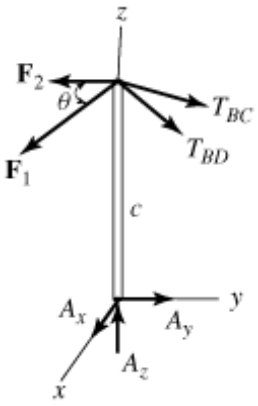
$F_2 = 75$ lb

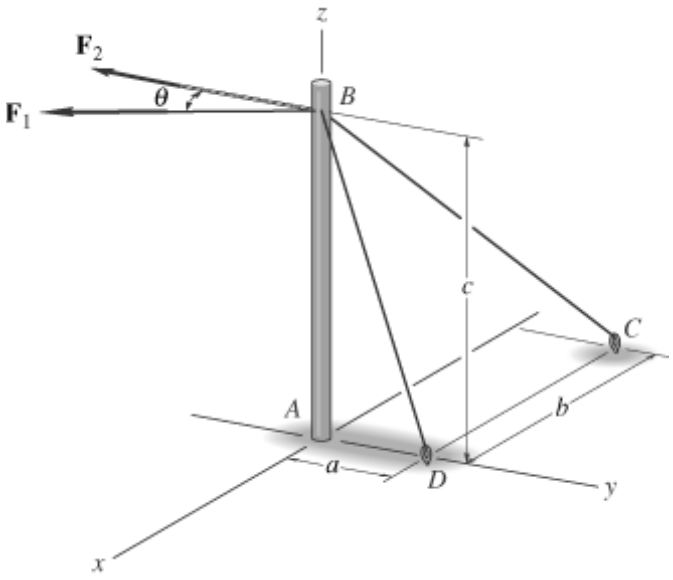
$\theta = 30$ deg

$a = 5$ ft

$b = 10$ ft

$c = 15$ ft





Solution:

The initial guesses are

$T_{BC} = 100$ lb $T_{BD} = 100$ lb $A_x = 100$ lb $A_y = 100$ lb $A_z = 100$ lb

Given

$$\left(F_1 \cos(\theta) + F_2\right)c - T_{BC}\,a\left(\frac{c}{\sqrt{a^2+b^2+c^2}}\right) - T_{BD}\,a\left(\frac{c}{\sqrt{a^2+c^2}}\right) = 0$$

$$F_1 \sin(\theta)c - b\,T_{BC}\left(\frac{c}{\sqrt{a^2+b^2+c^2}}\right) = 0$$

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$$A_x + F_1 \sin(\theta) - b\left(\frac{T_{BC}}{\sqrt{a^2 + b^2 + c^2}}\right) = 0$$

$$A_y - F_1 \cos(\theta) - F_2 + T_{BD}\left(\frac{a}{\sqrt{a^2 + c^2}}\right) + a\left(\frac{T_{BC}}{\sqrt{a^2 + b^2 + c^2}}\right) = 0$$

$$A_z - c\left(\frac{T_{BD}}{\sqrt{a^2 + c^2}}\right) - c\left(\frac{T_{BC}}{\sqrt{a^2 + b^2 + c^2}}\right) = 0$$

$$\begin{pmatrix} T_{BC} \\ T_{BD} \\ A_x \\ A_y \\ A_z \end{pmatrix} = \text{Find}(T_{BC}, T_{BD}, A_x, A_y, A_z) \qquad \begin{pmatrix} T_{BC} \\ T_{BD} \end{pmatrix} = \begin{pmatrix} 131.0 \\ 509.9 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} -0.0 \\ 0.0 \\ 588.7 \end{pmatrix} \text{lb}$$

Problem 5-90

The silo has a weight W, a center of gravity at G and a radius r. Determine the vertical component of force that each of the three struts at A, B, and C exerts on the silo if it is subjected to a resultant wind loading of **F** which acts in the direction shown.

Given:

$W = 3500$ lb

$F = 250$ lb

$\theta_1 = 30$ deg

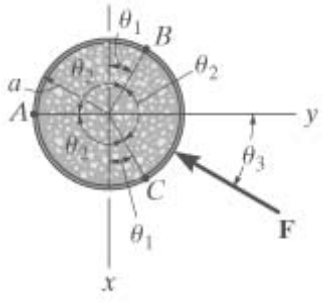
$\theta_2 = 120$ deg

$\theta_3 = 30$ deg

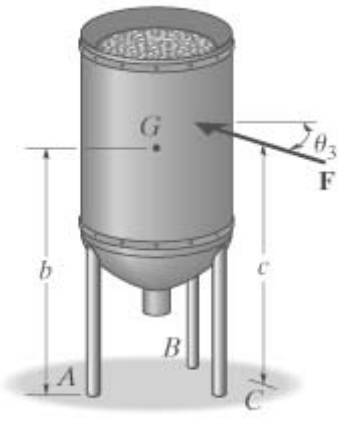
$r = 5$ ft

$b = 12$ ft

$c = 15$ ft



Top view



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Solution:

Initial Guesses: $A_z = 1$ lb $B_z = 2$ lb $C_z = 31$ lb

Given

$$\Sigma M_y = 0;\ B_z r\cos(\theta_1) - C_z r\cos(\theta_1) - F\sin(\theta_3)c = 0 \quad [1]$$

$$\Sigma M_x = 0;\ -B_z r\sin(\theta_1) - C_z r\sin(\theta_1) + A_z r - F\cos(\theta_3)c = 0 \quad [2]$$

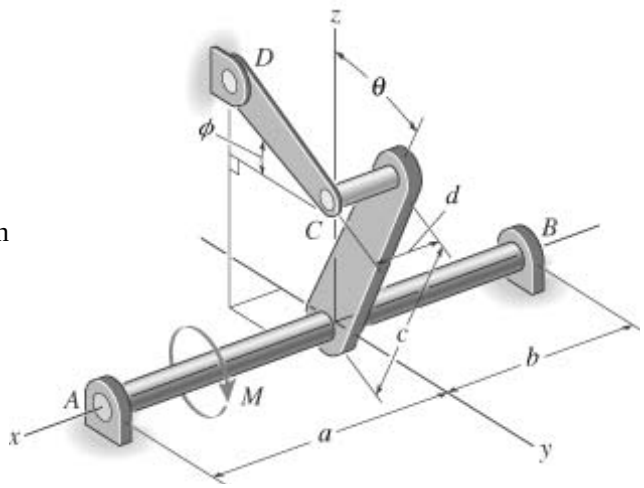
$$\Sigma F_z = 0;\ A_z + B_z + C_z = W \quad [3]$$

Solving Eqs.[1], [2] and [3] yields:

$$\begin{pmatrix} A_z \\ B_z \\ C_z \end{pmatrix} = \text{Find}(A_z, B_z, C_z) \qquad \begin{pmatrix} A_z \\ B_z \\ C_z \end{pmatrix} = \begin{pmatrix} 1600 \\ 1167 \\ 734 \end{pmatrix} \text{lb}$$

Problem 5-91

The shaft assembly is supported by two smooth journal bearings *A* and *B* and a short link *DC*. If a couple moment is applied to the shaft as shown, determine the components of force reaction at the bearings and the force in the link. The link lies in a plane parallel to the *y-z* plan and the bearings are properly aligned on the shaft.



Units Used:

$\text{kN} = 10^3$ N

Given:

$M = 250$ N·m

$a = 400$ mm

$b = 300$ mm

$c = 250$ mm

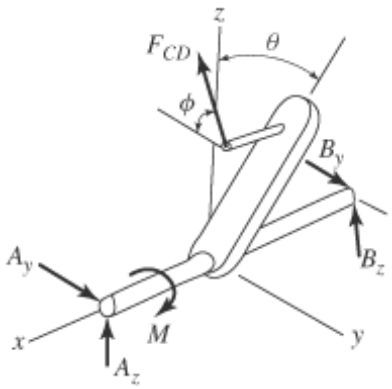
$d = 120$ mm

$\theta = 30$ deg

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$\phi = 20$ deg

Solution:



Initial Guesses:

$A_y = 1$ kN $\quad A_z = 1$ kN $\quad B_y = 1$ kN

$B_z = 1$ kN $\quad F_{CD} = 1$ kN

Given

$$\begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} + \begin{pmatrix} 0 \\ -F_{CD}\cos(\phi) \\ F_{CD}\sin(\phi) \end{pmatrix} = 0$$

$$\begin{pmatrix} d-a \\ c\sin(\theta) \\ c\cos(\theta) \end{pmatrix} \times \begin{pmatrix} 0 \\ -F_{CD}\cos(\phi) \\ F_{CD}\sin(\phi) \end{pmatrix} + \begin{pmatrix} -a-b \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} + \begin{pmatrix} -M \\ 0 \\ 0 \end{pmatrix} = 0$$

$$\begin{pmatrix} A_y \\ A_z \\ B_y \\ B_z \\ F_{CD} \end{pmatrix} = \text{Find}(A_y, A_z, B_y, B_z, F_{CD})$$

$$\begin{pmatrix} A_y \\ A_z \end{pmatrix} = \begin{pmatrix} 573 \\ -208 \end{pmatrix} \text{N}$$

$$\begin{pmatrix} B_y \\ B_z \end{pmatrix} = \begin{pmatrix} 382 \\ -139 \end{pmatrix} \text{N}$$

$$F_{CD} = 1.015\,\text{kN}$$

Problem 5-92

If neither the pin at *A* nor the roller at *B* can support a load no greater than F_{max}, determine the maximum intensity of the distributed load *w*, so that failure of a support does not occur.

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Units Used:

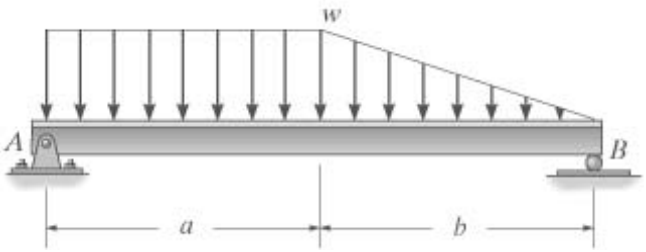
$kN = 10^3 N$

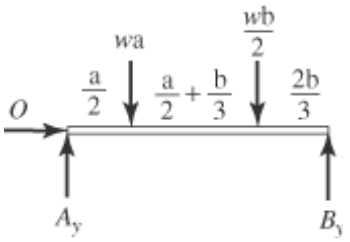
Given:

$F_{max} = 6\ kN$

$a = 3\ m$

$b = 3\ m$





Solution:

The greatest reaction is at A. Require

$$\Sigma M_B = 0; \quad -F_{max}(a+b) + w\,a\left(\frac{a}{2}+b\right) + \frac{1}{2}w\,b\,\frac{2}{3}b = 0$$

$$w = \frac{F_{max}(a+b)}{a\left(\frac{a}{2}+b\right)+\frac{b^2}{3}} \qquad w = 2.18\,\frac{kN}{m}$$

Problem 5-93

If the maximum intensity of the distributed load acting on the beam is w, determine the reactions at the pin A and roller B.

Units Used:

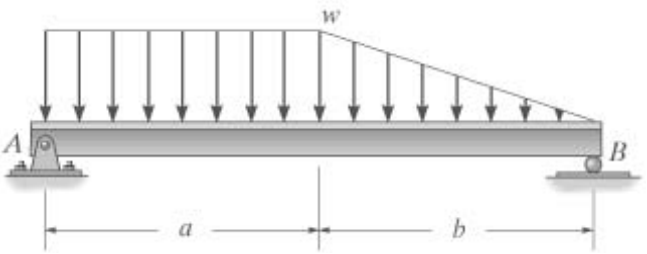
$kN = 10^3\ N$

Given:

$F = 6\ kN$

$a = 3\ m$

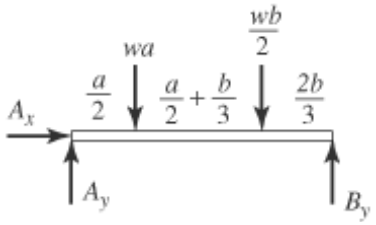
$b = 3\ m$



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$w = 4\ \frac{\text{kN}}{\text{m}}$

Solution:



$\Sigma F_x = 0;\qquad A_x = 0$

$\Sigma M_A = 0;\qquad -w\,a\frac{a}{2} - \frac{1}{2}\,w\,b\left(a + \frac{b}{3}\right) + B_y(a+b) = 0$

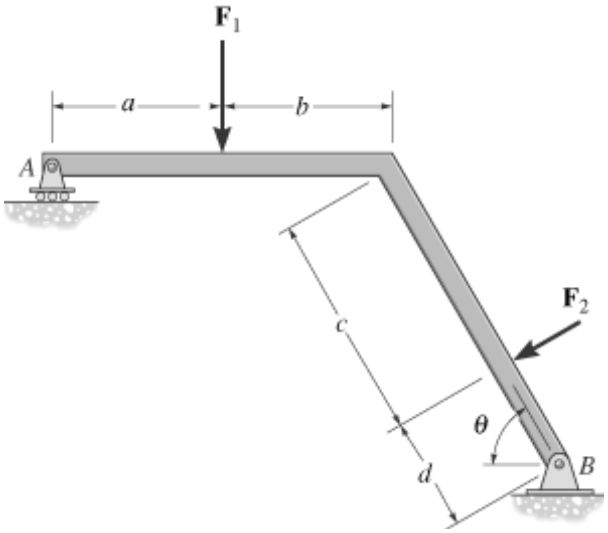
$B_y = \frac{1}{6}\,w\,\frac{3a^2 + 3\,a\,b + b^2}{a+b}\qquad B_y = 7\,\text{kN}$

$\Sigma F_y = 0;\qquad A_y + B_y - w\,a - \frac{1}{2}\,w\,b = 0$

$A_y = -B_y + w\,a + \frac{1}{2}\,w\,b\qquad A_y = 11\,\text{kN}$

Problem 5-94

Determine the normal reaction at the roller *A* and horizontal and vertical components at pin *B* for equilibrium of the member.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 10\ \text{kN}$

$F_2 = 6\ \text{kN}$

$a = 0.6\ \text{m}$

$b = 0.6\ \text{m}$

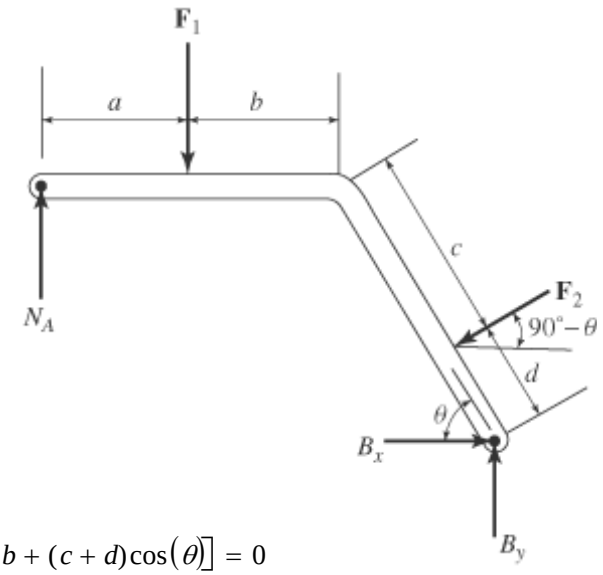
$c = 0.8\ \text{m}$

$d = 0.4\ \text{m}$

$\theta = 60\ \text{deg}$

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Solution:



Initial Guesses:

$N_A = 1$ kN

$B_x = 1$ kN

$B_y = 1$ kN

Given

$$B_x - F_2 \sin(\theta) = 0$$

$$B_y + N_A - F_1 - F_2 \cos(\theta) = 0$$

$$F_2 d + F_1[b + (c + d)\cos(\theta)] - N_A[a + b + (c + d)\cos(\theta)] = 0$$

$$\begin{pmatrix} N_A \\ B_x \\ B_y \end{pmatrix} = \text{Find}(N_A, B_x, B_y) \qquad \begin{pmatrix} N_A \\ B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 8 \\ 5.196 \\ 5 \end{pmatrix} \text{kN}$$

Problem 5-95

The symmetrical shelf is subjected to uniform pressure P. Support is provided by a bolt (or pin) located at each end A and A' and by the symmetrical brace arms, which bear against the smooth wall on both sides at B and B'. Determine the force resisted by each bolt at the wall and the normal force at B for equilibrium.

Units Used:

kPa $= 10^3$ Pa

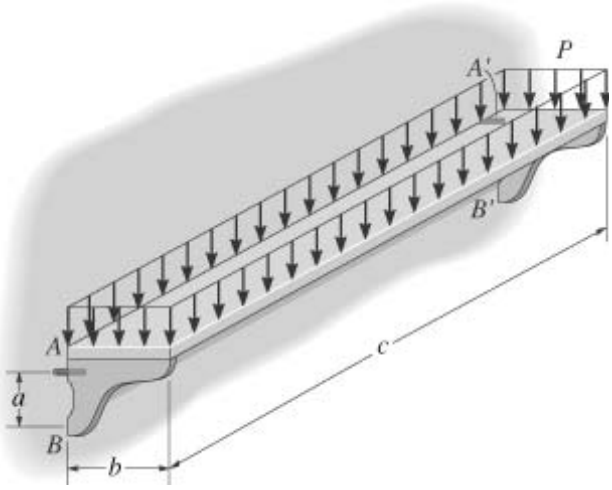
Given:

$P = 4$ kPa

$a = 0.15$ m

$b = 0.2$ m

$c = 1.5$ m



Solution:

$\Sigma M_A = 0;$

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$$N_B a - P\left(b\frac{c}{2}\right)\frac{b}{2} = 0$$

$$N_B = P\frac{b^2 c}{4\,a} \qquad N_B = 400 \text{ N}$$

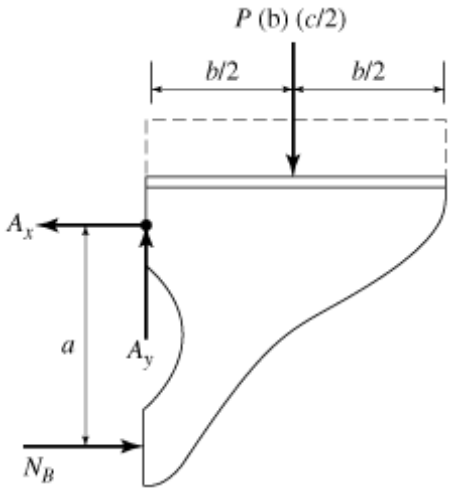
$\Sigma F_x = 0;$

$$A_x = N_B \qquad A_x = 400 \text{ N}$$

$\Sigma F_y = 0;$

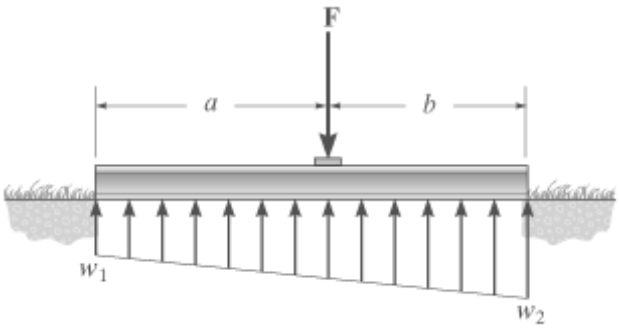
$$A_y = P\,b\frac{c}{2} \qquad A_y = 600 \text{ N}$$

$$F_A = \sqrt{A_x^2 + A_y^2} \qquad F_A = 721 \text{ N}$$



Problem 5-96

A uniform beam having a weight *W* supports a vertical load **F**. If the ground pressure varies linearly as shown, determine the load intensities w_1 and w_2 measured in lb/ft, necessary for equilibrium.



Given:

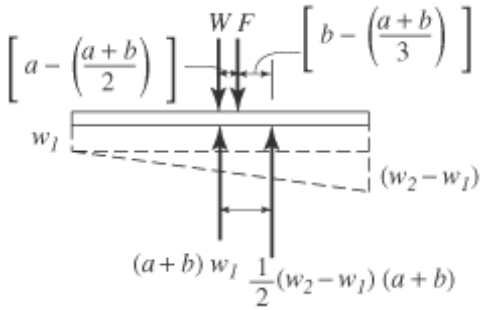
$W = 200$ lb

$F = 800$ lb

$a = 7$ ft

$b = 6$ ft

Solution:



Initial Guesses:

$$w_1 = 1\ \frac{\text{lb}}{\text{ft}} \qquad w_2 = 1\ \frac{\text{lb}}{\text{ft}}$$

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Given

$$w_1(a+b) + \frac{1}{2}(w_2 - w_1)(a+b) - F - W = 0$$

$$w_1(a+b)\frac{a+b}{2} + \frac{1}{2}(w_2 - w_1)(a+b)\frac{2}{3}(a+b) - W\frac{a+b}{2} - F\,a = 0$$

$$\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \text{Find}(w_1, w_2) \qquad \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 62.7 \\ 91.1 \end{pmatrix} \frac{\text{lb}}{\text{ft}}$$

Problem 5-97

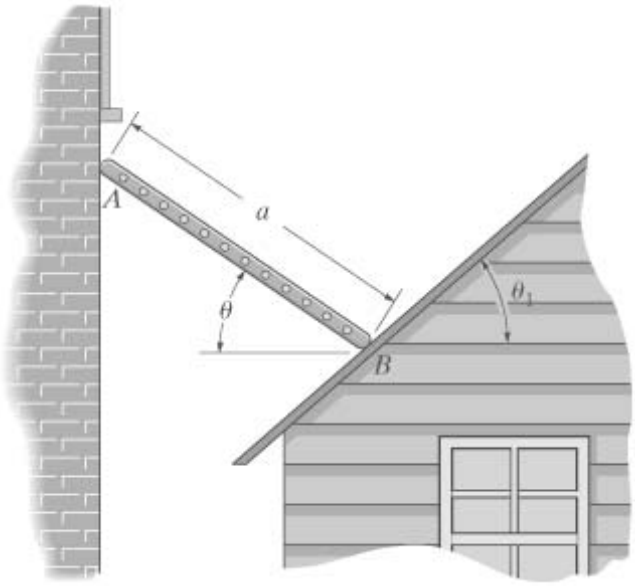
The uniform ladder rests along the wall of a building at A and on the roof at B. If the ladder has a weight W and the surfaces at A and B are assumed smooth, determine the angle θ for equilibrium.

Given:

$a = 18$ ft

$W = 25$ lb

$\theta_1 = 40$ deg



Solution:

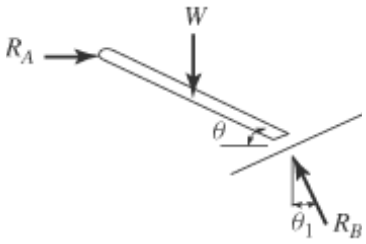
Initial guesses:

$R_A = 10$ lb

$R_B = 10$ lb

$\theta = 10$ deg

Given



$$\Sigma M_B = 0; \quad -R_A a \sin(\theta) + W\frac{a}{2}\cos(\theta) = 0$$

$$\Sigma F_x = 0; \quad R_A - R_B \sin(\theta_1) = 0$$

$$\Sigma F_y = 0; \quad R_B \cos(\theta_1) - W = 0$$

Solving,

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$$\begin{pmatrix} R_B \\ R_A \\ \theta \end{pmatrix} = \text{Find}(R_B, R_A, \theta) \qquad \begin{pmatrix} R_A \\ R_B \end{pmatrix} = \begin{pmatrix} 21 \\ 32.6 \end{pmatrix} \text{lb} \qquad \theta = 30.8\,\text{deg}$$

Problem 5-98

Determine the *x, y, z* components of reaction at the ball supports *B* and *C* and the ball-and-socket *A* (not shown) for the uniformly loaded plate.

Given:

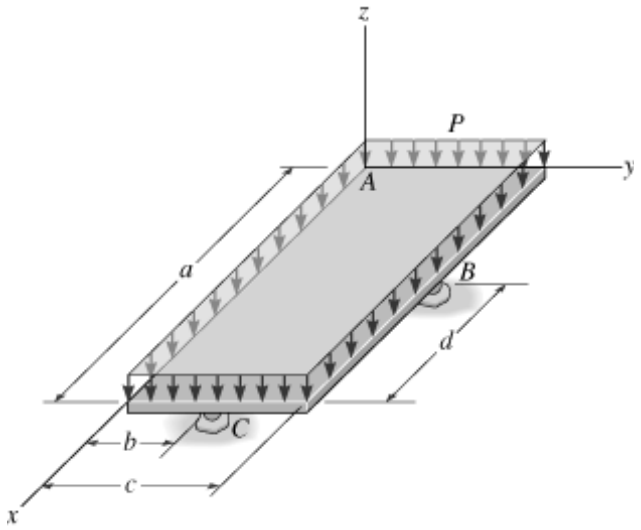
$P = 2\ \frac{\text{lb}}{\text{ft}^2}$

$a = 4$ ft

$b = 1$ ft

$c = 2$ ft

$d = 2$ ft



Solution:

The initial guesses are $A_x = 1$ lb $A_y = 1$ lb $A_z = 1$ lb $B_z = 1$ lb $C_z = 1$ lb

Given

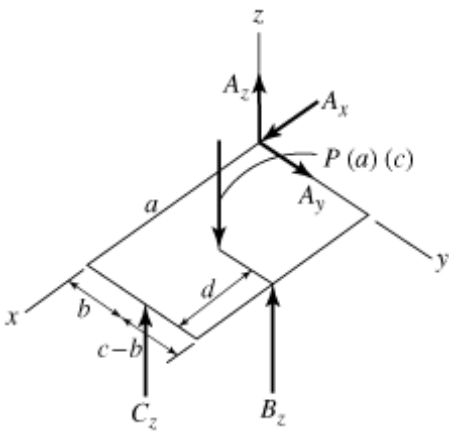
$\Sigma F_x = 0; \quad A_x = 0$

$\Sigma F_y = 0; \quad A_y = 0$

$\Sigma F_z = 0; \quad A_z + B_z + C_z - P\,a\,c = 0$

$\Sigma M_x = 0; \quad c\,B_z - P\,a\,c\left(\frac{c}{2}\right) + C_z\,b = 0$

$\Sigma M_y = 0; \quad -B_z\,(a-d) + P\,a\,c\left(\frac{a}{2}\right) - C_z\,a = 0$



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$$\begin{pmatrix} A_x \\ A_y \\ A_z \\ B_z \\ C_z \end{pmatrix} = \text{Find}(A_x, A_y, A_z, B_z, C_z) \qquad \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 5.333 \end{pmatrix} \text{lb} \qquad \begin{pmatrix} B_z \\ C_z \end{pmatrix} = \begin{pmatrix} 5.333 \\ 5.333 \end{pmatrix} \text{lb}$$

Problem 5-99

A vertical force **F** acts on the crankshaft. Determine the horizontal equilibrium force **P** that must be applied to the handle and the *x*, *y*, *z* components of force at the smooth journal bearing *A* and the thrust bearing *B*. The bearings are properly aligned and exert the force reactions on the shaft.

Given:

$F = 80$ lb

$a = 10$ in

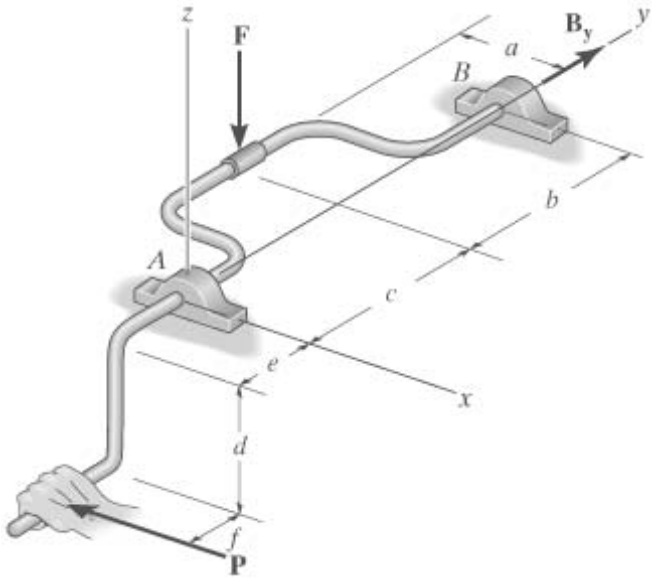
$b = 14$ in

$c = 14$ in

$d = 8$ in

$e = 6$ in

$f = 4$ in



Solution:

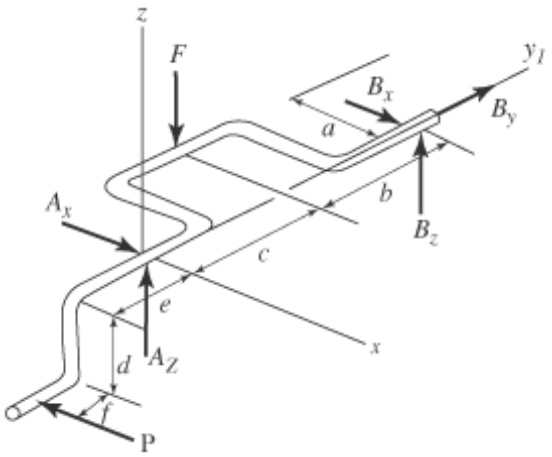
$\Sigma M_y = 0; \quad P\,d - F\,a = 0$

$$P = F\left(\frac{a}{d}\right) \qquad P = 100\,\text{lb}$$

$\Sigma M_x = 0; \quad B_z(b + c) - F\,c = 0$

$$B_z = F\left(\frac{c}{b + c}\right) \qquad B_z = 40\,\text{lb}$$

$\Sigma M_z = 0; \quad -B_x(b + c) - P(e + f) = 0$



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$$B_X = -P\left(\frac{e+f}{b+c}\right) \qquad B_X = -35.7\,\text{lb}$$

$\Sigma F_x = 0;$ $\quad A_X + B_X - P = 0$

$$A_X = -B_X + P \qquad A_X = 135.7\,\text{lb}$$

$\Sigma F_y = 0;$ $\quad B_y = 0$

$\Sigma F_z = 0;$ $\quad A_Z + B_Z - F = 0$

$$A_Z = -B_Z + F \qquad A_Z = 40\,\text{lb}$$

Problem 5-100

The horizontal beam is supported by springs at its ends. If the stiffness of the spring at A is k_A, determine the required stiffness of the spring at B so that if the beam is loaded with the force **F**, it remains in the horizontal position both before and after loading.

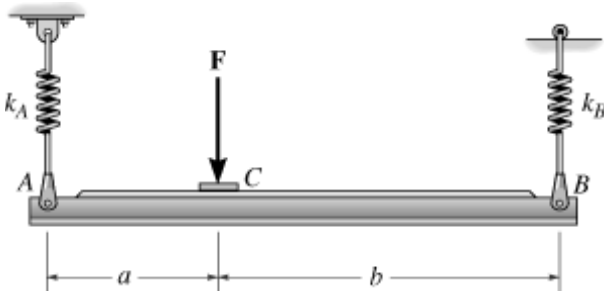
Units Used:

$$\text{kN} = 10^3\ \text{N}$$

Given:

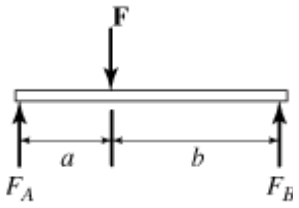
$$k_A = 5\ \frac{\text{kN}}{\text{m}} \qquad a = 1\ \text{m}$$

$$F = 800\ \text{N} \qquad b = 2\ \text{m}$$



Solution:

Equilibrium:



$\circlearrowleft + \Sigma M_A = 0;$ $\quad F_B(a+b) - F\,a = 0$

$$F_B = F\left(\frac{a}{a+b}\right)$$

$$F_B = 266.667\ \text{N}$$

$\circlearrowleft + \Sigma M_B = 0;$ $\quad F\,b - F_A(a+b) = 0$

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$$F_A = F\left(\frac{b}{a+b}\right)$$

$$F_A = 533.333 \text{ N}$$

Spring force formula:

$$x_A = x_B \qquad \frac{F_A}{k_A} = \frac{F_B}{k_B} \qquad k_B = \frac{F_B}{F_A}\,k_A \qquad k_B = 2.5\,\frac{\text{kN}}{\text{m}}$$

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